# STARFORGE

## OUTPOST STARBASE (TIER 0) — INTERIOR FIT-OUT DRAWINGS

Architectural and ventilation (HVAC) design at true scale, for modelling the full interior.

| SHEET | TITLE | SCALE |
|---|---|---|
| G-001 | COVER · INDEX · NOTES · LEGEND | — |
| G-002 | DESIGN CRITERIA · COMPARTMENTATION · EGRESS | NTS |
| A-101 | GENERAL ARRANGEMENT | 1:500 |
| A-102 | SECTION A-A | 1:200 |
| A-201 | RING DECK A · QUADRANT 1 | 1:100 |
| A-202 | RING DECK A · QUADRANT 2 | 1:100 |
| A-203 | RING DECK A · QUADRANT 3 | 1:100 |
| A-204 | RING DECK A · QUADRANT 4 | 1:100 |
| A-205 | RING DECK B · QUADRANT 1 | 1:100 |
| A-206 | RING DECK B · QUADRANT 2 | 1:100 |
| A-207 | RING DECK B · QUADRANT 3 | 1:100 |
| A-208 | RING DECK B · QUADRANT 4 | 1:100 |
| A-211 | COMPARTMENT C01 · PLANS 1:50 | 1:50 |
| A-212 | COMPARTMENT C02 · PLANS 1:50 | 1:50 |
| A-213 | COMPARTMENT C03 · PLANS 1:50 | 1:50 |
| A-214 | COMPARTMENT C04 · PLANS 1:50 | 1:50 |
| A-215 | COMPARTMENT C05 · PLANS 1:50 | 1:50 |
| A-216 | COMPARTMENT C06 · PLANS 1:50 | 1:50 |
| A-217 | COMPARTMENT C07 · PLANS 1:50 | 1:50 |
| A-218 | COMPARTMENT C08 · PLANS 1:50 | 1:50 |
| A-219 | COMPARTMENT C09 · PLANS 1:50 | 1:50 |
| A-220 | COMPARTMENT C10 · PLANS 1:50 | 1:50 |
| A-221 | COMPARTMENT C11 · PLANS 1:50 | 1:50 |
| A-222 | COMPARTMENT C12 · PLANS 1:50 | 1:50 |
| A-301 | RING SECTIONS · BULKHEAD | 1:25 · 1:50 |
| A-302 | SPOKE | 1:50 · 1:20 |
| A-401 | SPINE LEVELS | 1:50 |
| A-402 | SPINE LEVELS | 1:50 |
| A-403 | SPINE LEVELS | 1:50 |
| A-404 | SPINE LEVELS | 1:50 |
| A-405 | SPINE LEVELS | 1:50 |
| A-406 | DOCKING HUB | 1:100 |
| A-501 | ENLARGED PLANS | 1:25 · 1:50 |
| A-601 | INTERIOR CONCEPT SKETCHES | NTS |
| M-001 | ECLSS AND AIR DISTRIBUTION SCHEMATIC | NTS |
| M-002 | SCHEDULES AND CALCULATIONS | — |
| M-201 | RING VENTILATION · DECK A · QUADRANT 1 | 1:100 |
| M-202 | RING VENTILATION · DECK A · QUADRANT 2 | 1:100 |
| M-203 | RING VENTILATION · DECK A · QUADRANT 3 | 1:100 |
| M-204 | RING VENTILATION · DECK A · QUADRANT 4 | 1:100 |
| M-205 | RING VENTILATION · DECK B · QUADRANT 1 | 1:100 |
| M-206 | RING VENTILATION · DECK B · QUADRANT 2 | 1:100 |
| M-207 | RING VENTILATION · DECK B · QUADRANT 3 | 1:100 |
| M-208 | RING VENTILATION · DECK B · QUADRANT 4 | 1:100 |
| M-301 | HUB NODE · SPOKE CHASE · SPINE RISERS | 1:50 · 1:20 · NTS |

### GENERAL NOTES

1. REV B · ONE BUILD: THE EXTERIOR (assets/stations/outpost/build.py) READS docs/blueprints/openings.json, WRITTEN FROM THESE PLANS: EVERY WINDOW, SKYLIGHT, SPOKE PORTAL AND HANGAR DOOR OUTSIDE IS A ROOM INSIDE.
2. UNITS: DIMENSIONS IN MILLIMETRES; RADII IN mm FROM THE STATION AXIS; ANGLES IN DEGREES.
3. AXES: +Z = STATION AXIS TOWARD THE COMMAND DOME. θ = 0 ON +X (SPOKE 1), POSITIVE COUNTER-CLOCKWISE SEEN FROM +Z. SPOKES AT 0 / 90 / 180 / 270°.
4. GRAVITY IS GENERATED: 1.0 g, DOWN = −Z EVERYWHERE (SPINE, SPOKES, RING, HUB). ALL FLOORS ARE FLAT; THE RING'S CURVED FACES ARE WALLS. RING DECK A FFL z −200 (= SPOKE FLOOR = SPINE HUB DECK), DECK B FFL z −3 600.
5. RING PLANS ARE TRUE PLANS FROM +Z; THEIR GEOMETRY IS WRITTEN AT R 40 000 (x ALONG THE RING, y = r − 40 000).
6. 12 PRESSURE BULKHEADS B1–B12 (θ = 15° + 30°·(n−1)) MAKE 12 COMPARTMENTS C01–C12, EACH ABLE TO HOLD 101.3 kPa WITH ITS NEIGHBOURS EVACUATED. EVERY OPENING IN A BULKHEAD CLOSES AUTOMATICALLY.
7. ISO 128 LINES · ISO 129 DIMENSIONS · ISO 3098 LETTERING · ISO 5455 SCALES · ISO 5457 / 7200 SHEETS.
8. DXF UNDERLAYS (1:1, mm, LAYERED) ISSUED WITH THIS SET FOR MODELLING: docs/blueprints/outpost/dxf/.
9. ROOM NUMBERS: DECK · COMPARTMENT · ROOM, e.g. A05-03 = DECK A, C05, ROOM 03. SCHEDULE: room_schedule.csv.

### ABBREVIATIONS

AHU AIR HANDLING UNIT · BW BLACKWATER · CHWS/R CHILLED WATER SUPPLY/RETURN
EA EXHAUST AIR · ECLSS ENV. CONTROL & LIFE SUPPORT · EL EMERGENCY LOCKER
ER ELECTRICAL RISER · FFL FINISHED FLOOR LEVEL · FP FIRE POINT · FSD FIRE/SMOKE/PRESSURE DAMPER
GW GREYWATER · OA REVITALISED AIR FROM THE HUB ECLSS · PBA PORTABLE BREATHING APPARATUS
PD PRESSURE DOOR · PW POTABLE WATER · RA RETURN AIR · RH RETURN-TO-HUB AIR
RS RETURN-AIR SHAFT · SA SUPPLY AIR · SH SPOKE ISOLATION HATCH · TCCS TRACE CONTAM. CONTROL
WR WARDROBE / WATER RISER · 4BMS 4-BED MOLECULAR SIEVE ($CO_2$) · OGA $O_2$ GENERATION ASSEMBLY

### LEGEND

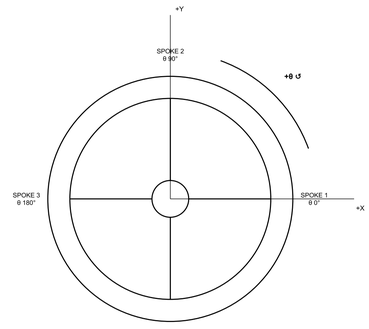
- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

### LEGEND · VENTILATION

- SUPPLY AIR DUCT (SA)
- RETURN AIR (RA) / RETURN-TO-HUB (RH)
- EXHAUST DUCT (EA) TO TCCS
- REVITALISED AIR FROM HUB (OA)
- DUCT IN SUBFLOOR PLENUM (BELOW)
- CEILING DIFFUSER (SUPPLY)
- RETURN / EXHAUST GRILLE
- ISOLATION DAMPER AT BULKHEAD (FSD)

### OCCUPANCY (COUNTED FROM THE PLANS)

| BERTHS | ROOMS | × | PERSONS |
|---|---|---|---|
| SINGLE CABINS · DECK A · WINDOWED | 28 | 1 | 28 |
| DOUBLE CABINS · DECK B | 28 | 2 | 56 |
| TRANSIENT BUNK ROOMS · DECK B C09 | 6 | 4 | 24 |
| DESIGN OCCUPANCY (ECLSS SIZED FOR 120) | | | 108 |




**STATION AXES · VIEWED FROM +Z**

| | | | |
|---|---|---|---|
| LEGAL OWNER / PROJECT: **STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT** | | | |
| TITLE: **COVER · INDEX · NOTES · LEGEND** — Outpost starbase interior fit-out · drawing set | | | |
| DISCIPLINE: General | DOCUMENT STATUS: ISSUED FOR MODELLING | SCALE: NTS | SIZE / UNITS: A1 · mm |
| CREATED BY: Claude (AI) | APPROVED BY: — | DATE OF ISSUE: 2026-10-17 | PROJECTION |
| IDENTIFICATION NUMBER: **SF-OUT-G-001** | | REV: **A** | SHEET: **G-001** |

## GRAVITY AND LEVELS (GENERATED 1.0 g, DOWN = −Z)

| LEVEL | z mm | CLEAR | NOTE |
|---|---|---|---|
| RING ROOF SKIN | +4 600 | | SKYLIGHTS OVER THE CORRIDOR |
| DECK A CEILING | +3 000 | | ROOF VOID 1 000 ABOVE |
| DECK A FFL · SPOKE FLOOR · SPINE HUB DECK | -200 | 3 200 | MAIN HABITAT · W1 / W3 WINDOWS |
| DECK B CEILING | -900 | | INTERSTITIAL VOID 400 + SLAB 300 |
| DECK B FFL | -3 500 | 2 600 | SERVICES / BERTHING · W2 · HANGARS |
| RING FLOOR SKIN | -4 600 | | SUBFLOOR PLENUM 500 ABOVE |

ONE GRAVITY FOR THE WHOLE STATION: WALK LEVEL FROM THE SPINE HUB DECK ALONG A SPOKE INTO RING DECK A.
THE RING STILL TURNS SLOWLY (COSMETIC / STATION-KEEPING); IT CARRIES NO LOAD FOR GRAVITY.
STAIRS DECK A ↔ B: 18 RISERS 183, GOINGS 260, IN EVERY SPOKE JUNCTION, BESIDE THE LIFT.

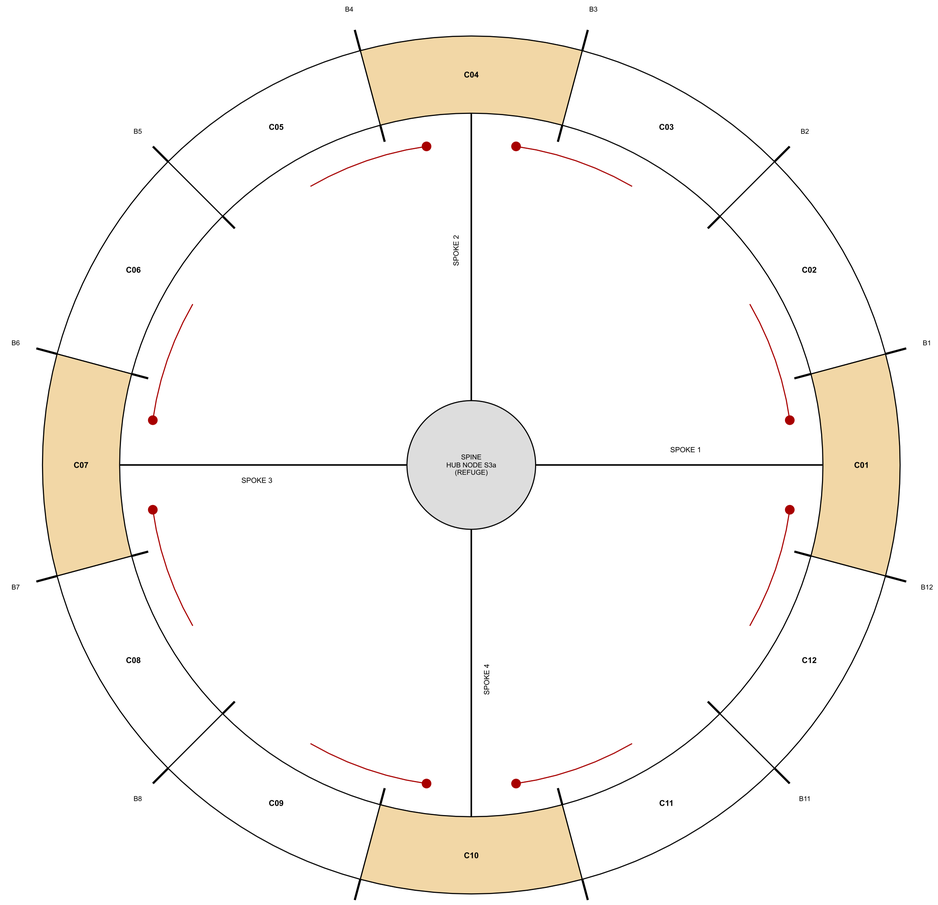
## ATMOSPHERE AND ENVIRONMENT

| PARAMETER | DESIGN VALUE |
|---|---|
| TOTAL PRESSURE | 101.3 kPa |
| $O_2$ PARTIAL PRESSURE | 21.3 kPa (19.5–23.1) |
| $CO_2$ PARTIAL PRESSURE | ≤ 0.4 kPa (3 mmHg) |
| TEMPERATURE | 18–27 °C · 22 °C NOMINAL |
| RELATIVE HUMIDITY | 25–70 % · 45 % NOMINAL |
| VENTILATION | 6 AIR CHANGES / h, ALL HABITABLE ZONES |
| AIR VELOCITY, OCCUPIED ZONE | 0.08–0.20 m/s |
| FILTRATION | HEPA H13 AT EVERY AHU |
| NOISE | ≤ NC-40 BERTHS · ≤ NC-50 WORK AREAS |
| REVITALISED AIR EXCHANGE | 12 L/s PER PERSON (HUB ECLSS) |

## COMPARTMENT PROGRAMME

| COMP. | θ | DECK A | DECK B | SERVED BY |
|---|---|---|---|---|
| C01 | −15–15° | SPOKE JUNCTION · CONCOURSE · SAFE HAVEN | SPOKE LOBBY · AHU PLANT · SUBSTATION | AHU-R1 |
| C02 | 15–45° | HYDROPONICS | NUTRIENTS · HARVEST · SEED BANK | AHU-R1 |
| C03 | 45–75° | HYDROPONICS | WATER STORE · ALGAE · PUMPS | AHU-R2 |
| C04 | 75–105° | SPOKE JUNCTION · CONCOURSE · SAFE HAVEN | SPOKE LOBBY · AHU PLANT · SUBSTATION | AHU-R2 |
| C05 | 105–135° | QUARTERS (SINGLES) | QUARTERS (DOUBLES) | AHU-R2 |
| C06 | 135–165° | QUARTERS (SINGLES) | QUARTERS (DOUBLES) | AHU-R3 |
| C07 | 165–195° | SPOKE JUNCTION · CONCOURSE · SAFE HAVEN | SPOKE LOBBY · AHU PLANT · SUBSTATION | AHU-R3 |
| C08 | 195–225° | GALLEY · WARDROOM | COLD / DRY STORES · LAUNDRY | AHU-R3 |
| C09 | 225–255° | GYM · LOUNGE · HANGAR AIRLOCK | TRANSIENT BERTHING | AHU-R4 |
| C10 | 255–285° | SPOKE JUNCTION · CONCOURSE · SAFE HAVEN | SPOKE LOBBY · AHU PLANT · SUBSTATION | AHU-R4 |
| C11 | 285–315° | FABRICATION WORKSHOP | STORES · DATA NODE | AHU-R4 |
| C12 | 315–345° | MEDICAL CENTRE | LABORATORY | AHU-R1 |

**AREAS FROM THE PLANS: DECK A 1678 m² · DECK B 1567 m² · RING 3245 m² · SPINE DECK PLATE ≈ 1619 m²**



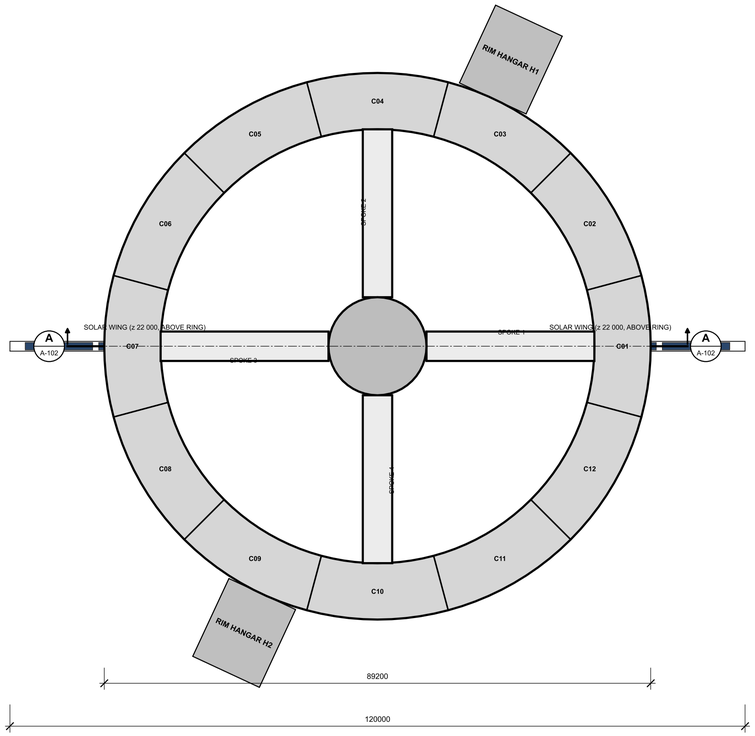

**PRESSURE COMPARTMENTATION AND EGRESS · DIAGRAM VIEWED FROM +Z · NOT TO SCALE**

- 12 RING COMPARTMENTS, EACH A PRESSURE ZONE BETWEEN TWO 600 BULKHEADS AND THE HULL.
- EACH BULKHEAD HAS A SLIDING PRESSURE DOOR ON EACH DECK (PD=A 1200×2100 · PD=B 1000×2000) THAT CLOSES ON 2 kPa DIFFERENTIAL OR ON COMMAND AND OPENS MANUALLY FROM EITHER SIDE.
- SHADED: SPOKE JUNCTION COMPARTMENTS — LIFT AND LADDER TO THE HUB, AND A 30-PERSON SAFE HAVEN ($O_2$ + LiOH, 4 h) ON DECK A.
- RED: PRIMARY EGRESS FROM EACH COMPARTMENT TO THE NEAREST JUNCTION (ONE BULKHEAD). SECONDARY: THE OTHER WAY ROUND.
- SPOKE ISOLATION HATCHES SH AT BOTH ENDS OF EVERY SPOKE. HUB NODE S3a IS THE LAST REFUGE; DOCKED CRAFT ARE THE LIFEBOATS.
- SMOKE: THE ZONE'S AHU CLOSES ITS DAMPERS AND RUNS EXHAUST-ONLY. DETECTION IN EVERY ROOM AND PLENUM.

LEGAL OWNER / PROJECT
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

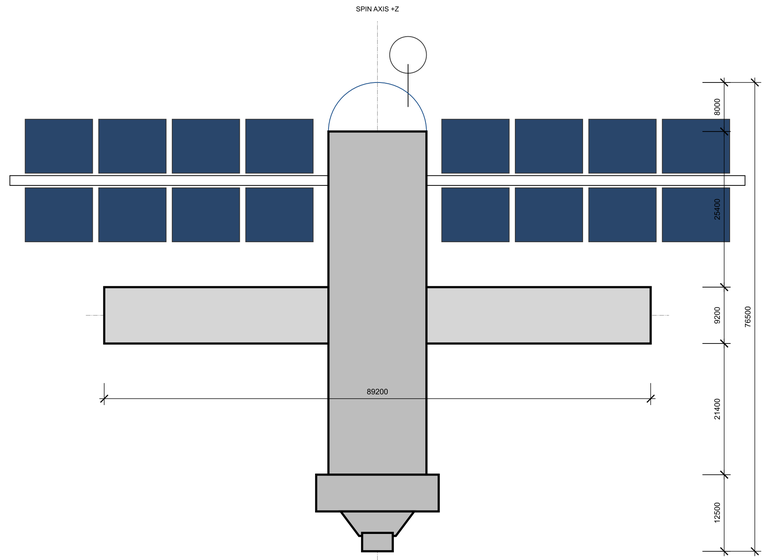
TITLE
# DESIGN CRITERIA · COMPARTMENTATION · EGRESS
Spin, gravity, atmosphere, areas; pressure compartments, safe havens

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| General | ISSUED FOR MODELLING | NTS | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-G-002 | | A | G-002 |




## 1 STATION PLAN · VIEWED FROM +Z

SCALE 1:500




## 2 STATION ELEVATION · VIEWED ALONG +Y

SCALE 1:500

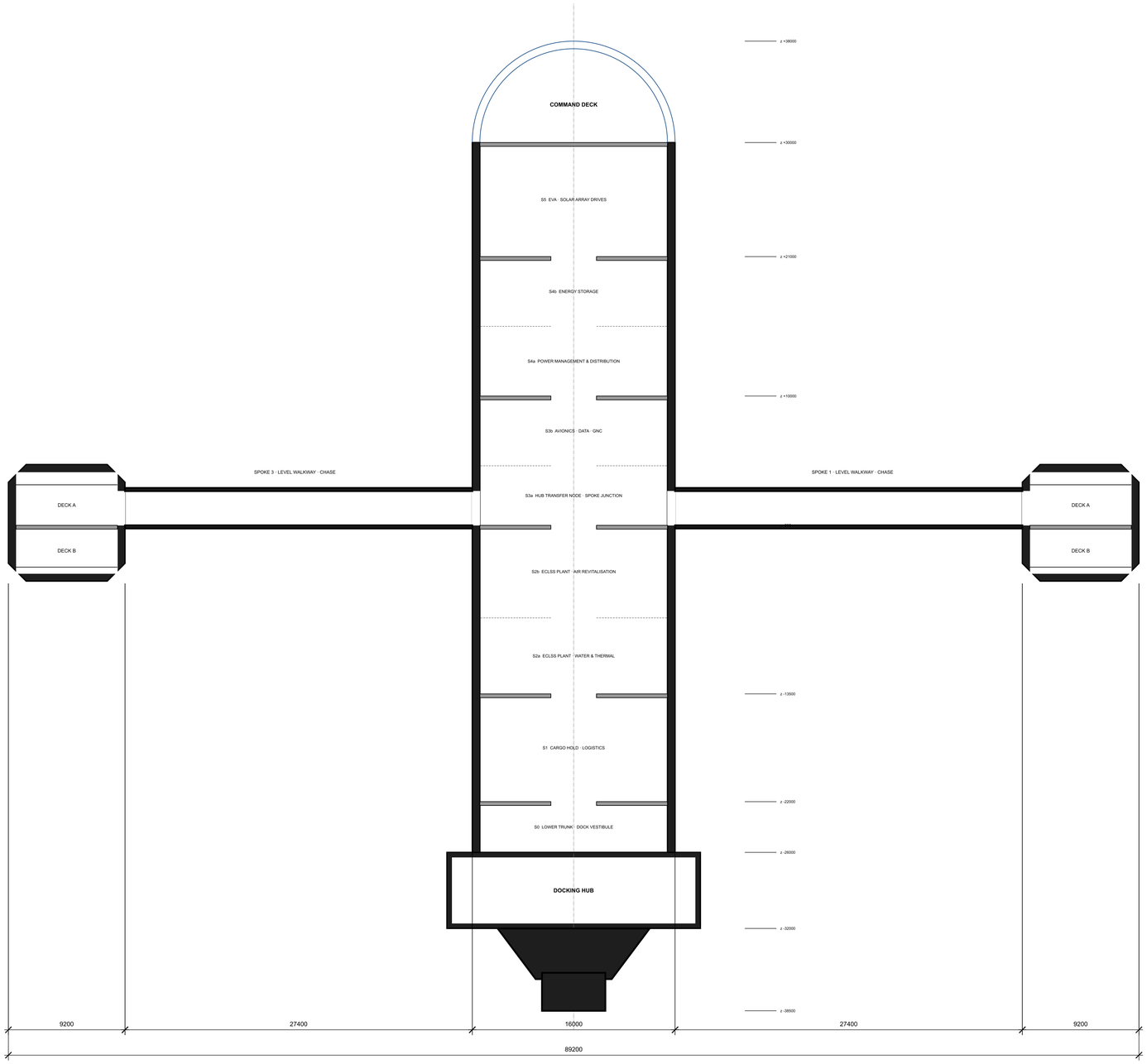
**ENVELOPE (FROM build.py)**

RING: OUTER RADIUS 44 600 · HUB SIDE 36 400 · AXIAL WIDTH 9 200 · CHAMFERS 1 400 · 12 BULKHEADS · 96 FRAMES.
SPINE: Ø16 000 OUTSIDE, Ø14 800 INSIDE, z −26 000 TO +30 000; COMMAND DOME Ø16 000 HEMISPHERE ON TOP.
SPOKES: 4 × 4 800 × 3 200 TUBES AT 0/90/180/270°, FLOOR z −200, FROM THE SPINE (r 8 000) TO THE RING (r 35 400).
DOCKING HUB: Ø20 000 × 6 000 BELOW THE SPINE, 4 ARMS TO PORTS AT r 25 350, AXIAL PORT P0 BELOW.
SOLAR WINGS ±X AT z 22 000 (8 PANELS 11 × 9 m EACH SIDE) · RADIATORS ±Y AT z −12 000 · TANK FARM z −18 000.
RIM HANGARS H1 (@ 65°) / H2 (@ 245°): 12 000 W × 6 000 H × 14 000 DEEP, OUT FROM THE RING'S FLOOR SIDE.

LEGAL OWNER / PROJECT
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

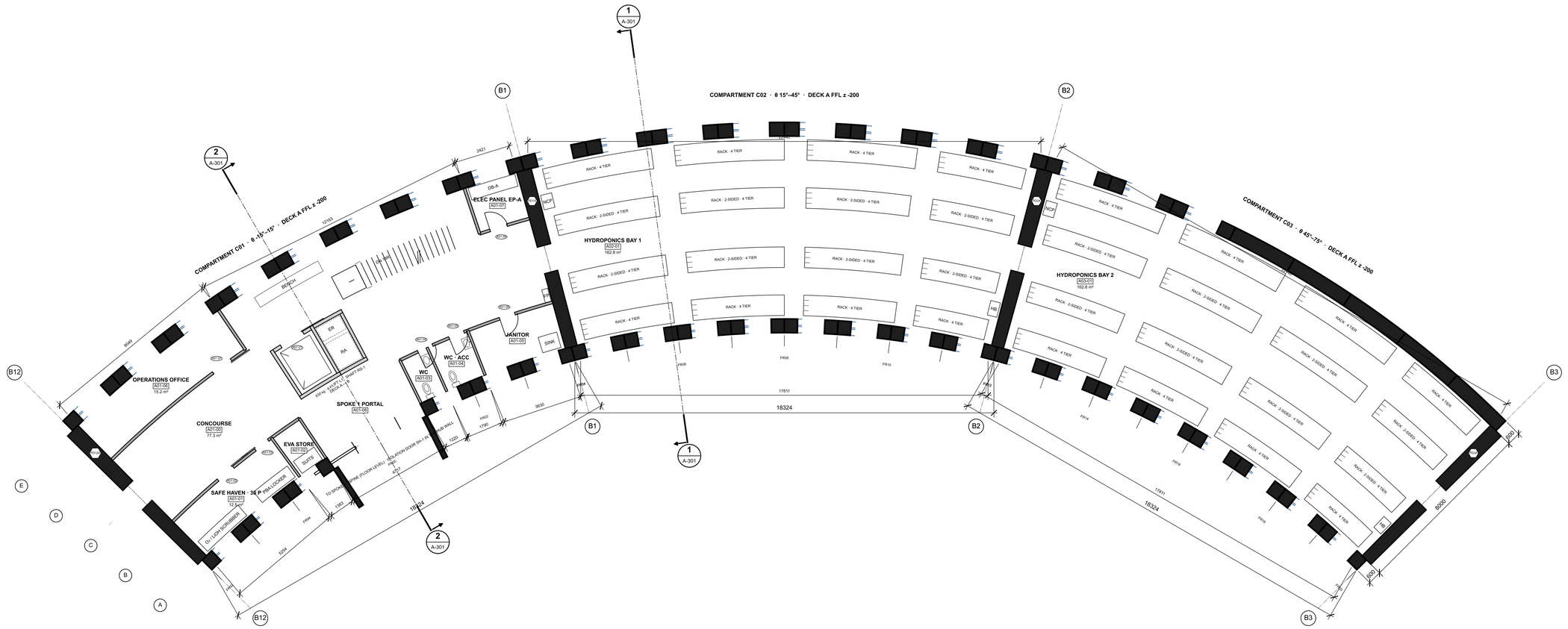
TITLE
# GENERAL ARRANGEMENT
Station plan and elevation · exterior envelope

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-101 | | A | A-101 |




**A** **SECTION A-A**
SCALE 1:200

| | |
|---|---|
| LEGAL OWNER / PROJECT | **STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT** |
| TITLE | **SECTION A-A**<br>Meridional section on X–Z through Spokes 1 and 3 |
| DISCIPLINE: Architecture | DOCUMENT STATUS: ISSUED FOR MODELLING |
| SCALE: AS SHOWN | SIZE / UNITS: A1 · mm |
| CREATED BY: Claude (AI) | APPROVED BY: — |
| DATE OF ISSUE: 2026-10-17 | PROJECTION |
| IDENTIFICATION NUMBER: **SF-OUT-A-102** | REV: **A** |
| | SHEET: **A-102** |




## 1 DECK A (MAIN HABITAT) · FFL Z −200 · QUADRANT 1

SCALE 1:100

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

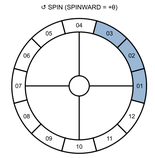
### NOTES

1. TRUE PLAN FROM +Z. GRAVITY 1.0 g DOWN = −Z (GENERATED); FLOORS FLAT, THE HULL'S CURVED FACES ARE WALLS.
2. OUTWARD (+r) IS UP THE SHEET AT THE MIDDLE OF THE VIEW. ONE WINDOW PER FRAME BAY (3.75°), SEE openings.json.
3. PLAN CUT 1 200 ABOVE FLOOR. DIMENSIONS mm TO FACE OF STRUCTURE; ALONG THE RING, CHORDS AT THE HUB-WALL SKIN.
4. PARTITIONS 120 LIGHTWEIGHT, NON-STRUCTURAL; BULKHEADS ARE PRESSURE BOUNDARIES (SEE G-002).
5. EVERY BULKHEAD PENETRATION (DUCT, PIPE, CABLE) IS FITTED WITH AN ISOLATION DEVICE (SEE M-20x).
6. CABIN DOORS 800 CLEAR, SLIDING, 25 mm UNDERCUT FOR RETURN AIR TRANSFER.
7. WINDOWS: W1 OUTER WALL DECK A 1 600×1 800 · W2 OUTER WALL DECK B 1 400×1 200 · W3 HUB WALL DECK A 1 200×1 600.

GRAPHIC SCALE 1:100

0 2 4 6 8 10 12 14 16 18 20 m

↺ SPIN (SPINWARD = +θ)




KEY PLAN · VIEWED FROM +Z (DOME)

LEGAL OWNER / PROJECT

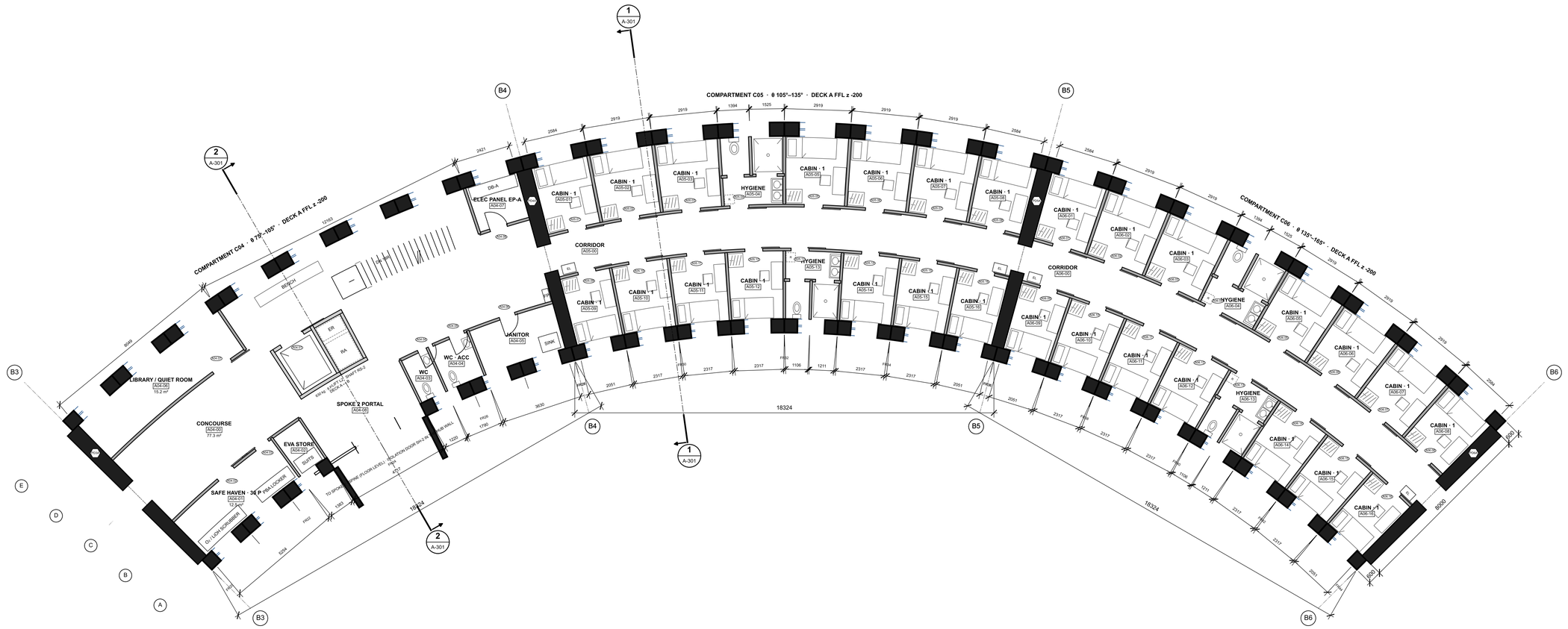
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE

**RING DECK A · QUADRANT 1**

Compartments C01–C03 · plan

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-201 | | A | A-201 |




## 1 DECK A (MAIN HABITAT) · FFL Z −200 · QUADRANT 2

SCALE 1:100

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

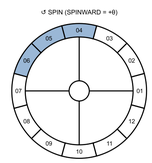
### NOTES

1. TRUE PLAN FROM +Z. GRAVITY 1.0 g DOWN = −Z (GENERATED); FLOORS FLAT, THE HULL'S CURVED FACES ARE WALLS.
2. OUTWARD (+r) IS UP THE SHEET AT THE MIDDLE OF THE VIEW. ONE WINDOW PER FRAME BAY (3.75°), SEE openings.json.
3. PLAN CUT 1 200 ABOVE FLOOR. DIMENSIONS mm TO FACE OF STRUCTURE; ALONG THE RING, CHORDS AT THE HUB-WALL SKIN.
4. PARTITIONS 120 LIGHTWEIGHT, NON-STRUCTURAL; BULKHEADS ARE PRESSURE BOUNDARIES (SEE G-002).
5. EVERY BULKHEAD PENETRATION (DUCT, PIPE, CABLE) IS FITTED WITH AN ISOLATION DEVICE (SEE M-20x).
6. CABIN DOORS 800 CLEAR, SLIDING, 25 mm UNDERCUT FOR RETURN AIR TRANSFER.
7. WINDOWS: W1 OUTER WALL DECK A 1 600×1 800 · W2 OUTER WALL DECK B 1 400×1 200 · W3 HUB WALL DECK A 1 200×1 600.

GRAPHIC SCALE 1:100

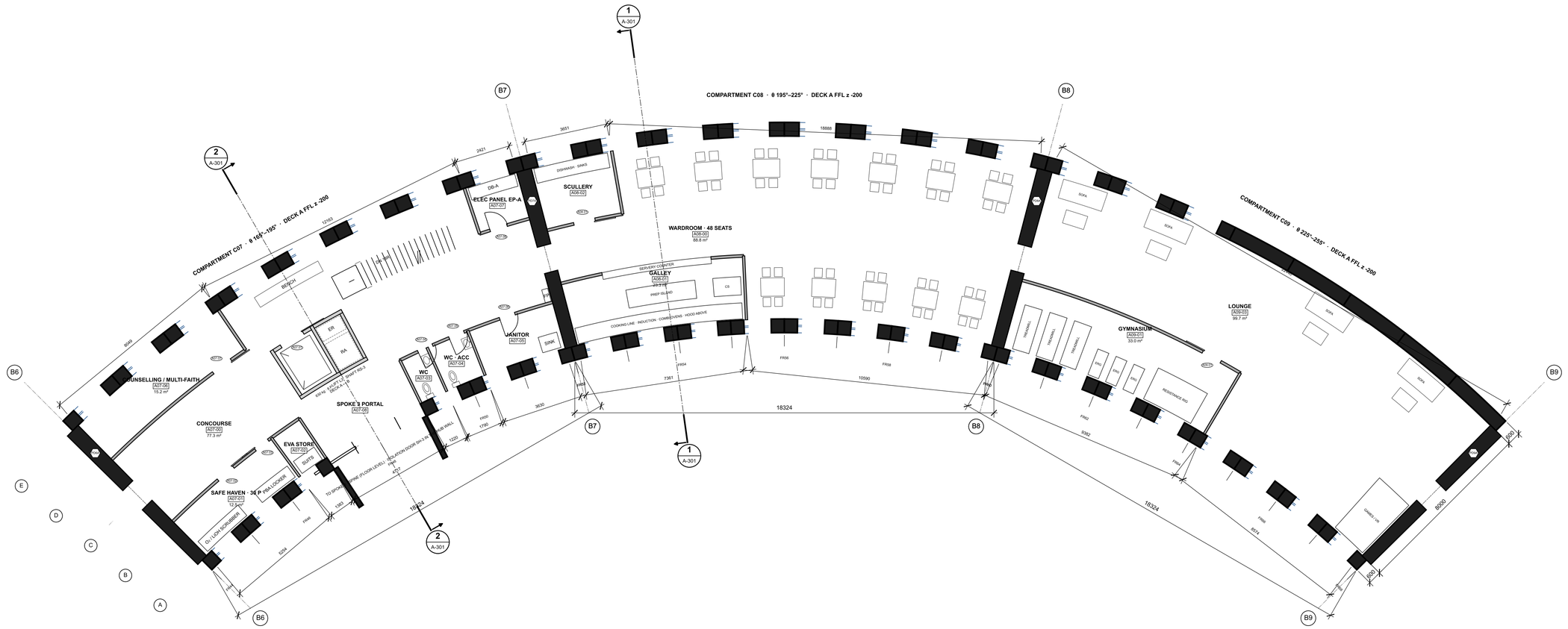
0 2 4 6 8 10 12 14 16 18 20 m

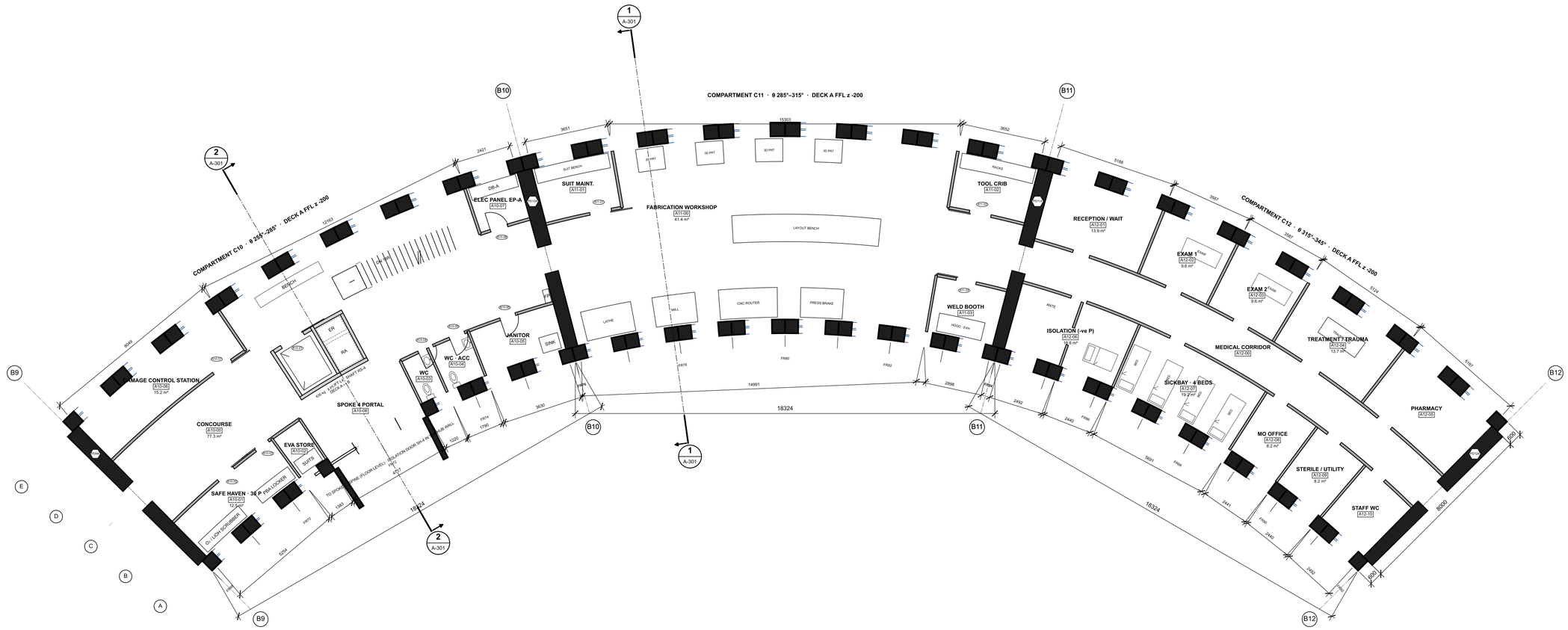
↺ SPIN (SPINWARD = +θ)




KEY PLAN · VIEWED FROM +Z (DOME)

| | |
|---|---|
| LEGAL OWNER / PROJECT | STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT |
| TITLE | RING DECK A · QUADRANT 2<br>Compartments C04–C06 · plan |
| DISCIPLINE | Architecture |
| DOCUMENT STATUS | ISSUED FOR MODELLING |
| SCALE | AS SHOWN |
| SIZE / UNITS | A1 · mm |
| CREATED BY | Claude (AI) |
| APPROVED BY | — |
| DATE OF ISSUE | 2026-10-17 |
| PROJECTION | |
| IDENTIFICATION NUMBER | SF-OUT-A-202 |
| REV | A |
| SHEET | A-202 |


COMPARTMENT C07 · θ 165°–195° · DECK A FFL z -200
COMPARTMENT C08 · θ 195°–225° · DECK A FFL z -200
COMPARTMENT C09 · θ 225°–255° · DECK A FFL z -200
COUNSELLING / MULTI-FAITH
A07-06
15.2 m²
CONCOURSE
A07-00
77.3 m²
SPOKE 3 PORTAL
A07-08
SAFE HAVEN · 36 P
A07-01
13.3 m²
EVA STORE
A07-02
SUITS
ER
RA
WC
A07-03
WC · ACC
A07-04
JANITOR
A07-05
SCULLERY
A08-02
ELEC PANEL EP-A
A07-01
WARDROOM · 48 SEATS
A08-00
88.8 m²
GALLEY
A08-01
23.5 m²
SERVERY COUNTER
PREP ISLAND
COOKING LINE · INDUCTION · COMBI OVENS · HOOD ABOVE
SINK
DISHWASH · SINKS
GYMNASIUM
A09-01
33.0 m²
LOUNGE
A09-03
99.7 m²
SOFA
TREADMILL
RESISTANCE RIG
GUARD 1.4m
BENCH
DB-A
FR54
FR56
FR58
FR60
FR62
FR64
FR66
FR68
FR74
7361
10590
18324
3630
1790
1220
4727
5254
1363
8383
8674
8005
8000
2421
3651
18589
1763
8040
1400
1200
1000
800
600
TO SPOKE 3 PORTAL (FLOOR LEVEL) · SLIDING DOOR SK1.3 m
FLOOR WELL
FLOOR WELL
FLOOR WELL




## 1 DECK A (MAIN HABITAT) · FFL Z −200 · QUADRANT 4

SCALE 1:100

### LEGEND

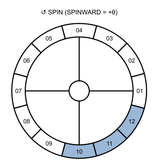
- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

### NOTES

1. TRUE PLAN FROM +Z. GRAVITY 1.0 g DOWN = −Z (GENERATED); FLOORS FLAT, THE HULL'S CURVED FACES ARE WALLS.
2. OUTWARD (+r) IS UP THE SHEET AT THE MIDDLE OF THE VIEW. ONE WINDOW PER FRAME BAY (3.75°), SEE openings.json.
3. PLAN CUT 1 200 ABOVE FLOOR. DIMENSIONS mm TO FACE OF STRUCTURE; ALONG THE RING, CHORDS AT THE HUB-WALL SKIN.
4. PARTITIONS 120 LIGHTWEIGHT, NON-STRUCTURAL; BULKHEADS ARE PRESSURE BOUNDARIES (SEE G-002).
5. EVERY BULKHEAD PENETRATION (DUCT, PIPE, CABLE) IS FITTED WITH AN ISOLATION DEVICE (SEE M-20x).
6. CABIN DOORS 800 CLEAR, SLIDING, 25 mm UNDERCUT FOR RETURN AIR TRANSFER.
7. WINDOWS: W1 OUTER WALL DECK A 1 600×1 800 · W2 OUTER WALL DECK B 1 400×1 200 · W3 HUB WALL DECK A 1 200×1 600.

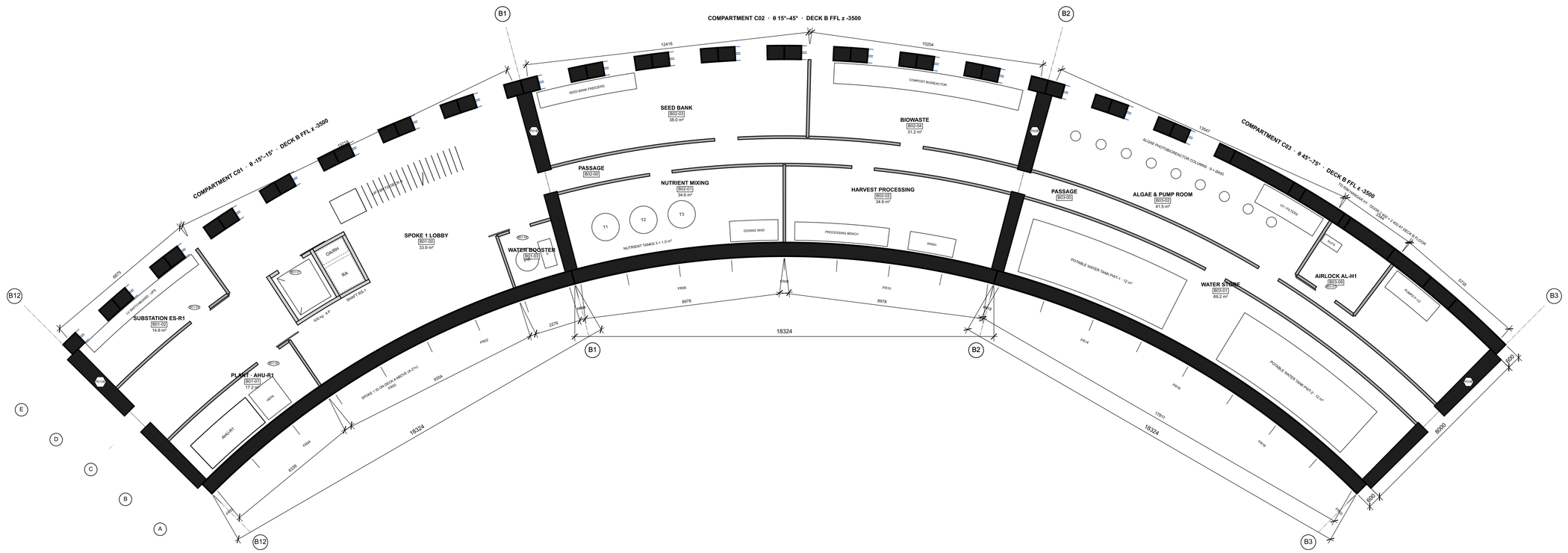
0 2 4 6 8 10 12 14 16 18 20 m

GRAPHIC SCALE 1:100




KEY PLAN · VIEWED FROM +Z (DOME)

| | | | |
|---|---|---|---|
| LEGAL OWNER / PROJECT: STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT | | | |
| TITLE: RING DECK A · QUADRANT 4 — Compartments C10–C12 · plan | | | |
| DISCIPLINE: Architecture | DOCUMENT STATUS: ISSUED FOR MODELLING | SCALE: AS SHOWN | SIZE / UNITS: A1 · mm |
| CREATED BY: Claude (AI) | APPROVED BY: — | DATE OF ISSUE: 2026-10-17 | PROJECTION |
| IDENTIFICATION NUMBER: SF-OUT-A-204 | | REV: A | SHEET: A-204 |




## 1 DECK B (SERVICES & BERTHING) · FFL Z −3 500 · QUADRANT 1

SCALE 1:100

### LEGEND

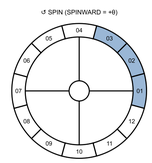
- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

### NOTES

1. TRUE PLAN FROM +Z. GRAVITY 1.0 g DOWN = −Z (GENERATED); FLOORS FLAT, THE HULL'S CURVED FACES ARE WALLS.
2. OUTWARD (+r) IS UP THE SHEET AT THE MIDDLE OF THE VIEW. ONE WINDOW PER FRAME BAY (3.75°), SEE openings.json.
3. PLAN CUT 1 200 ABOVE FLOOR. DIMENSIONS mm TO FACE OF STRUCTURE; ALONG THE RING, CHORDS AT THE HUB-WALL SKIN.
4. PARTITIONS 120 LIGHTWEIGHT, NON-STRUCTURAL; BULKHEADS ARE PRESSURE BOUNDARIES (SEE G-002).
5. EVERY BULKHEAD PENETRATION (DUCT, PIPE, CABLE) IS FITTED WITH AN ISOLATION DEVICE (SEE M-20x).
6. CABIN DOORS 800 CLEAR, SLIDING, 25 mm UNDERCUT FOR RETURN AIR TRANSFER.
7. WINDOWS: W1 OUTER WALL DECK A 1 600×1 800 · W2 OUTER WALL DECK B 1 400×1 200 · W3 HUB WALL DECK A 1 200×1 600.

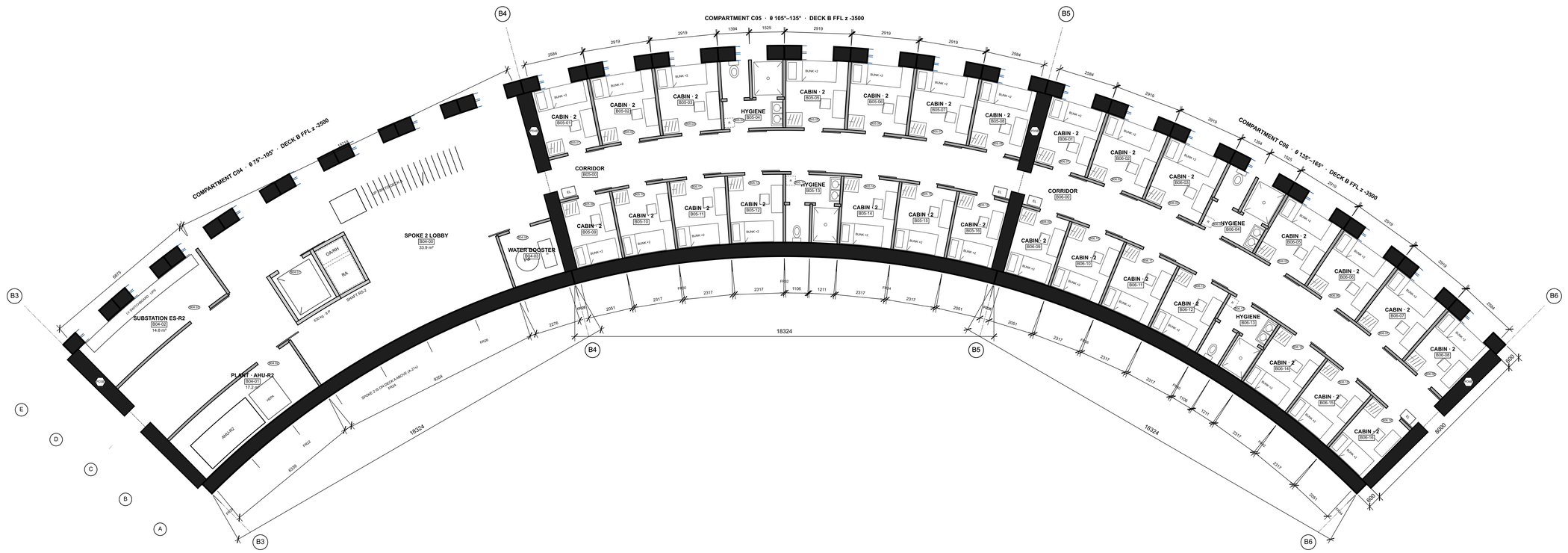
0 2 4 6 8 10 12 14 16 18 20 m

GRAPHIC SCALE 1:100




KEY PLAN · VIEWED FROM +Z (DOME)

| | |
|---|---|
| LEGAL OWNER / PROJECT | STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT |
| TITLE | RING DECK B · QUADRANT 1<br>Compartments C01–C03 · plan |
| DISCIPLINE | Architecture |
| DOCUMENT STATUS | ISSUED FOR MODELLING |
| SCALE | AS SHOWN |
| SIZE / UNITS | A1 · mm |
| CREATED BY | Claude (AI) |
| APPROVED BY | — |
| DATE OF ISSUE | 2026-10-17 |
| PROJECTION | |
| IDENTIFICATION NUMBER | SF-OUT-A-205 |
| REV | A |
| SHEET | A-205 |




## 1 DECK B (SERVICES & BERTHING) · FFL Z −3 500 · QUADRANT 2

SCALE 1:100

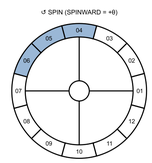
### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

### NOTES

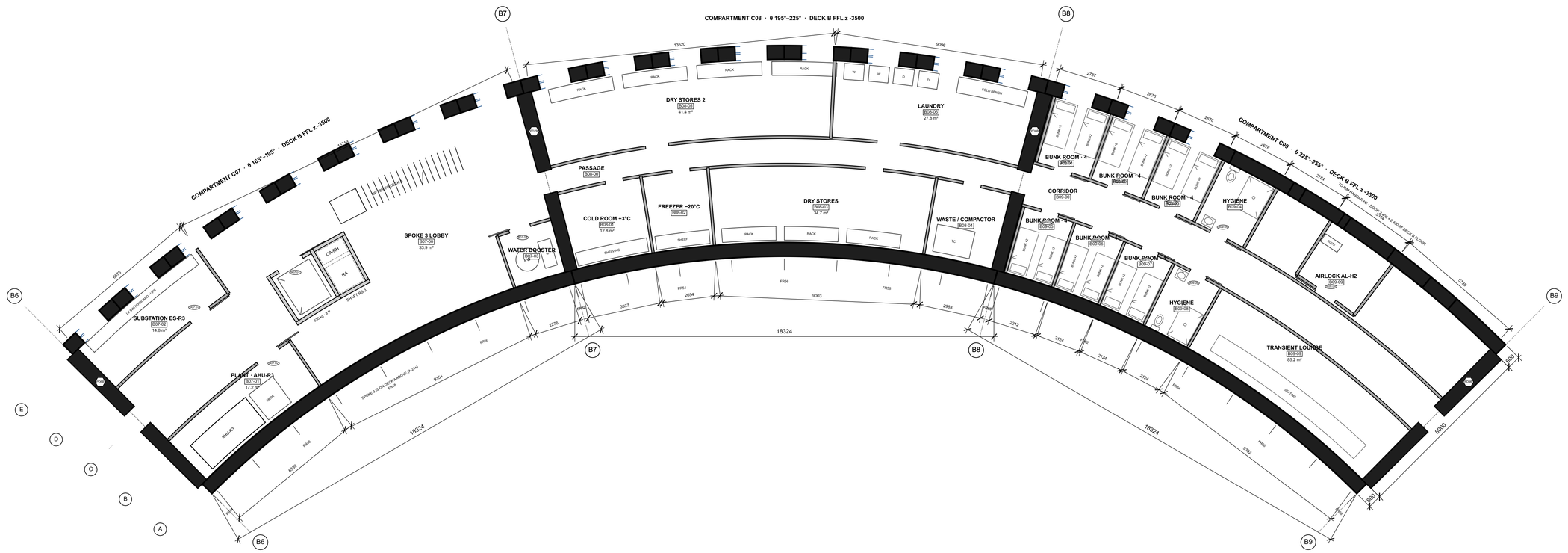
1. TRUE PLAN FROM +Z. GRAVITY 1.0 g DOWN = −Z (GENERATED); FLOORS FLAT, THE HULL'S CURVED FACES ARE WALLS.
2. OUTWARD (+r) IS UP THE SHEET AT THE MIDDLE OF THE VIEW. ONE WINDOW PER FRAME BAY (3.75°), SEE openings.json.
3. PLAN CUT 1 200 ABOVE FLOOR. DIMENSIONS mm TO FACE OF STRUCTURE; ALONG THE RING, CHORDS AT THE HUB-WALL SKIN.
4. PARTITIONS 120 LIGHTWEIGHT, NON-STRUCTURAL; BULKHEADS ARE PRESSURE BOUNDARIES (SEE G-002).
5. EVERY BULKHEAD PENETRATION (DUCT, PIPE, CABLE) IS FITTED WITH AN ISOLATION DEVICE (SEE M-20x).
6. CABIN DOORS 800 CLEAR, SLIDING, 25 mm UNDERCUT FOR RETURN AIR TRANSFER.
7. WINDOWS: W1 OUTER WALL DECK A 1 600×1 800 · W2 OUTER WALL DECK B 1 400×1 200 · W3 HUB WALL DECK A 1 200×1 600.

GRAPHIC SCALE 1:100




KEY PLAN · VIEWED FROM +Z (DOME)

| | | | |
|---|---|---|---|
| LEGAL OWNER / PROJECT: STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT | | | |
| TITLE: RING DECK B · QUADRANT 2 — Compartments C04–C06 · plan | | | |
| DISCIPLINE: Architecture | DOCUMENT STATUS: ISSUED FOR MODELLING | SCALE: AS SHOWN | SIZE / UNITS: A1 · mm |
| CREATED BY: Claude (AI) | APPROVED BY: — | DATE OF ISSUE: 2026-10-17 | PROJECTION |
| IDENTIFICATION NUMBER: SF-OUT-A-206 | | REV: A | SHEET: A-206 |




## 1 DECK B (SERVICES & BERTHING) · FFL Z −3 500 · QUADRANT 3

SCALE 1:100

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

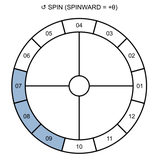
### NOTES

1. TRUE PLAN FROM +Z. GRAVITY 1.0 g DOWN = −Z (GENERATED); FLOORS FLAT, THE HULL'S CURVED FACES ARE WALLS.
2. OUTWARD (+r) IS UP THE SHEET AT THE MIDDLE OF THE VIEW. ONE WINDOW PER FRAME BAY (3.75°), SEE openings.json.
3. PLAN CUT 1 200 ABOVE FLOOR. DIMENSIONS mm TO FACE OF STRUCTURE; ALONG THE RING, CHORDS AT THE HUB-WALL SKIN.
4. PARTITIONS 120 LIGHTWEIGHT, NON-STRUCTURAL; BULKHEADS ARE PRESSURE BOUNDARIES (SEE G-002).
5. EVERY BULKHEAD PENETRATION (DUCT, PIPE, CABLE) IS FITTED WITH AN ISOLATION DEVICE (SEE M-20x).
6. CABIN DOORS 800 CLEAR, SLIDING, 25 mm UNDERCUT FOR RETURN AIR TRANSFER.
7. WINDOWS: W1 OUTER WALL DECK A 1 600×1 800 · W2 OUTER WALL DECK B 1 400×1 200 · W3 HUB WALL DECK A 1 200×1 600.

0 2 4 6 8 10 12 14 16 18 20 m

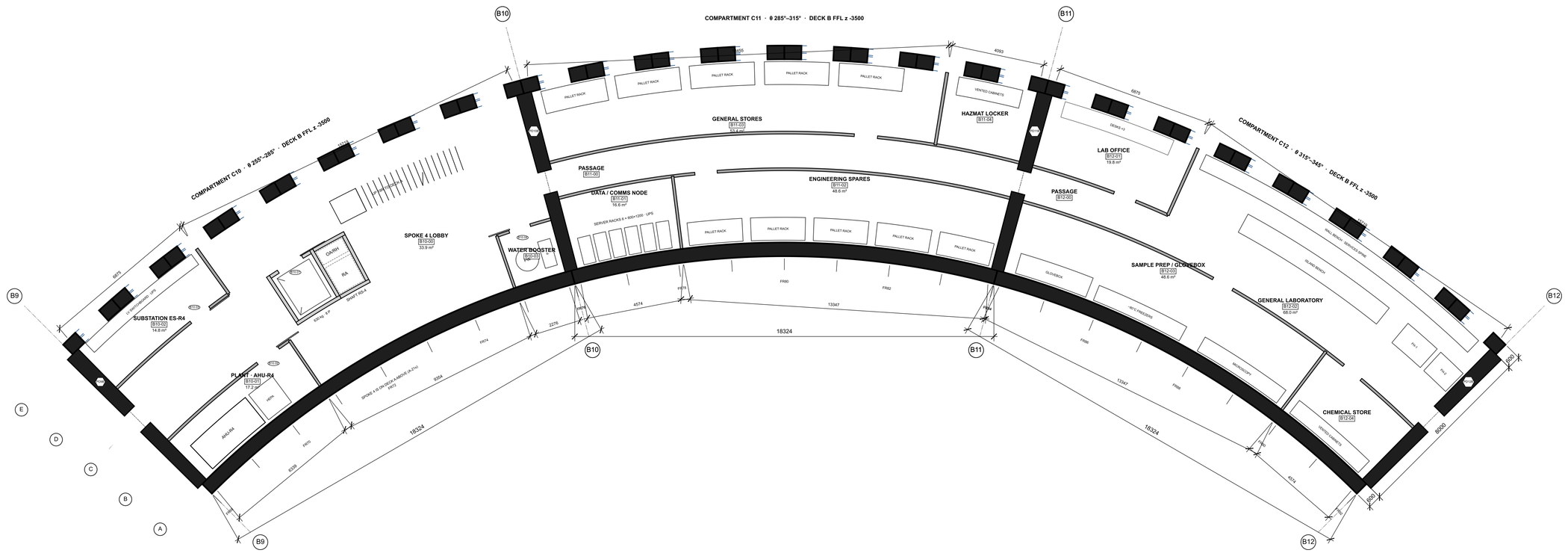
GRAPHIC SCALE 1:100

↺ SPIN (SPINWARD = +θ)



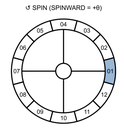

KEY PLAN · VIEWED FROM +Z (DOME)

| | |
|---|---|
| LEGAL OWNER / PROJECT | STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT |
| TITLE | RING DECK B · QUADRANT 3<br>Compartments C07–C09 · plan |
| DISCIPLINE | Architecture |
| DOCUMENT STATUS | ISSUED FOR MODELLING |
| SCALE | AS SHOWN |
| SIZE / UNITS | A1 · mm |
| CREATED BY | Claude (AI) |
| APPROVED BY | — |
| DATE OF ISSUE | 2026-10-17 |
| PROJECTION | |
| IDENTIFICATION NUMBER | SF-OUT-A-207 |
| REV | A |
| SHEET | A-207 |


COMPARTMENT C11 · θ 285°–315° · DECK B FFL z -3500
COMPARTMENT C10 · θ 255°–285° · DECK B FFL z -3500
COMPARTMENT C12 · θ 315°–345° · DECK B FFL z -3500
B9
B10
B11
B12
GENERAL STORES
B11-03
PASSAGE
B11-00
DATA / COMMS NODE
B11-01
16.6 m²
SERVER RACKS 6 × 600×1200 · UPS
ENGINEERING SPARES
B11-02
48.6 m²
HAZMAT LOCKER
B11-04
VENTED CABINETS
LAB OFFICE
B12-01
19.8 m²
DESKS ×3
PASSAGE
B12-00
SAMPLE PREP / GLOVEBOX
B12-03
48.6 m²
GLOVEBOX
FUME HOOD
GENERAL LABORATORY
B12-02
66.0 m²
WALL BENCH · SERVICES MARK
ISLAND BENCH
FUME HOOD
CHEMICAL STORE
B12-04
VENTED CABINETS
SPOKE 4 LOBBY
B10-00
33.9 m²
WATER BOOSTER
B10-03
SUBSTATION ES-R4
B10-02
14.8 m²
11 kV SWITCHBOARD · UPS
PLANT · AHU-R4
B10-01
17.2 m²
AHU-R4
HWR
SPOKE 4 RA DUCT
SHAFT SD-4
SPOKE 4 ID ON DECK ABOVE (A-21x)
18324
13347
4574
2276
6329
6870
4093
8005
FR74
FR76
FR78
FR80
FR82
FR84
FR86
FR88
FR90
FR92
FR94
FR96
FR98
FR100
FR102
FR104
FR106
FR108
FR110
FR112
FR114
FR116
FR118
FR120
FR122
FR124
FR126
FR128
FR130
FR132
FR134
FR136
FR138
FR140
FR142
FR144
FR146
FR148
FR150
FR152




↺ SPIN (SPINWARD = +θ)




KEY PLAN

## 1 DECK B PLAN · FFL Z −3 500 · SPOKE LOBBY · AHU PLANT · SUBSTATION

SCALE 1:50

### ROOM SCHEDULE · DECK B

| No. | ROOM | m² |
|---|---|---|
| B01-00 | SPOKE 1 LOBBY | 33.9 |
| B01-01 | PLANT · AHU-R1 | 17.2 |
| B01-02 | SUBSTATION ES-R1 | 14.8 |
| B01-03 | WATER BOOSTER | 6.2 |

### ROOM SCHEDULE · DECK A

| No. | ROOM | m² |
|---|---|---|
| A01-00 | CONCOURSE | 77.3 |
| A01-01 | SAFE HAVEN · 30 P | 12.5 |
| A01-02 | EVA STORE | 3.3 |
| A01-03 | WC | 2.9 |
| A01-04 | WC · ACC | 4.2 |
| A01-05 | JANITOR | 8.6 |
| A01-06 | OPERATIONS OFFICE | 15.2 |
| A01-07 | ELEC PANEL EP-A | 4.6 |
| A01-08 | SPOKE 1 PORTAL | 11.2 |

## 2 DECK A PLAN · FFL Z −200 · SPOKE JUNCTION · CONCOURSE · SAFE HAVEN

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.nn: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY; EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

GRAPHIC SCALE 1:50

LEGAL OWNER / PROJECT

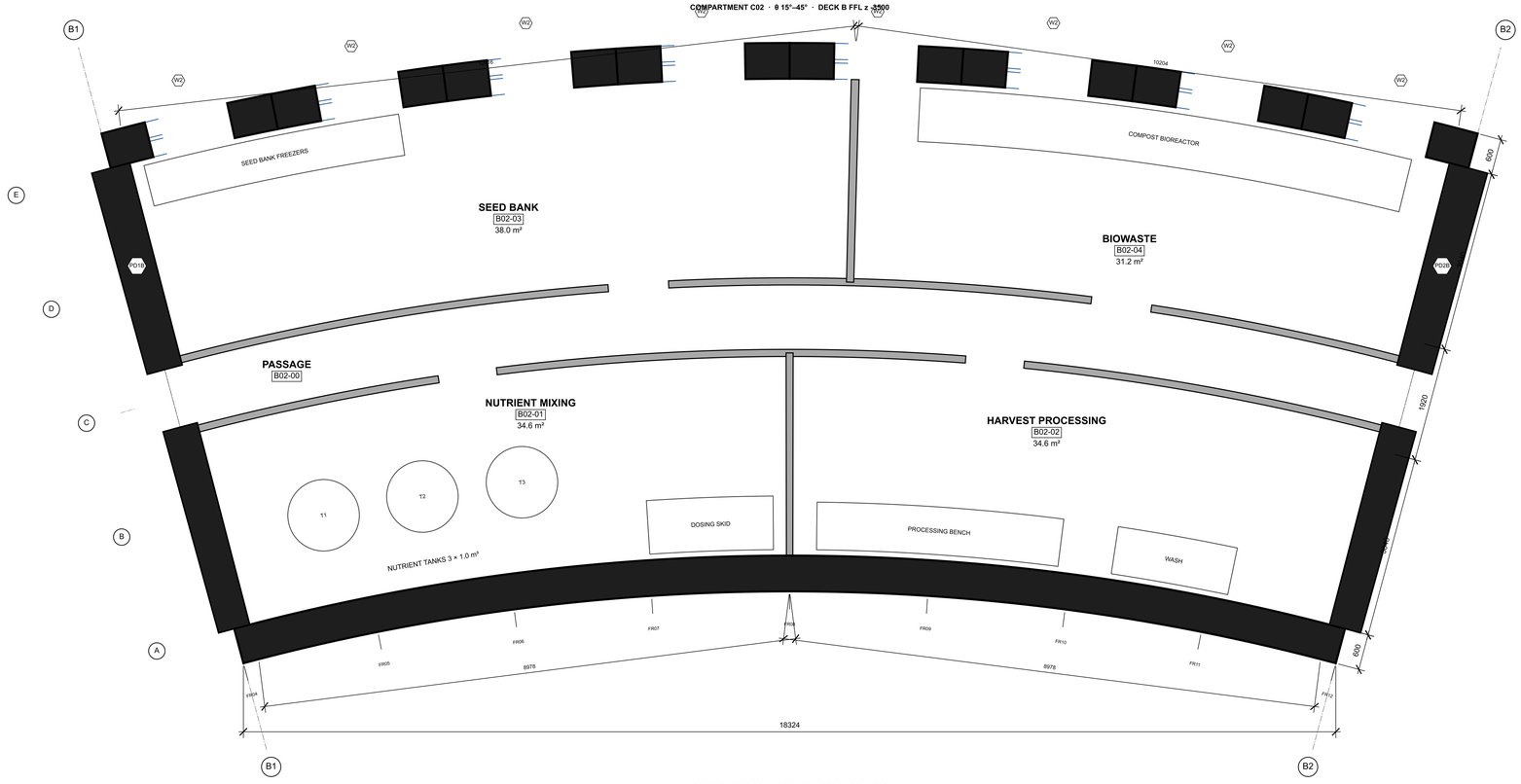
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE

# COMPARTMENT C01 · PLANS 1:50

0 -15°–15° · Deck A spoke junction · concourse · safe haven · Deck B spoke lobby · ahu plant · substation

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-211 | | A | A-211 |




## 1 DECK B PLAN · FFL Z −3 500 · NUTRIENTS · HARVEST · SEED BANK

SCALE 1:50



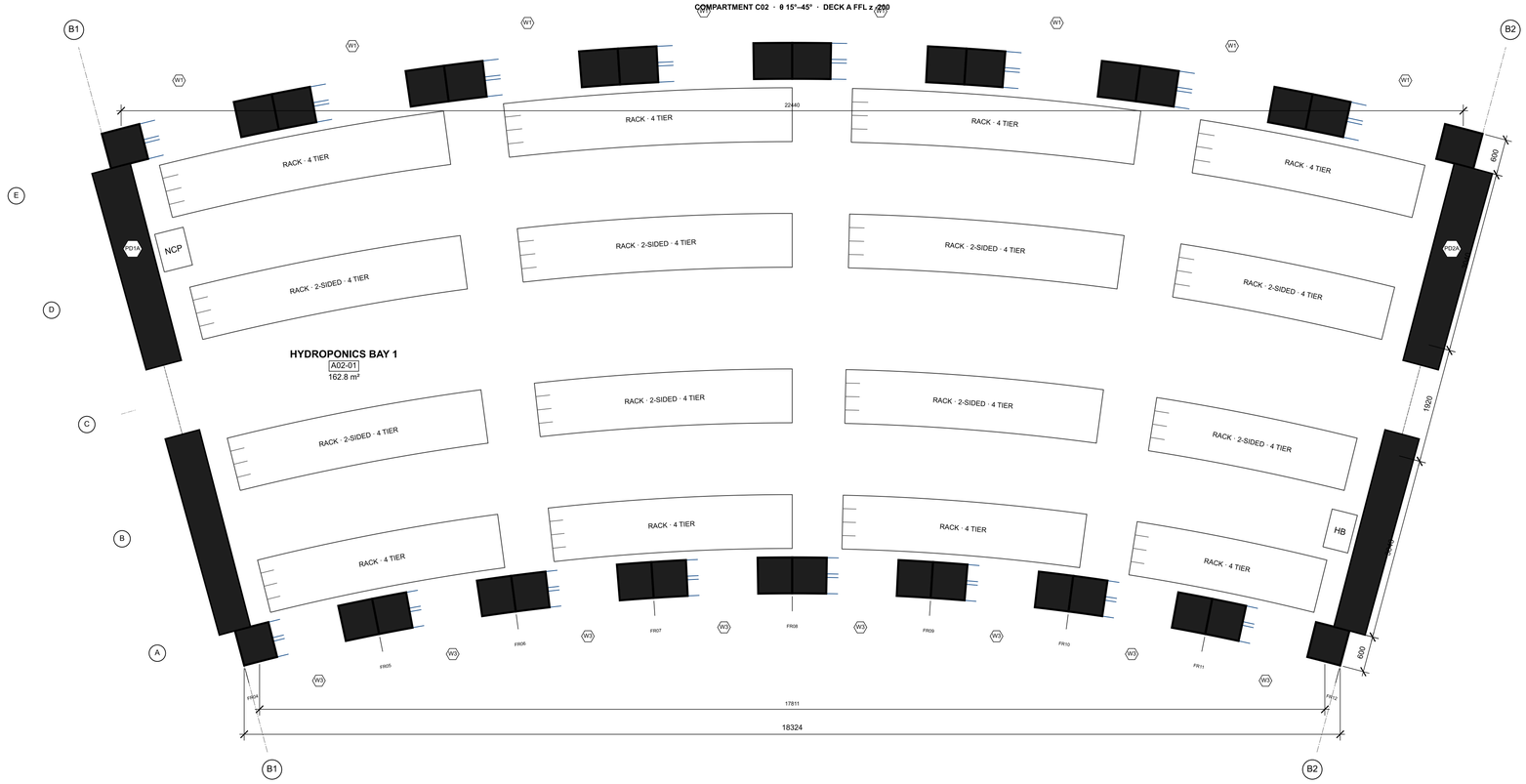

KEY PLAN

**ROOM SCHEDULE · DECK B**

| No. | ROOM | m² |
|---|---|---|
| B02-00 | PASSAGE | 24.4 |
| B02-01 | NUTRIENT MIXING | 34.6 |
| B02-02 | HARVEST PROCESSING | 34.6 |
| B02-03 | SEED BANK | 38.0 |
| B02-04 | BIOWASTE | 31.2 |

**ROOM SCHEDULE · DECK A**

| No. | ROOM | m² |
|---|---|---|
| A02-01 | HYDROPONICS BAY 1 | 162.8 |




## 2 DECK A PLAN · FFL Z −200 · HYDROPONICS

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.nn: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY; EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

GRAPHIC SCALE 1:50

LEGAL OWNER / PROJECT

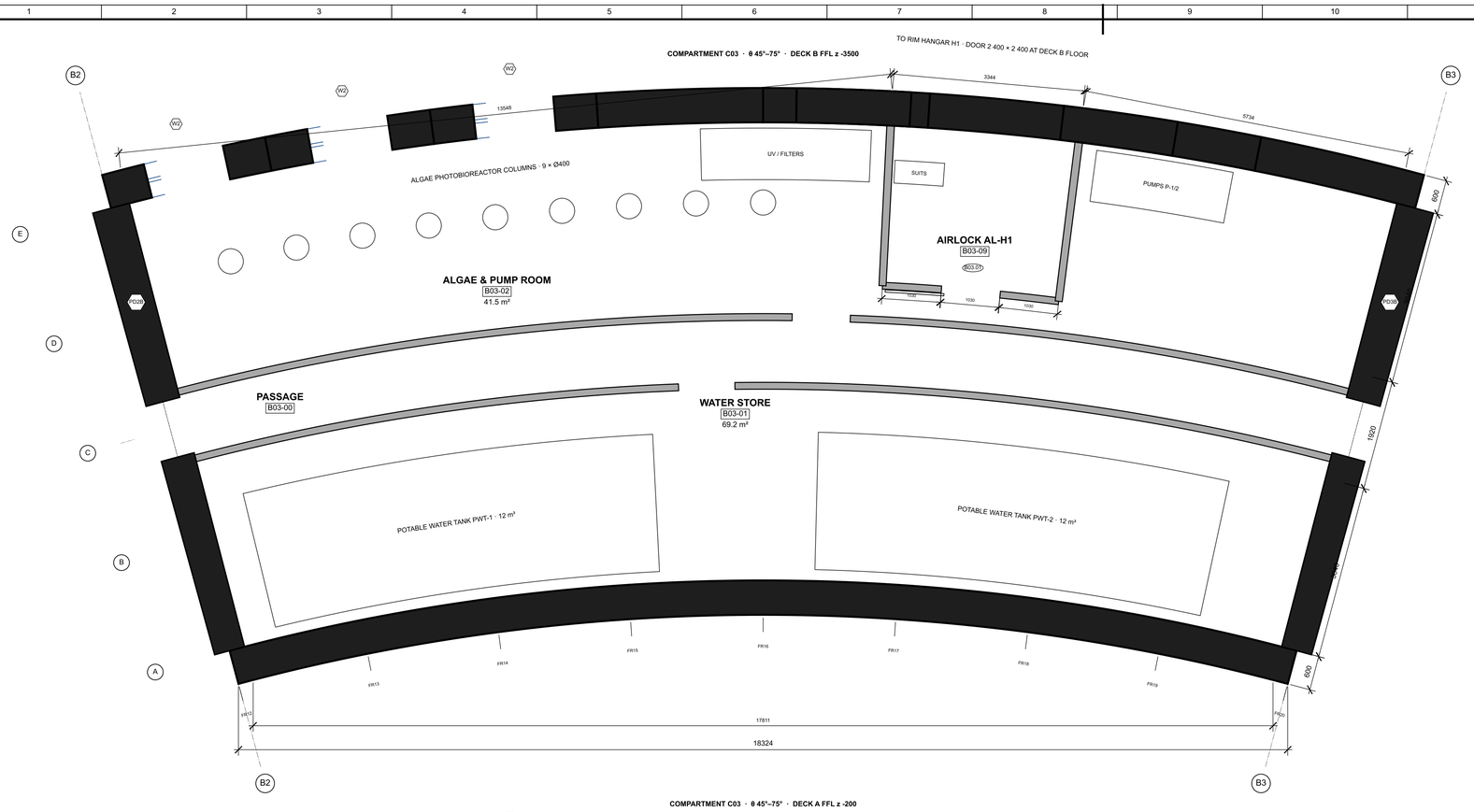
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

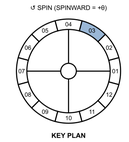
TITLE

# COMPARTMENT C02 · PLANS 1:50

θ 15°–45° · Deck A hydroponics · Deck B nutrients · harvest · seed bank

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-212 | | A | A-212 |




## 1 DECK B PLAN · FFL Z −3 500 · WATER STORE · ALGAE · PUMPS

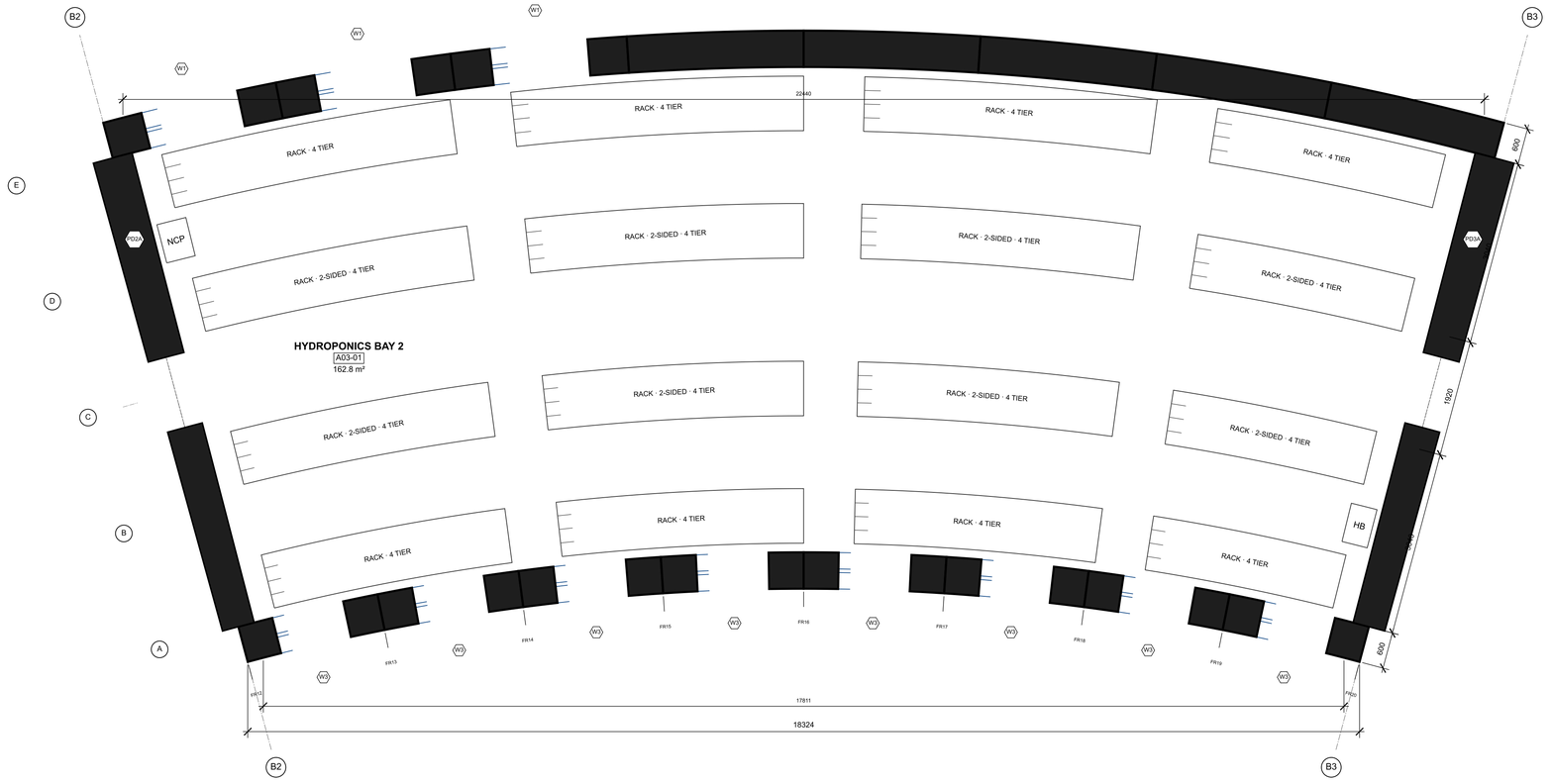
SCALE 1:50

### ROOM SCHEDULE · DECK B

| No. | ROOM | m² |
|---|---|---|
| B03-00 | PASSAGE | 24.4 |
| B03-01 | WATER STORE | 69.2 |
| B03-02 | ALGAE & PUMP ROOM | 41.5 |
| B03-09 | AIRLOCK AL-H1 | 8.4 |

### ROOM SCHEDULE · DECK A

| No. | ROOM | m² |
|---|---|---|
| A03-01 | HYDROPONICS BAY 2 | 162.8 |




## 2 DECK A PLAN · FFL Z −200 · HYDROPONICS

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.m: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY, EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

GRAPHIC SCALE 1:50

LEGAL OWNER / PROJECT

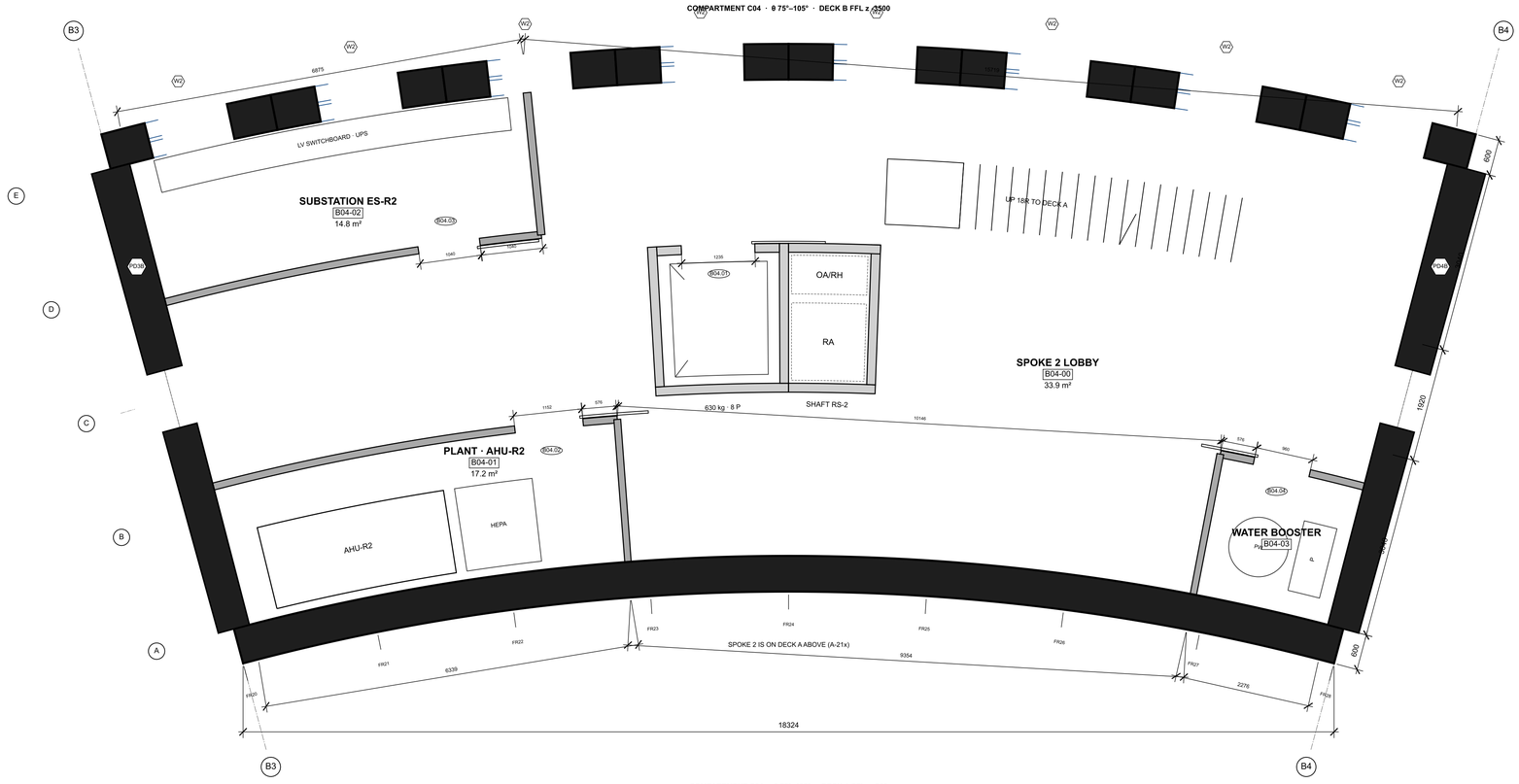
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

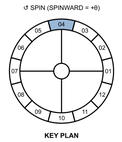
TITLE

# COMPARTMENT C03 · PLANS 1:50

θ 45°–75° · Deck A hydroponics · Deck B water store · algae · pumps

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| **CREATED BY** | **APPROVED BY** | **DATE OF ISSUE** | **PROJECTION** |
| Claude (AI) | — | 2026-10-17 | |
| **IDENTIFICATION NUMBER** | | **REV** | **SHEET** |
| **SF-OUT-A-213** | | **A** | **A-213** |




## 1 DECK B PLAN · FFL Z −3 500 · SPOKE LOBBY · AHU PLANT · SUBSTATION

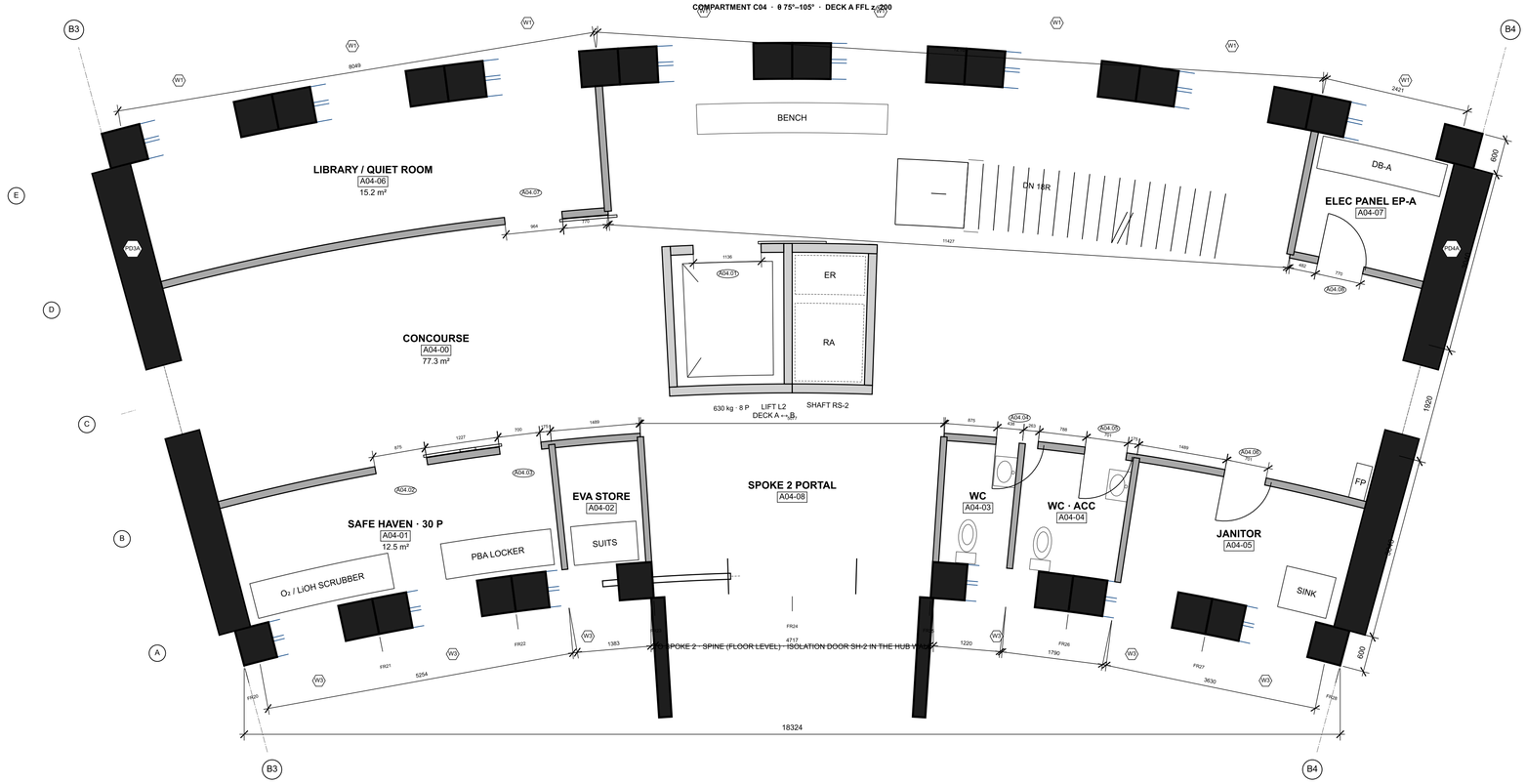
SCALE 1:50

### ROOM SCHEDULE · DECK B

| No. | ROOM | m² |
|---|---|---|
| B04-00 | SPOKE 2 LOBBY | 33.9 |
| B04-01 | PLANT · AHU-R2 | 17.2 |
| B04-02 | SUBSTATION ES-R2 | 14.8 |
| B04-03 | WATER BOOSTER | 6.2 |

### ROOM SCHEDULE · DECK A

| No. | ROOM | m² |
|---|---|---|
| A04-00 | CONCOURSE | 77.3 |
| A04-01 | SAFE HAVEN · 30 P | 12.5 |
| A04-02 | EVA STORE | 3.3 |
| A04-03 | WC | 2.9 |
| A04-04 | WC · ACC | 4.2 |
| A04-05 | JANITOR | 8.6 |
| A04-06 | LIBRARY / QUIET ROOM | 15.2 |
| A04-07 | ELEC PANEL EP-A | 4.6 |
| A04-08 | SPOKE 2 PORTAL | 11.2 |




## 2 DECK A PLAN · FFL Z −200 · SPOKE JUNCTION · CONCOURSE · SAFE HAVEN

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.nn: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY; EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

GRAPHIC SCALE 1:50

LEGAL OWNER / PROJECT

**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

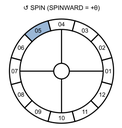
TITLE

# COMPARTMENT C04 · PLANS 1:50

θ 75°–105° · Deck A spoke junction · concourse · safe haven · Deck B spoke lobby · ahu plant · substation

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-214 | | A | A-214 |




## 1 DECK B PLAN · FFL Z −3 500 · QUARTERS · DOUBLE CABINS

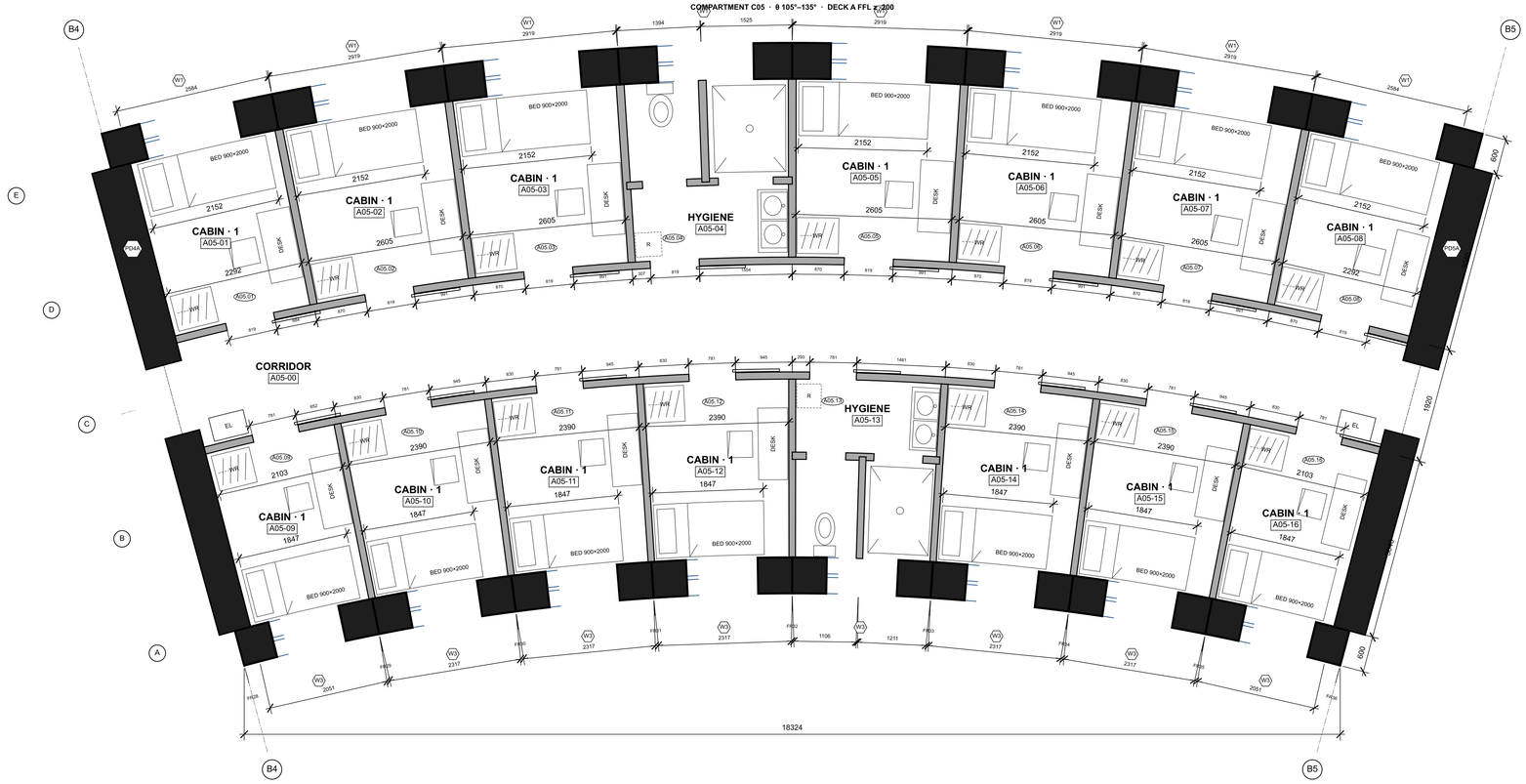
SCALE 1:50

### ROOM SCHEDULE · DECK B

| No. | ROOM | m² |
|---|---|---|
| B05-00 | CORRIDOR | 36.6 |
| B05-01 | CABIN · 2 | 6.9 |
| B05-02 | CABIN · 2 | 7.8 |
| B05-03 | CABIN · 2 | 7.8 |
| B05-04 | HYGIENE | 7.8 |
| B05-05 | CABIN · 2 | 7.8 |
| B05-06 | CABIN · 2 | 7.8 |
| B05-07 | CABIN · 2 | 7.8 |
| B05-08 | CABIN · 2 | 6.9 |
| B05-09 | CABIN · 2 | 6.9 |
| B05-10 | CABIN · 2 | 7.8 |
| B05-11 | CABIN · 2 | 7.8 |
| B05-12 | CABIN · 2 | 7.8 |
| B05-13 | HYGIENE | 7.8 |
| B05-14 | CABIN · 2 | 7.8 |
| B05-15 | CABIN · 2 | 7.8 |
| B05-16 | CABIN · 2 | 6.9 |

### ROOM SCHEDULE · DECK A

| No. | ROOM | m² |
|---|---|---|
| A05-00 | CORRIDOR | 36.6 |
| A05-01 | CABIN · 1 | 6.9 |
| A05-02 | CABIN · 1 | 7.8 |
| A05-03 | CABIN · 1 | 7.8 |
| A05-04 | HYGIENE | 7.8 |
| A05-05 | CABIN · 1 | 7.8 |
| A05-06 | CABIN · 1 | 7.8 |
| A05-07 | CABIN · 1 | 7.8 |
| A05-08 | CABIN · 1 | 6.9 |
| A05-09 | CABIN · 1 | 6.9 |
| A05-10 | CABIN · 1 | 7.8 |
| A05-11 | CABIN · 1 | 7.8 |
| A05-12 | CABIN · 1 | 7.8 |
| A05-13 | HYGIENE | 7.8 |
| A05-14 | CABIN · 1 | 7.8 |
| A05-15 | CABIN · 1 | 7.8 |
| A05-16 | CABIN · 1 | 6.9 |




## 2 DECK A PLAN · FFL Z −200 · QUARTERS · SINGLE CABINS

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.nn: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY; EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

GRAPHIC SCALE 1:50

LEGAL OWNER / PROJECT

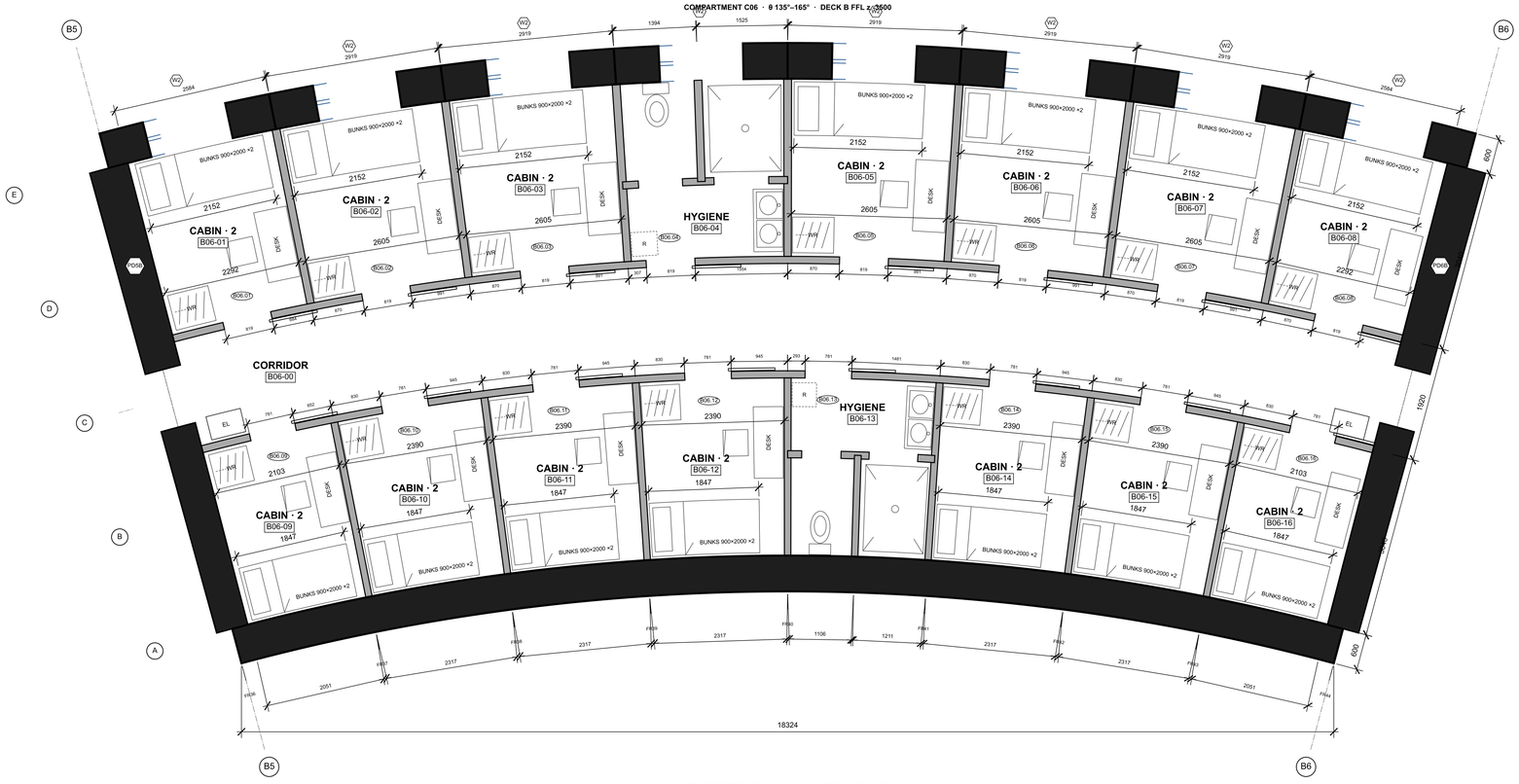
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE

**COMPARTMENT C05 · PLANS 1:50**

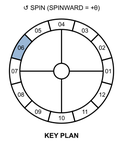
θ 105°–135° · Deck A quarters · single cabins · Deck B quarters · double cabins

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-215 | | A | A-215 |




## 1 DECK B PLAN · FFL Z −3 500 · QUARTERS · DOUBLE CABINS

SCALE 1:50

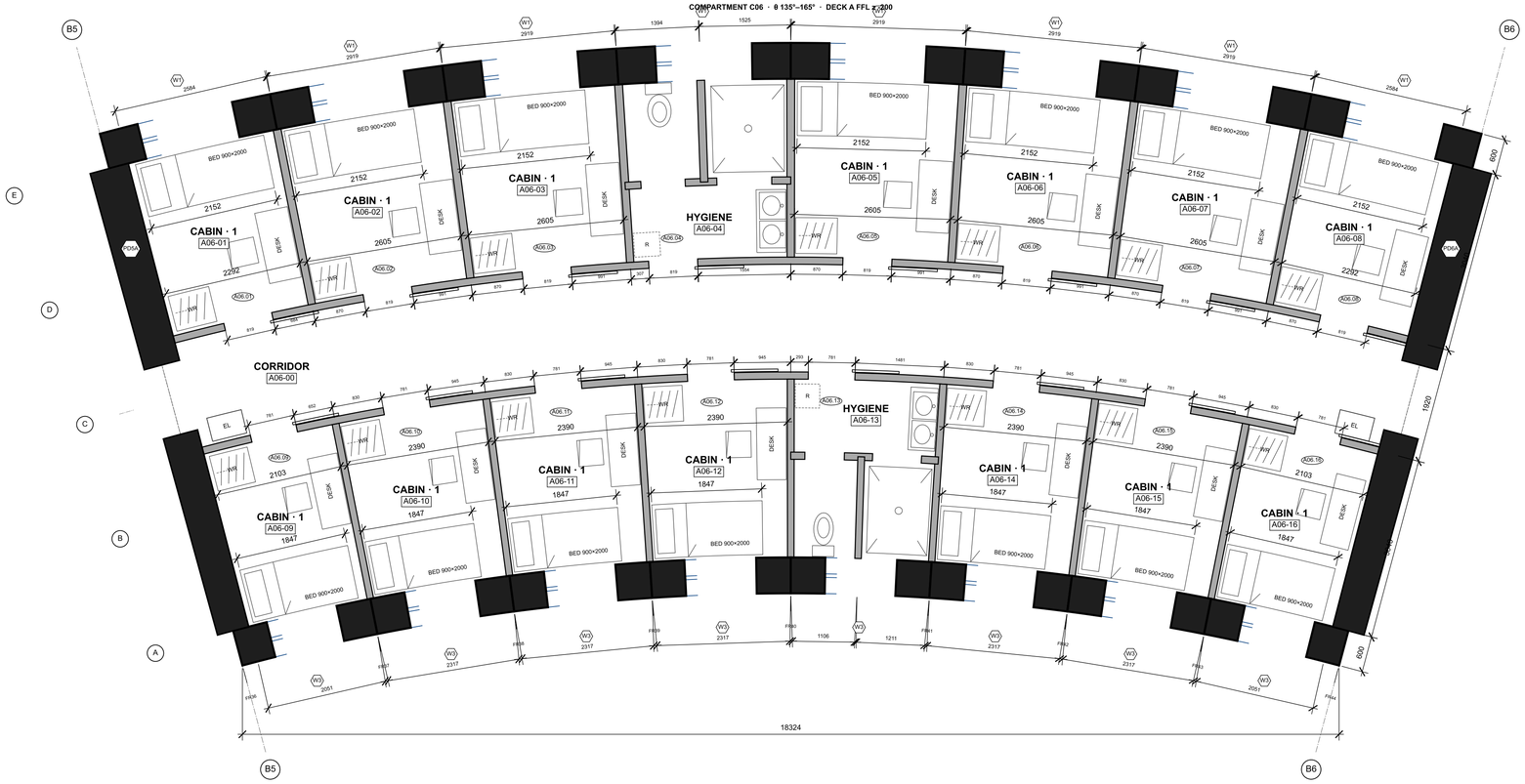



### ROOM SCHEDULE · DECK B

| No. | ROOM | m² |
|---|---|---|
| B06-00 | CORRIDOR | 36.6 |
| B06-01 | CABIN · 2 | 6.9 |
| B06-02 | CABIN · 2 | 7.8 |
| B06-03 | CABIN · 2 | 7.8 |
| B06-04 | HYGIENE | 7.8 |
| B06-05 | CABIN · 2 | 7.8 |
| B06-06 | CABIN · 2 | 7.8 |
| B06-07 | CABIN · 2 | 7.8 |
| B06-08 | CABIN · 2 | 6.9 |
| B06-09 | CABIN · 2 | 6.9 |
| B06-10 | CABIN · 2 | 7.8 |
| B06-11 | CABIN · 2 | 7.8 |
| B06-12 | CABIN · 2 | 7.8 |
| B06-13 | HYGIENE | 7.8 |
| B06-14 | CABIN · 2 | 7.8 |
| B06-15 | CABIN · 2 | 7.8 |
| B06-16 | CABIN · 2 | 6.9 |

### ROOM SCHEDULE · DECK A

| No. | ROOM | m² |
|---|---|---|
| A06-00 | CORRIDOR | 36.6 |
| A06-01 | CABIN · 1 | 6.9 |
| A06-02 | CABIN · 1 | 7.8 |
| A06-03 | CABIN · 1 | 7.8 |
| A06-04 | HYGIENE | 7.8 |
| A06-05 | CABIN · 1 | 7.8 |
| A06-06 | CABIN · 1 | 7.8 |
| A06-07 | CABIN · 1 | 7.8 |
| A06-08 | CABIN · 1 | 6.9 |
| A06-09 | CABIN · 1 | 6.9 |
| A06-10 | CABIN · 1 | 7.8 |
| A06-11 | CABIN · 1 | 7.8 |
| A06-12 | CABIN · 1 | 7.8 |
| A06-13 | HYGIENE | 7.8 |
| A06-14 | CABIN · 1 | 7.8 |
| A06-15 | CABIN · 1 | 7.8 |
| A06-16 | CABIN · 1 | 6.9 |




## 2 DECK A PLAN · FFL Z −200 · QUARTERS · SINGLE CABINS

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.m: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY; EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

GRAPHIC SCALE 1:50

LEGAL OWNER / PROJECT

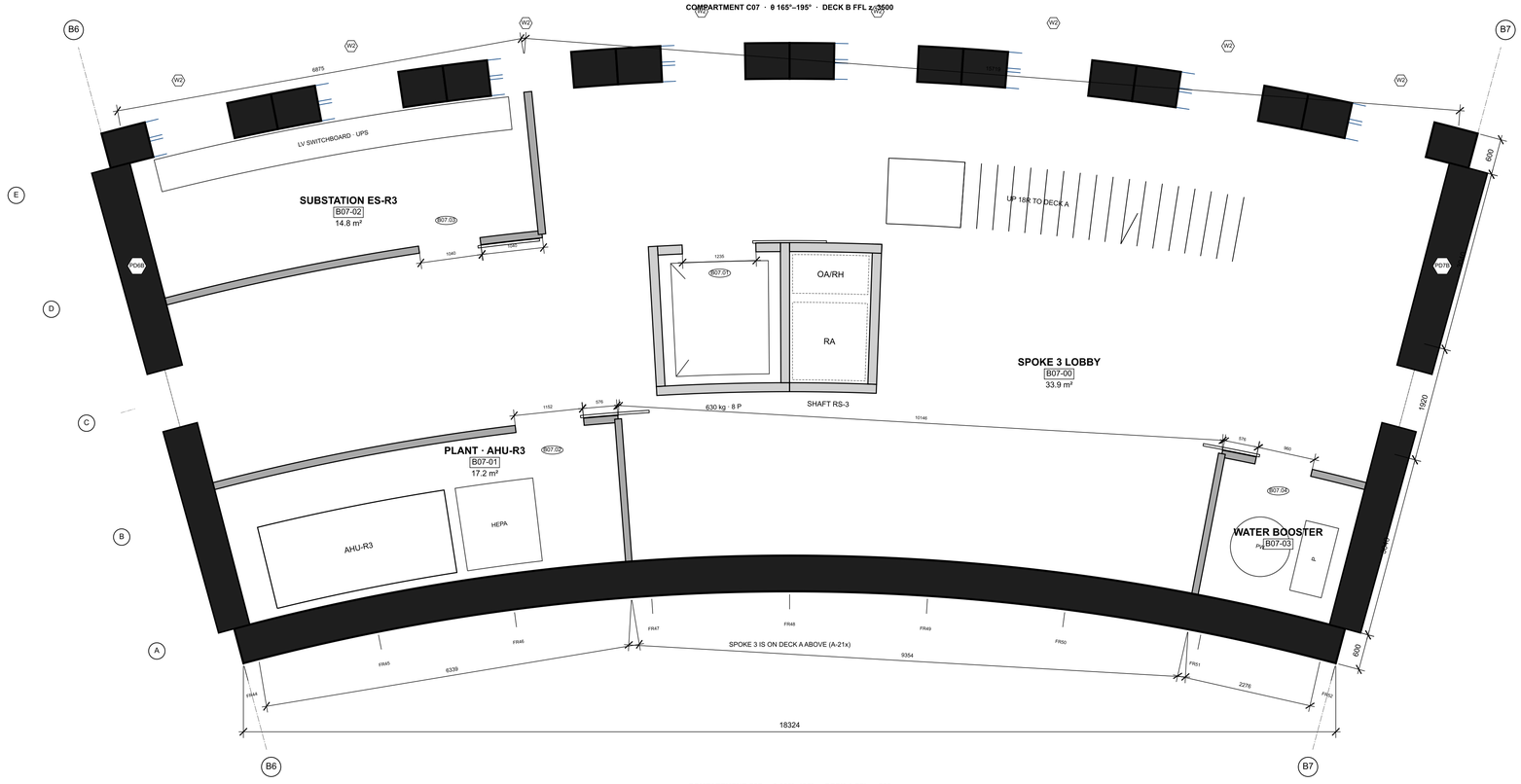
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE

# COMPARTMENT C06 · PLANS 1:50

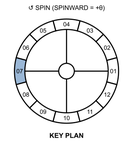
θ 135°–165° · Deck A quarters · single cabins · Deck B quarters · double cabins

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-216 | | A | A-216 |




## 1 DECK B PLAN · FFL Z −3 500 · SPOKE LOBBY · AHU PLANT · SUBSTATION

SCALE 1:50

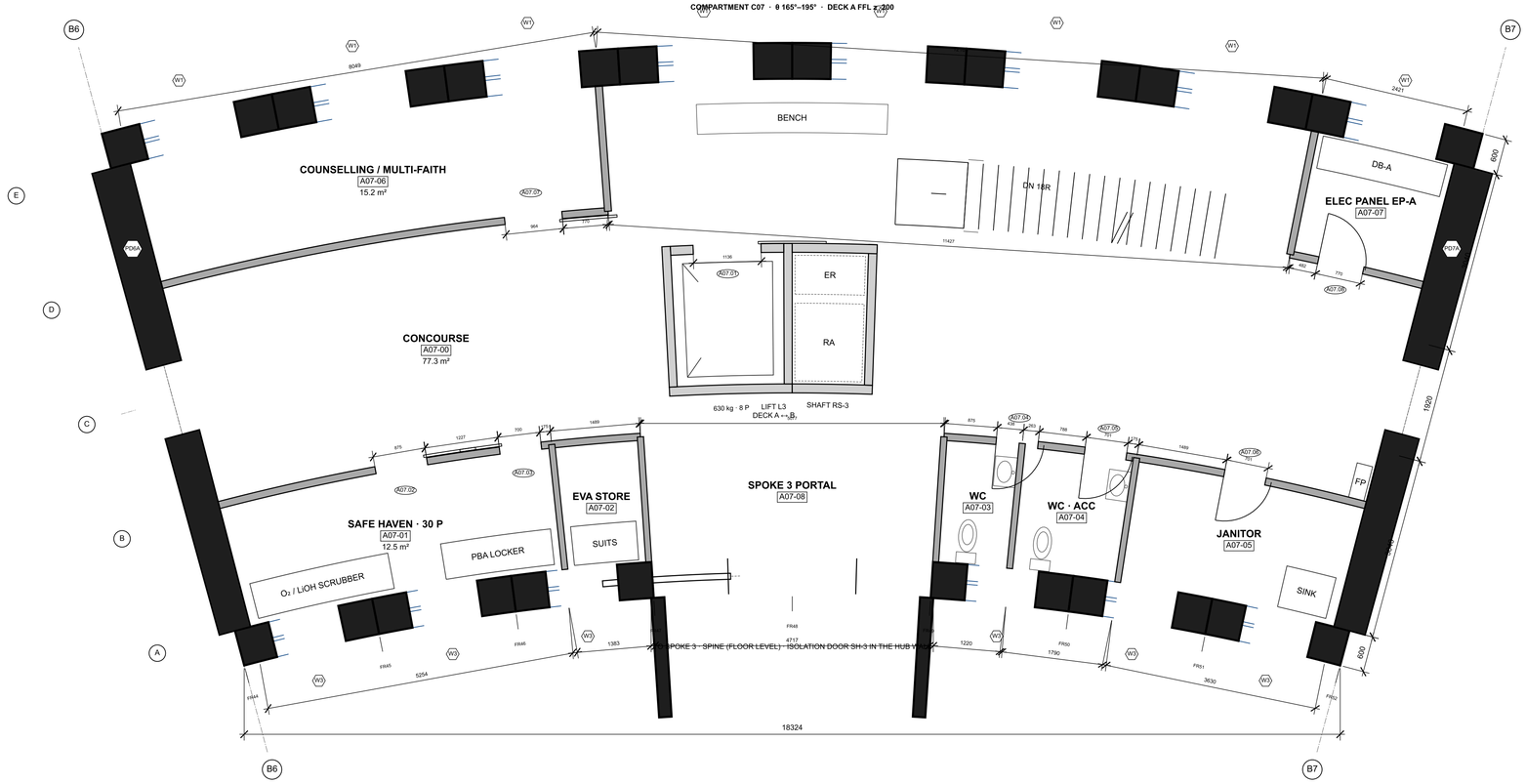



### ROOM SCHEDULE · DECK B

| No. | ROOM | m² |
|---|---|---|
| B07-00 | SPOKE 3 LOBBY | 33.9 |
| B07-01 | PLANT · AHU-R3 | 17.2 |
| B07-02 | SUBSTATION ES-R3 | 14.8 |
| B07-03 | WATER BOOSTER | 6.2 |

### ROOM SCHEDULE · DECK A

| No. | ROOM | m² |
|---|---|---|
| A07-00 | CONCOURSE | 77.3 |
| A07-01 | SAFE HAVEN · 30 P | 12.5 |
| A07-02 | EVA STORE | 3.3 |
| A07-03 | WC | 2.9 |
| A07-04 | WC · ACC | 4.2 |
| A07-05 | JANITOR | 8.6 |
| A07-06 | COUNSELLING / MULTI-FAITH | 15.2 |
| A07-07 | ELEC PANEL EP-A | 4.6 |
| A07-08 | SPOKE 3 PORTAL | 11.2 |




## 2 DECK A PLAN · FFL Z −200 · SPOKE JUNCTION · CONCOURSE · SAFE HAVEN

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.nn: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY; EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

GRAPHIC SCALE 1:50

LEGAL OWNER / PROJECT

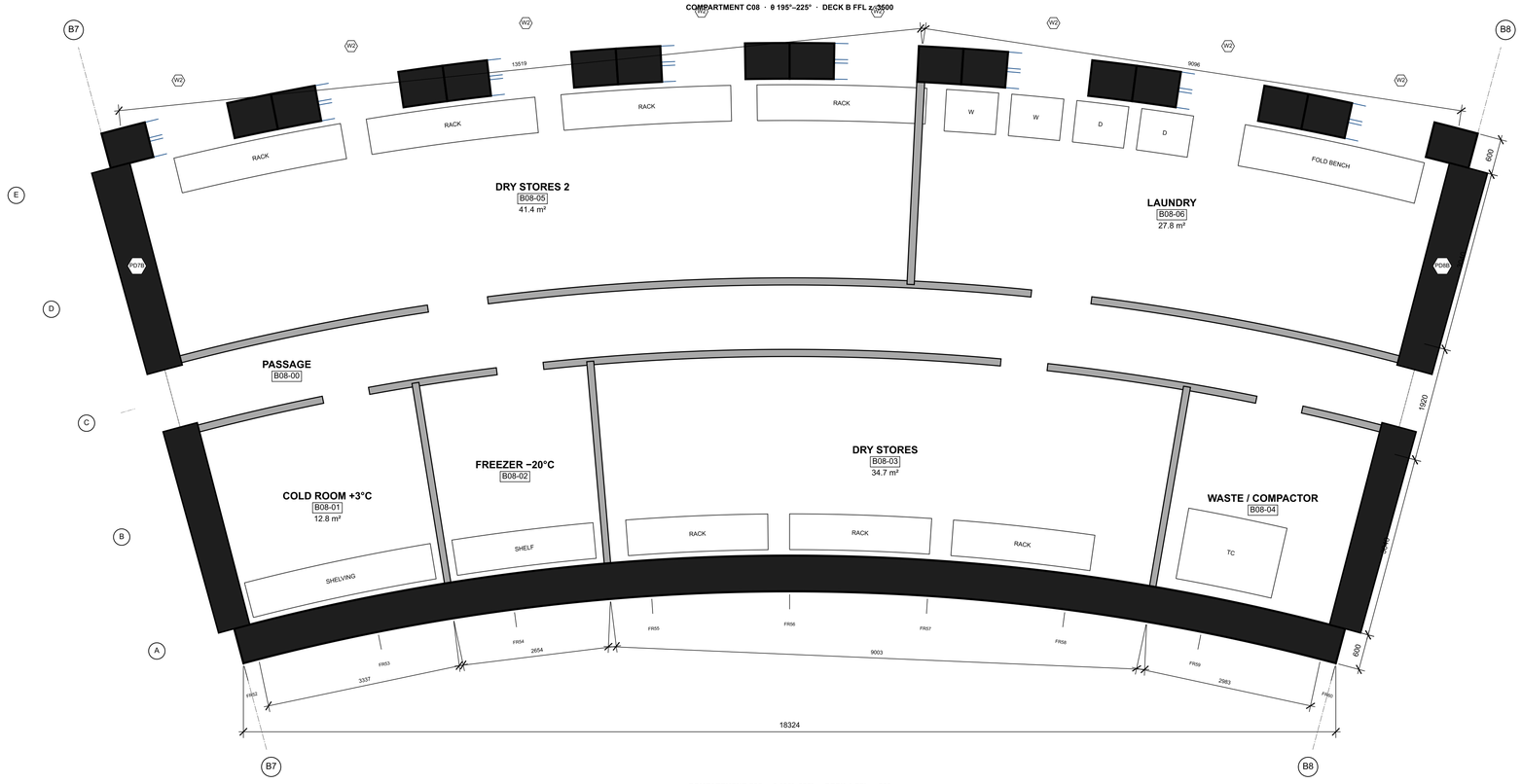
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

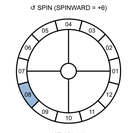
TITLE

# COMPARTMENT C07 · PLANS 1:50

θ 165°–195° · Deck A spoke junction · concourse · safe haven · Deck B spoke lobby · ahu plant · substation

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| **CREATED BY** | **APPROVED BY** | **DATE OF ISSUE** | **PROJECTION** |
| Claude (AI) | — | 2026-10-17 | |
| **IDENTIFICATION NUMBER** | | **REV** | **SHEET** |
| SF-OUT-A-217 | | A | A-217 |




## 1 DECK B PLAN · FFL Z −3 500 · COLD / DRY STORES · LAUNDRY

SCALE 1:50

### ROOM SCHEDULE · DECK B

| No. | ROOM | m² |
|---|---|---|
| B08-00 | PASSAGE | 24.4 |
| B08-01 | COLD ROOM +3°C | 12.8 |
| B08-02 | FREEZER −20°C | 10.2 |
| B08-03 | DRY STORES | 34.7 |
| B08-04 | WASTE / COMPACTOR | 11.5 |
| B08-05 | DRY STORES 2 | 41.4 |
| B08-06 | LAUNDRY | 27.8 |

### ROOM SCHEDULE · DECK A

| No. | ROOM | m² |
|---|---|---|
| A08-00 | WARDROOM · 48 SEATS | 88.8 |
| A08-01 | GALLEY | 23.3 |
| A08-02 | SCULLERY | 9.2 |




## 2 DECK A PLAN · FFL Z −200 · GALLEY · WARDROOM

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.nn: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY; EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

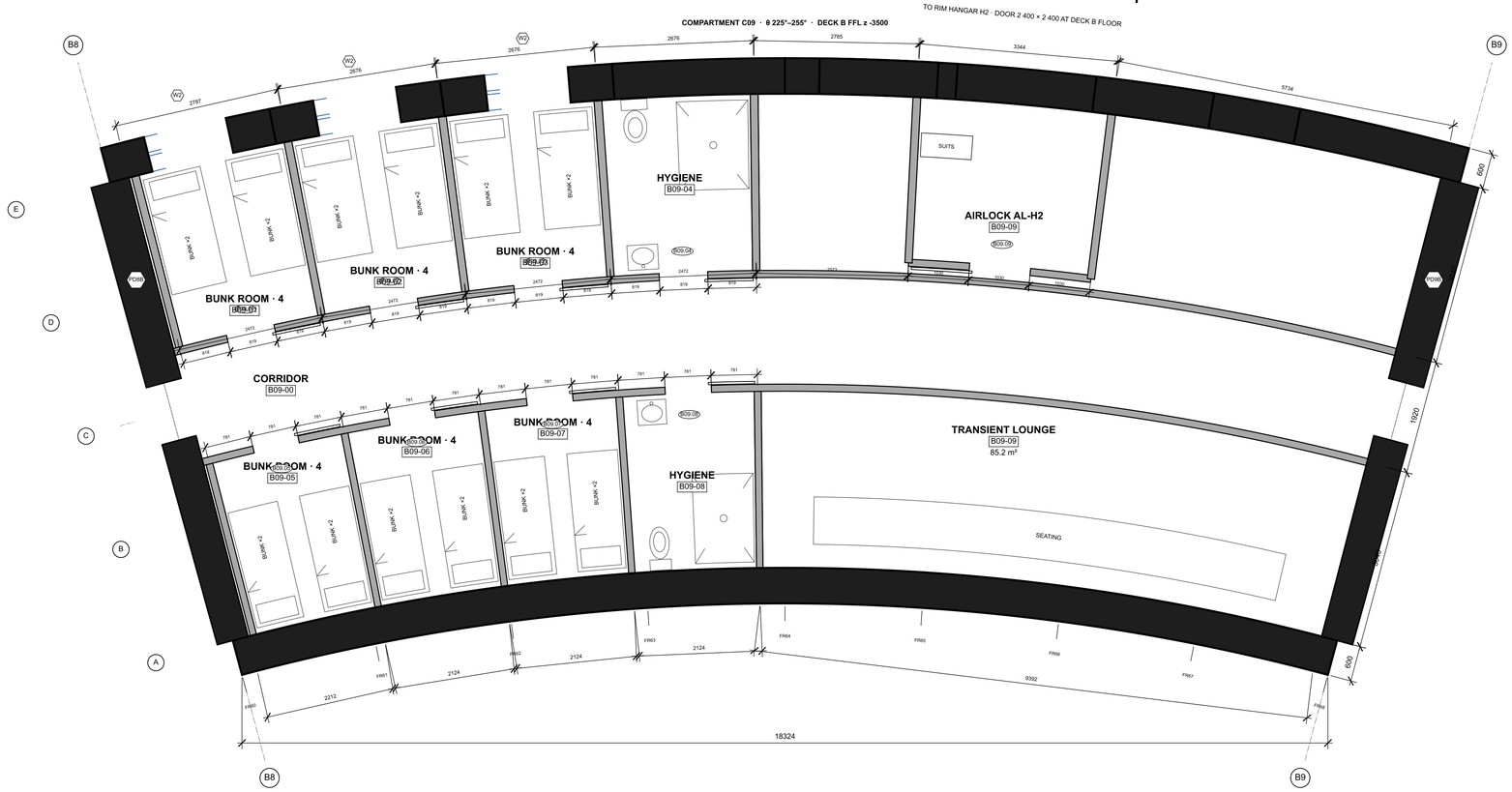
### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

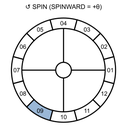
GRAPHIC SCALE 1:50

| | | | |
|---|---|---|---|
| LEGAL OWNER / PROJECT: STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT | | | |
| TITLE: COMPARTMENT C08 · PLANS 1:50<br>θ 195°–225° · Deck A galley · wardroom · Deck B cold / dry stores · laundry | | | |
| DISCIPLINE: Architecture | DOCUMENT STATUS: ISSUED FOR MODELLING | SCALE: AS SHOWN | SIZE / UNITS: A1 · mm |
| CREATED BY: Claude (AI) | APPROVED BY: — | DATE OF ISSUE: 2026-10-17 | PROJECTION |
| IDENTIFICATION NUMBER: SF-OUT-A-218 | | REV: A | SHEET: A-218 |




## 1 DECK B PLAN · FFL Z −3 500 · TRANSIENT BERTHING

SCALE 1:50




### ROOM SCHEDULE · DECK B

| No. | ROOM | m² |
|---|---|---|
| B09-00 | CORRIDOR | 17.5 |
| B09-01 | BUNK ROOM · 4 | 7.2 |
| B09-02 | BUNK ROOM · 4 | 7.2 |
| B09-03 | BUNK ROOM · 4 | 7.2 |
| B09-04 | HYGIENE | 7.2 |
| B09-05 | BUNK ROOM · 4 | 7.2 |
| B09-06 | BUNK ROOM · 4 | 7.2 |
| B09-07 | BUNK ROOM · 4 | 7.2 |
| B09-08 | HYGIENE | 7.2 |
| B09-09 | TRANSIENT LOUNGE | 85.2 |
| B09-09 | AIRLOCK AL-H2 | 8.4 |

### ROOM SCHEDULE · DECK A

| No. | ROOM | m² |
|---|---|---|
| A09-01 | GYMNASIUM | 33.0 |
| A09-03 | LOUNGE | 99.7 |




## 2 DECK A PLAN · FFL Z −200 · GYM · LOUNGE

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.m: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY, EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

GRAPHIC SCALE 1:50

LEGAL OWNER / PROJECT

**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE

**COMPARTMENT C09 · PLANS 1:50**

θ 225°–255° · Deck A gym · lounge · Deck B transient berthing

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| **CREATED BY** | **APPROVED BY** | **DATE OF ISSUE** | **PROJECTION** |
| Claude (AI) | — | 2026-10-17 | |
| **IDENTIFICATION NUMBER** | | **REV** | **SHEET** |
| **SF-OUT-A-219** | | **A** | **A-219** |

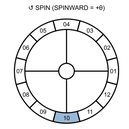
COMPARTMENT C10 · θ 255°–285° · DECK B FFL z −3500




## 1 DECK B PLAN · FFL Z −3 500 · SPOKE LOBBY · AHU PLANT · SUBSTATION

SCALE 1:50

↺ SPIN (SPINWARD = +θ)




KEY PLAN

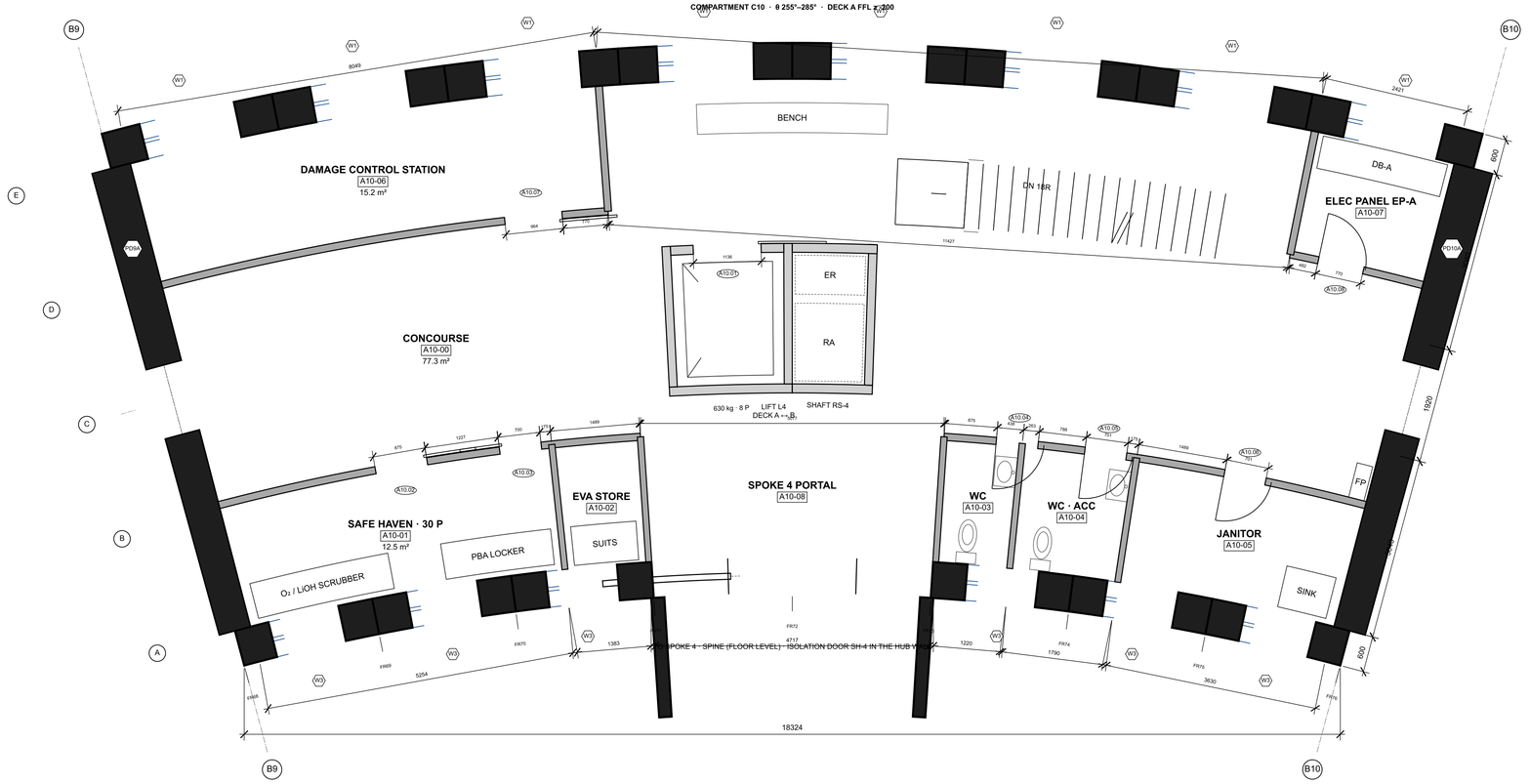
### ROOM SCHEDULE · DECK B

| No. | ROOM | m² |
|---|---|---|
| B10-00 | SPOKE 4 LOBBY | 33.9 |
| B10-01 | PLANT · AHU-R4 | 17.2 |
| B10-02 | SUBSTATION ES-R4 | 14.8 |
| B10-03 | WATER BOOSTER | 6.2 |

### ROOM SCHEDULE · DECK A

| No. | ROOM | m² |
|---|---|---|
| A10-00 | CONCOURSE | 77.3 |
| A10-01 | SAFE HAVEN · 30 P | 12.5 |
| A10-02 | EVA STORE | 3.3 |
| A10-03 | WC | 2.9 |
| A10-04 | WC · ACC | 4.2 |
| A10-05 | JANITOR | 8.6 |
| A10-06 | DAMAGE CONTROL STATION | 15.2 |
| A10-07 | ELEC PANEL EP-A | 4.6 |
| A10-08 | SPOKE 4 PORTAL | 11.2 |

COMPARTMENT C10 · θ 255°–285° · DECK A FFL z −200




## 2 DECK A PLAN · FFL Z −200 · SPOKE JUNCTION · CONCOURSE · SAFE HAVEN

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.nn: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY; EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

GRAPHIC SCALE 1:50

LEGAL OWNER / PROJECT

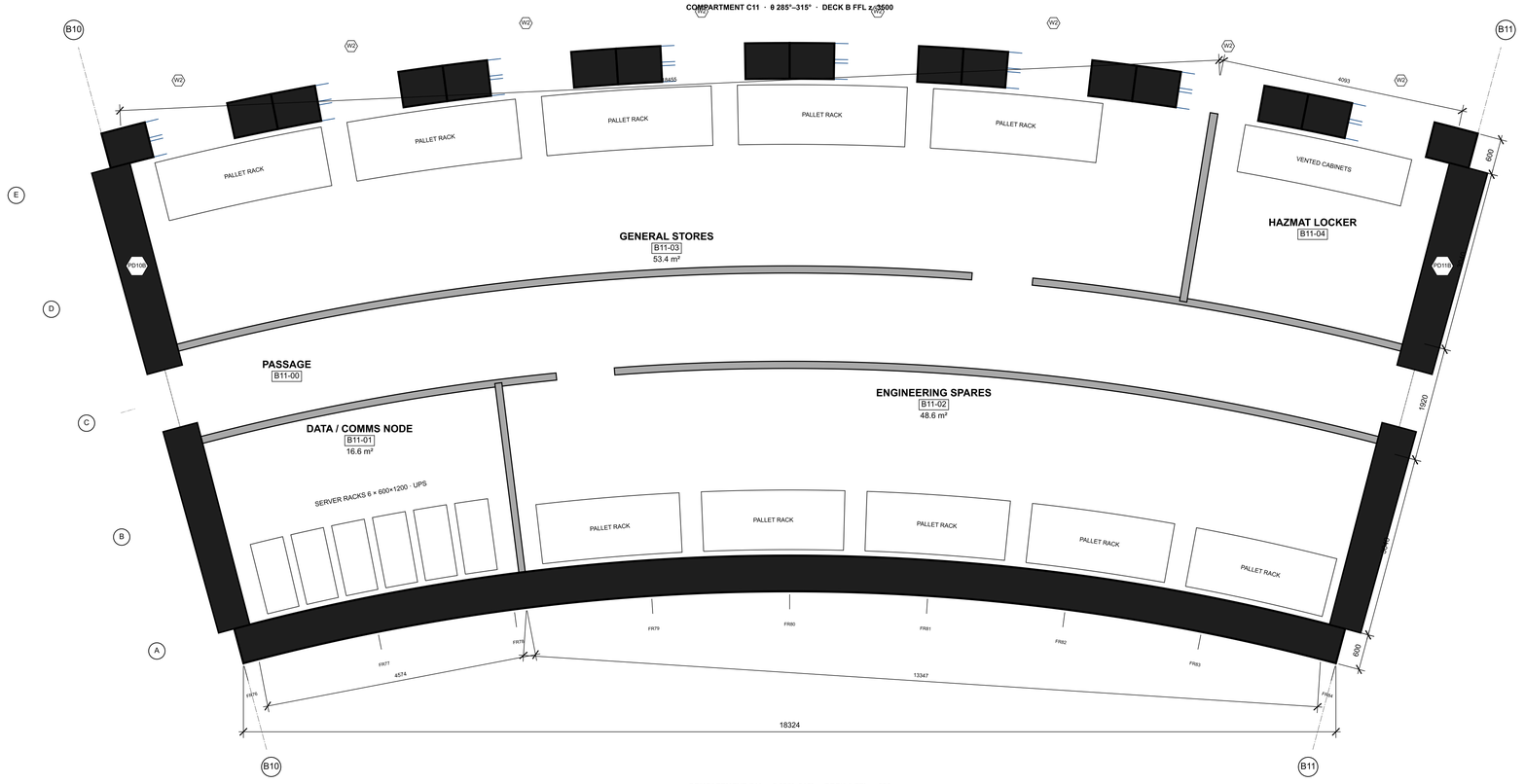
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

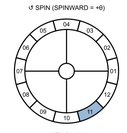
TITLE

# COMPARTMENT C10 · PLANS 1:50

θ 255°–285° · Deck A spoke junction · concourse · safe haven · Deck B spoke lobby · ahu plant · substation

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-220 | | A | A-220 |




## 1 DECK B PLAN · FFL Z −3 500 · STORES · DATA NODE

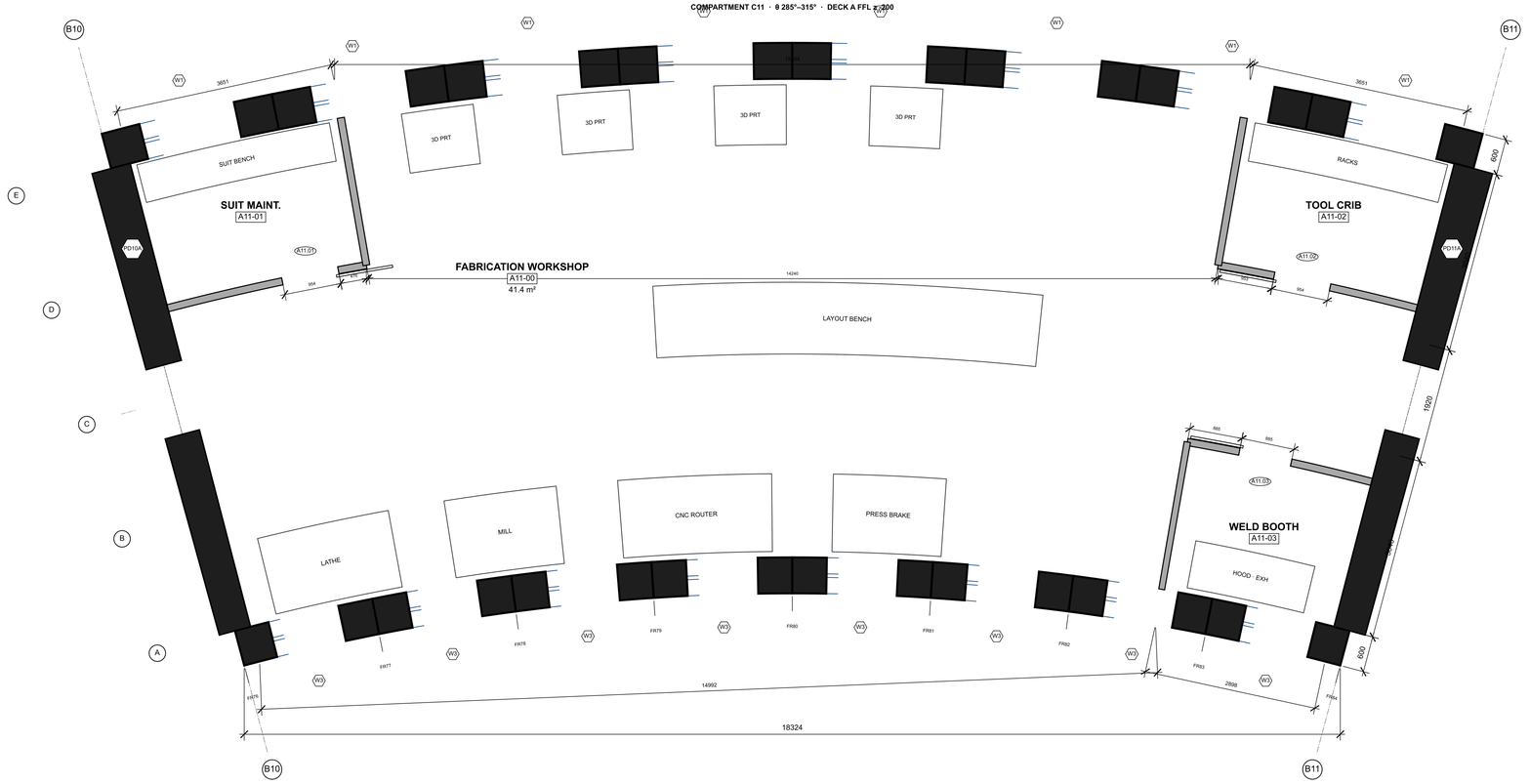
SCALE 1:50

### ROOM SCHEDULE · DECK B

| No. | ROOM | m² |
|---|---|---|
| B11-00 | PASSAGE | 32.6 |
| B11-01 | DATA / COMMS NODE | 16.6 |
| B11-02 | ENGINEERING SPARES | 48.6 |
| B11-03 | GENERAL STORES | 53.4 |
| B11-04 | HAZMAT LOCKER | 11.8 |

### ROOM SCHEDULE · DECK A

| No. | ROOM | m² |
|---|---|---|
| A11-00 | FABRICATION WORKSHOP | 41.4 |
| A11-01 | SUIT MAINT. | 8.2 |
| A11-02 | TOOL CRIB | 8.2 |
| A11-03 | WELD BOOTH | 8.2 |




## 2 DECK A PLAN · FFL Z −200 · FABRICATION WORKSHOP

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.nn: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY; EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

GRAPHIC SCALE 1:50

LEGAL OWNER / PROJECT

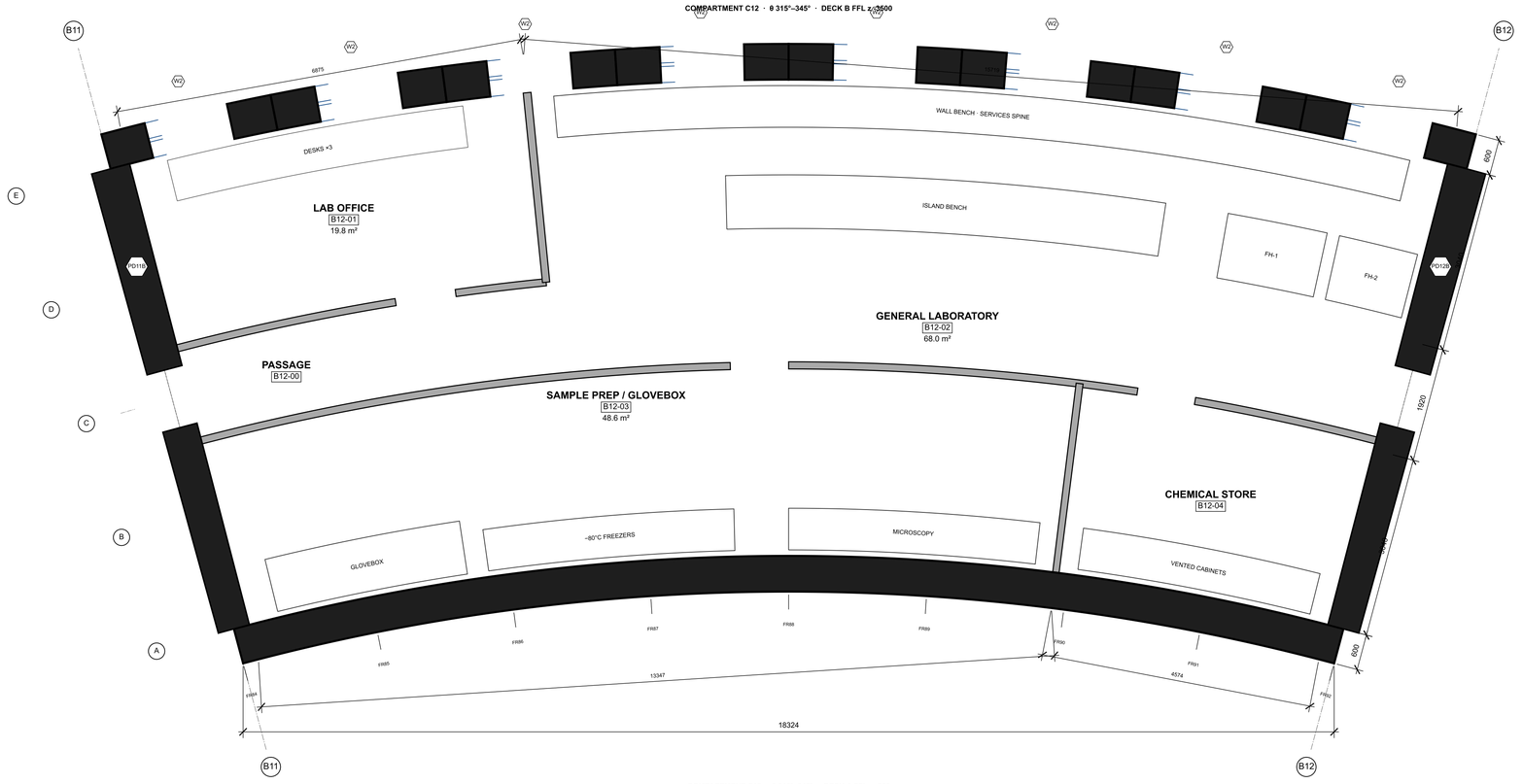
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

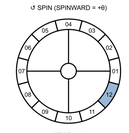
TITLE

# COMPARTMENT C11 · PLANS 1:50

θ 285°–315° · Deck A fabrication workshop · Deck B stores · data node

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-221 | | A | A-221 |




## 1 DECK B PLAN · FFL Z −3 500 · LABORATORY

SCALE 1:50

### ROOM SCHEDULE · DECK B

| No. | ROOM | m² |
|---|---|---|
| B12-00 | PASSAGE | 9.9 |
| B12-01 | LAB OFFICE | 19.8 |
| B12-02 | GENERAL LABORATORY | 68.0 |
| B12-03 | SAMPLE PREP / GLOVEBOX | 48.6 |
| B12-04 | CHEMICAL STORE | 16.6 |

### ROOM SCHEDULE · DECK A

| No. | ROOM | m² |
|---|---|---|
| A12-00 | MEDICAL CORRIDOR | 36.6 |
| A12-01 | RECEPTION / WAIT | 13.9 |
| A12-02 | EXAM 1 | 9.6 |
| A12-03 | EXAM 2 | 9.6 |
| A12-04 | TREATMENT / TRAUMA | 13.7 |
| A12-05 | PHARMACY | 13.3 |
| A12-06 | ISOLATION (-ve P) | 16.6 |
| A12-07 | SICKBAY · 4 BEDS | 19.2 |
| A12-08 | MO OFFICE | 8.2 |
| A12-09 | STERILE / UTILITY | 8.2 |
| A12-10 | STAFF WC | 8.4 |




## 2 DECK A PLAN · FFL Z −200 · MEDICAL CENTRE

SCALE 1:50

### NOTES

1. TRUE PLANS FROM +Z; OUTWARD (+r) IS UP THE SHEET. GRAVITY 1.0 g, DOWN = −Z.
2. PLAN CUT 1 200 ABOVE FFL. DIMENSIONS mm TO FACE OF STRUCTURE.
3. DOOR TAGS C.nn: SEE DOOR TYPES, A-501. PD = PRESSURE DOOR.
4. CEILING +3 200 (DECK A) / +2 600 (DECK B). VENTILATION: M-20x.
5. ONE CABIN PER FRAME BAY; EACH WITH ITS OWN WINDOW (W1/W2 OUTER, W3 HUB).

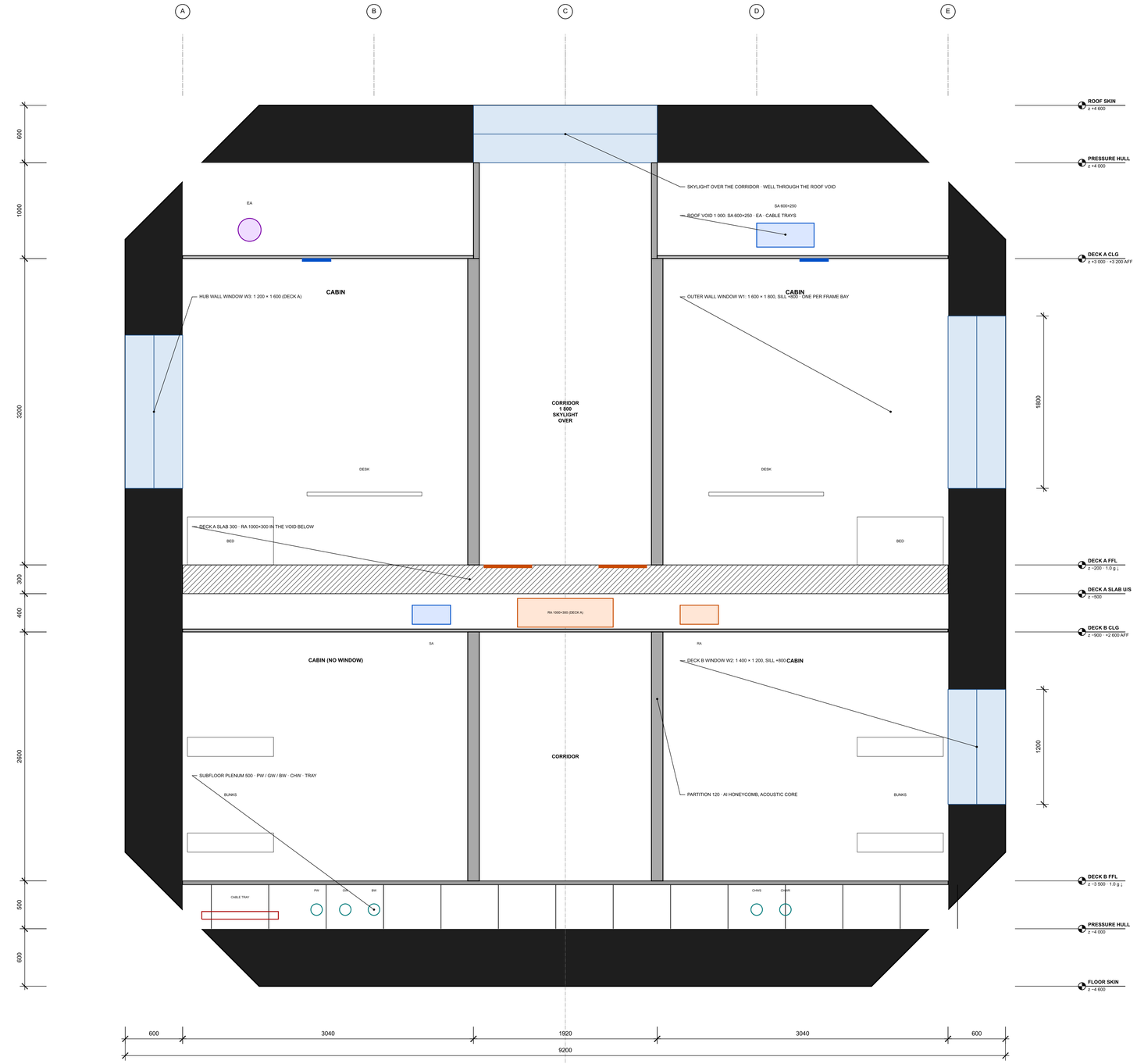
### LEGEND

- PRESSURE HULL (CUT) / WINDOW BAND
- PRESSURE BULKHEAD 600 + SLIDING PRESSURE DOOR
- PARTITION 120 (CUT)
- RATED SHAFT ENCLOSURE 150
- OVERHEAD (SPOKE / HATCH ABOVE)
- HIDDEN / BELOW
- BULKHEAD GRID B1–B12 · AXIAL GRID A–E
- SLIDING DOOR (SHOWN OPEN) / HINGED DOOR

0 1 2 3 4 5 6 7 8 9 10 m

GRAPHIC SCALE 1:50

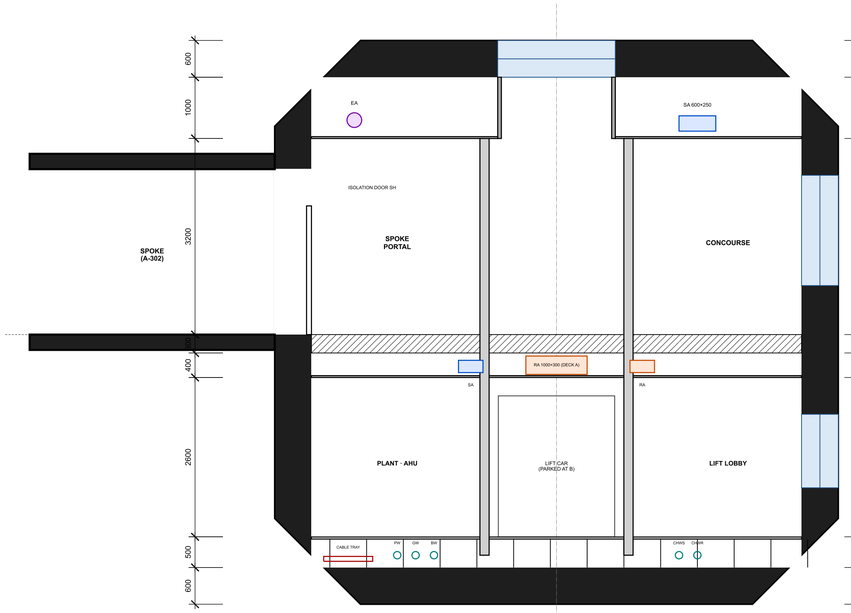
| | | | |
|---|---|---|---|
| LEGAL OWNER / PROJECT: STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT | | | |
| TITLE: COMPARTMENT C12 · PLANS 1:50 — θ 315°–345° · Deck A medical centre · Deck B laboratory | | | |
| DISCIPLINE: Architecture | DOCUMENT STATUS: ISSUED FOR MODELLING | SCALE: AS SHOWN | SIZE / UNITS: A1 · mm |
| CREATED BY: Claude (AI) | APPROVED BY: — | DATE OF ISSUE: 2026-10-17 | PROJECTION |
| IDENTIFICATION NUMBER: SF-OUT-A-222 | | REV: A | SHEET: A-222 |




## 1 TYPICAL RING SECTION · QUARTERS (C05)

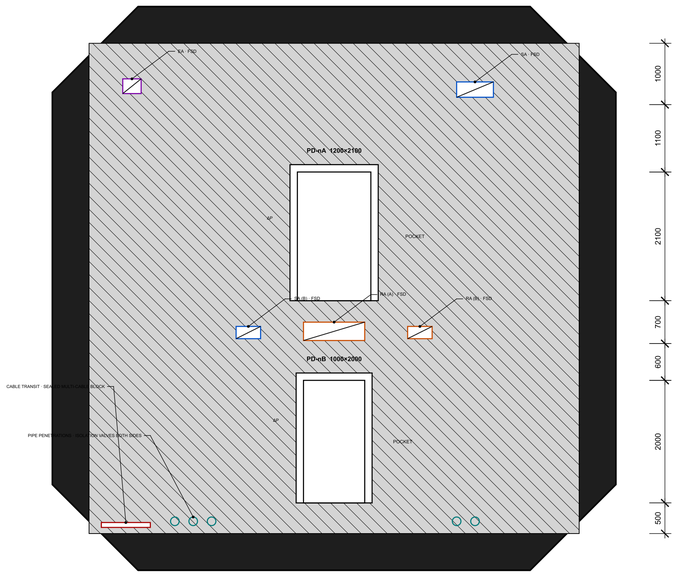
SCALE 1:25

SECTIONS LOOK ALONG THE CIRCUMFERENCE (SPINWARD); HUB WALL LEFT, OUTER WALL RIGHT; 'UP' IS +Z (GRAVITY 1.0 g DOWN).
REV B: TWO FLAT DECKS; WINDOWS IN THE CURVED WALLS (W1/W2 OUTER, W3 HUB), SKYLIGHTS IN THE ROOF OVER THE CORRIDOR.
VOIDS: SUBFLOOR PLENUM 500 · DECK B / DECK A INTERSTITIAL VOID 400 + SLAB 300 · ROOF VOID 1 000.
FLOORS: RAISED ACCESS FLOOR ON PEDESTALS @ 600 (DECK B); SANDWICH SLAB ON BEAMS (DECK A).




## 2 SECTION THROUGH SPOKE JUNCTION (C04)

SCALE 1:50




## 3 PRESSURE BULKHEAD · ELEVATION LOOKING SPINWARD

SCALE 1:50

| | |
|---|---|
| LEGAL OWNER / PROJECT | STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT |
| TITLE | RING SECTIONS · BULKHEAD<br>Typical tube section · junction section · pressure bulkhead elevation |
| DISCIPLINE | Architecture |
| DOCUMENT STATUS | ISSUED FOR MODELLING |
| SCALE | AS SHOWN |
| SIZE / UNITS | A1 · mm |
| CREATED BY | Claude (AI) |
| APPROVED BY | — |
| DATE OF ISSUE | 2026-10-17 |
| PROJECTION | |
| IDENTIFICATION NUMBER | SF-OUT-A-301 |
| REV. | A |
| SHEET | A-301 |

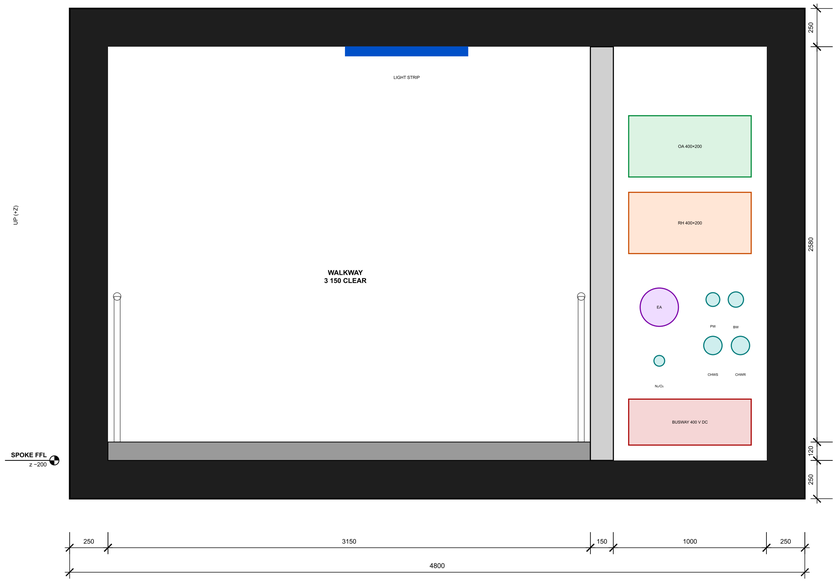



## 1 SPOKE PLAN · CUT 1 200 ABOVE THE WALKWAY · TYPICAL OF SPOKES 1–4

SCALE 1:50

**SPOKE**

TANGENTIAL




## 2 SPOKE CROSS-SECTION · LOOKING OUTWARD

SCALE 1:20

1. A LEVEL WALKWAY: ITS FLOOR (z −200) IS THE SPINE HUB DECK AND RING DECK A, SO THE CREW WALK STRAIGHT THROUGH.
2. 28.0 m FROM THE SPINE WALL TO THE RING'S HUB WALL; HANDRAILS BOTH SIDES; FLOOR LIGHT STRIPS; 2 VIEW PORTS EACH SIDE.
3. ISOLATION DOORS AT BOTH ENDS SEAL THE SPOKE FROM THE HUB AND FROM THE RING.
4. THE CHASE CARRIES OA 400×200 (HUB→AHU), RH 400×200 (AHU→HUB), EA Ø250, PW, BW, CHWS/R, $N_2/O_2$ MAKE-UP, 400 V DC BUSWAY AND DATA; ALL PASS THE END BULKHEADS THROUGH ISOLATING FITTINGS.
5. AT THE RING THE CHASE DROPS INTO SHAFT RS-n TO THE DECK B PLANT ROOM (A-21x).

LEGAL OWNER / PROJECT

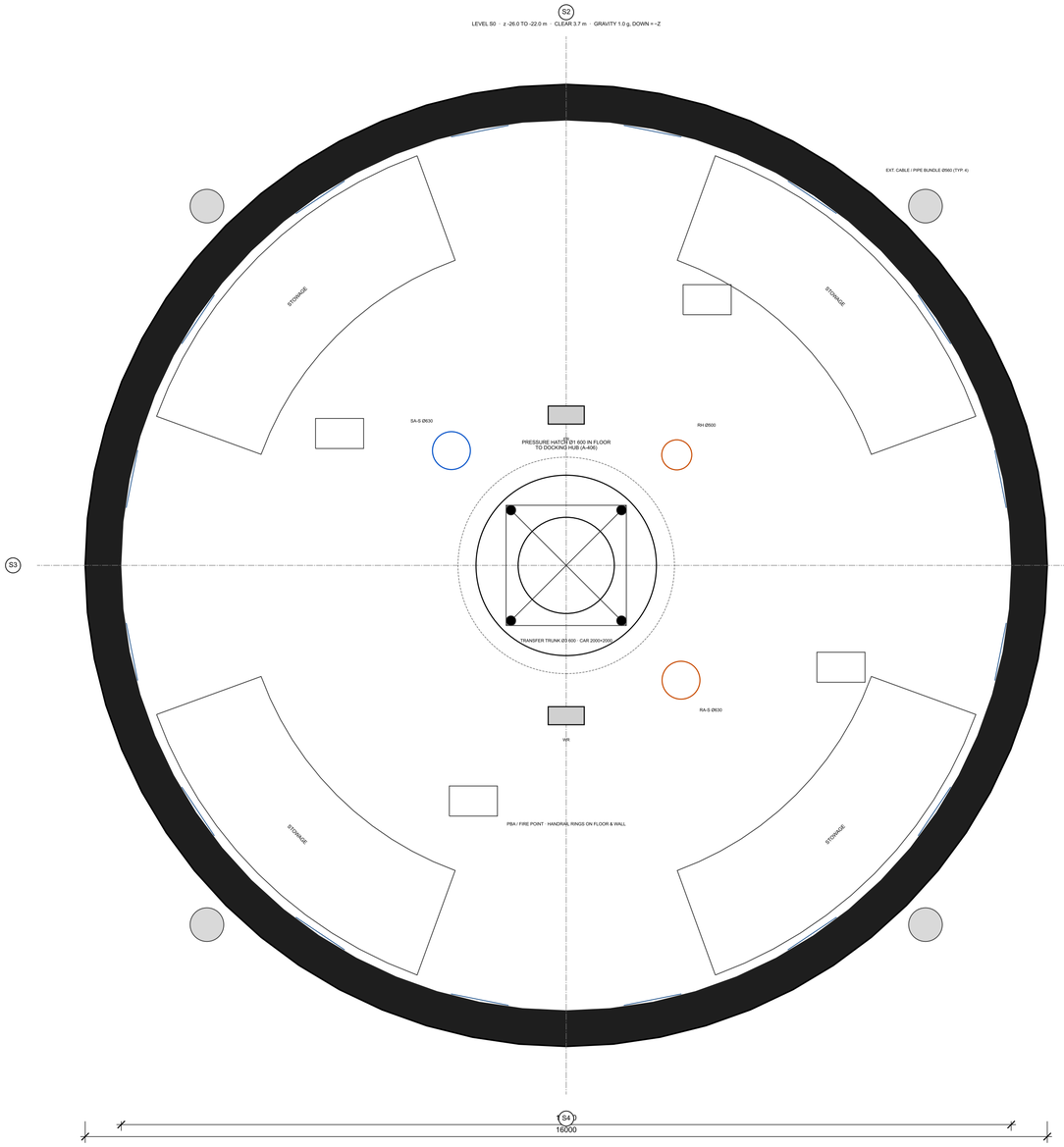
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE

**SPOKE**

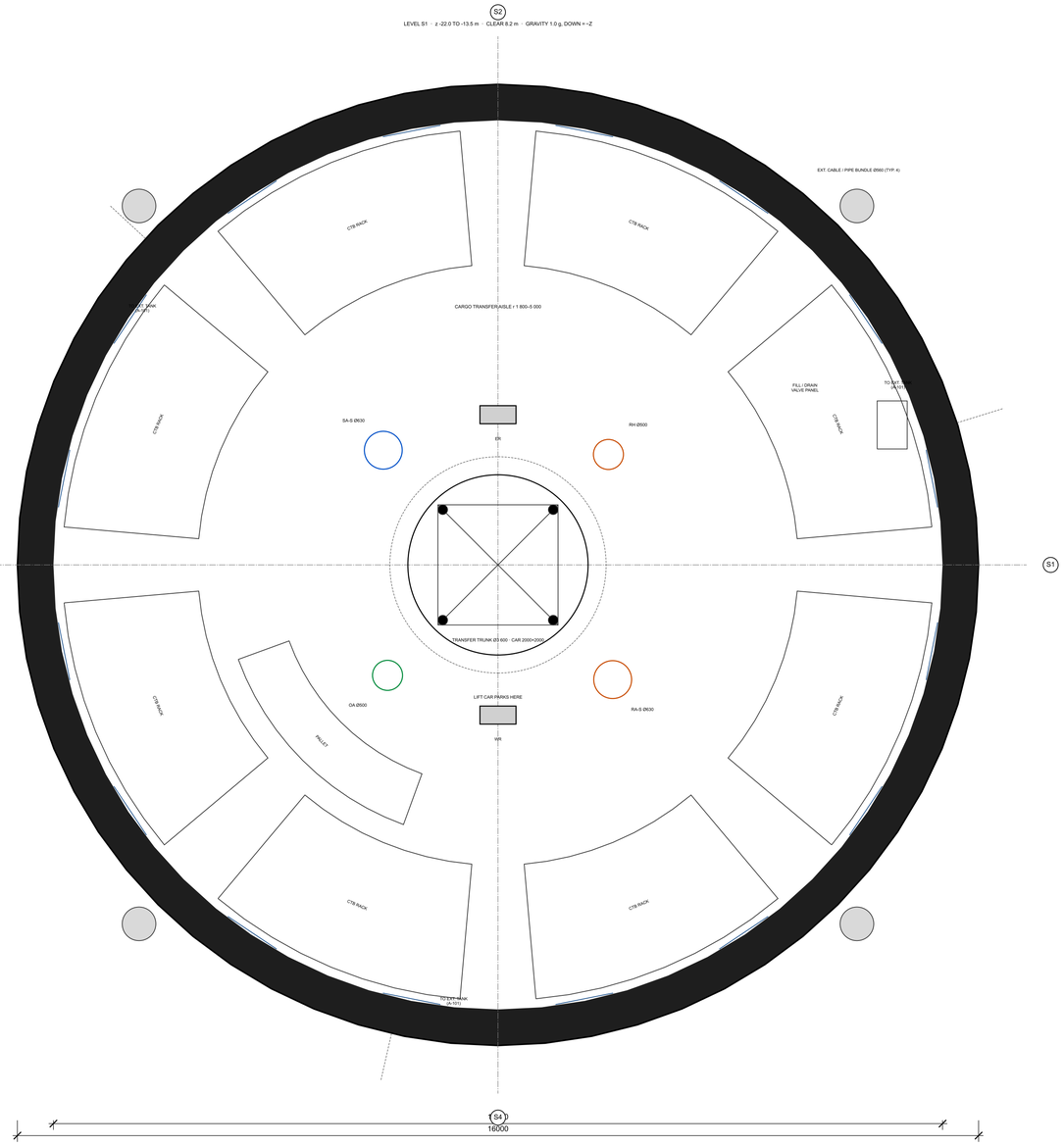
Spoke plan from +Z and cross-section

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV. | SHEET |
| **SF-OUT-A-302** | | **A** | **A-302** |




## 1 LEVEL S0 · LOWER TRUNK · DOCK VESTIBULE

SCALE 1:50




## 2 LEVEL S1 · CARGO HOLD · LOGISTICS

SCALE 1:50

1. GRAVITY 1.0 g DOWN = −Z, AS IN THE RING: EVERY SPINE LEVEL IS AN ORDINARY FLOOR. S3a (z −200) IS LEVEL WITH THE SPOKES AND THE RING DECK A, SO THE HUB IS ONE CONTINUOUS FLOOR WITH ALL FOUR SPOKES.
2. CENTRAL TRANSFER TRUNK Ø3 600 THROUGH EVERY DECK; THE TRANSFER CAR RUNS ON FOUR RAILS AT r 1 300.
3. RISERS AT EVERY LEVEL: SA-S / RA-S Ø630 (SPINE AHU), OA / RH Ø500 (HUB +- ECLSS), ELECTRICAL (ER), WATER (WR).
4. 16 PORTHOLES Ø1 400 PER LEVEL AT MID-HEIGHT (NOT ON S3a, WHICH HAS THE SPOKE PORTALS).

LEGAL OWNER / PROJECT

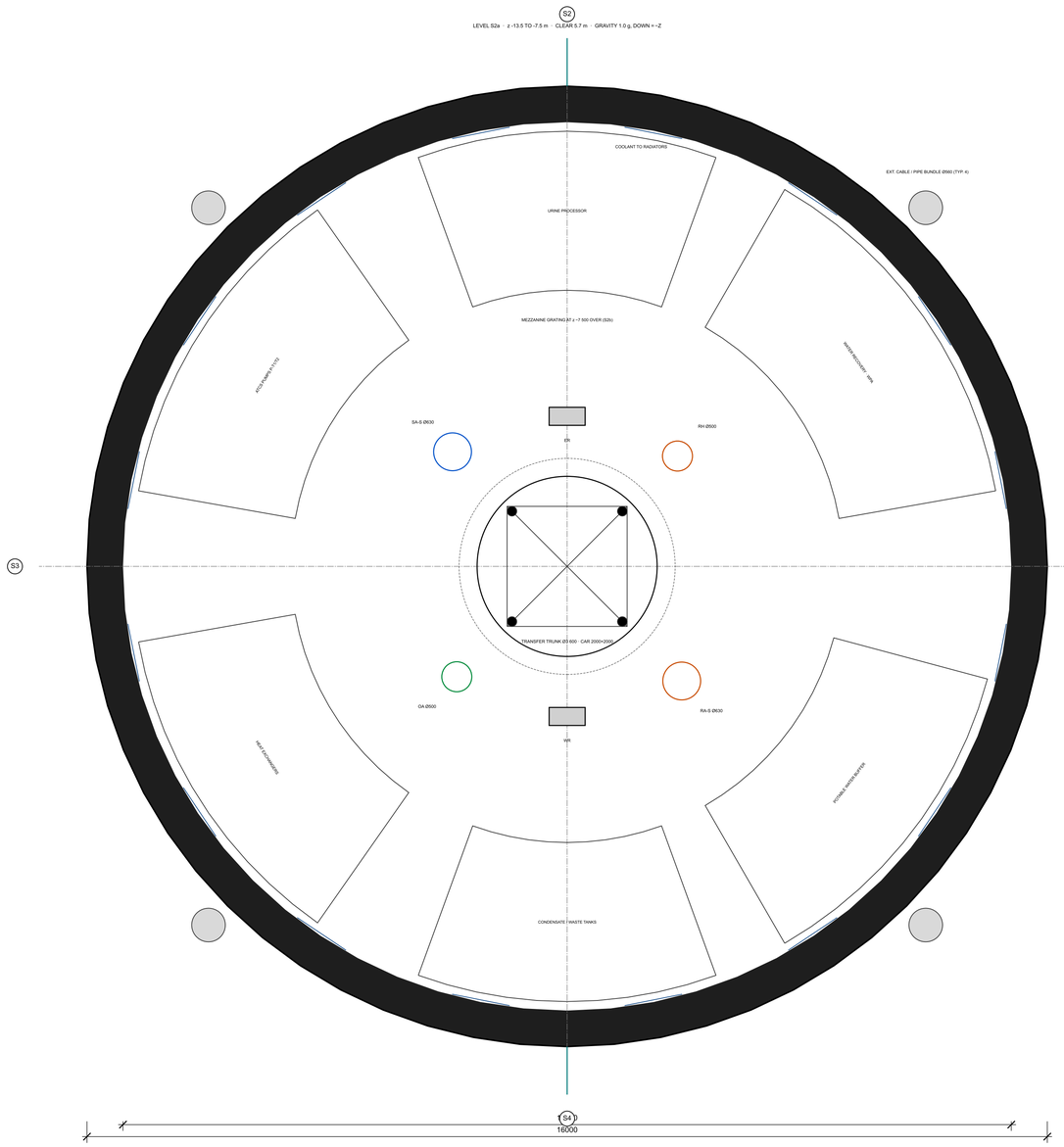
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE

# SPINE LEVELS

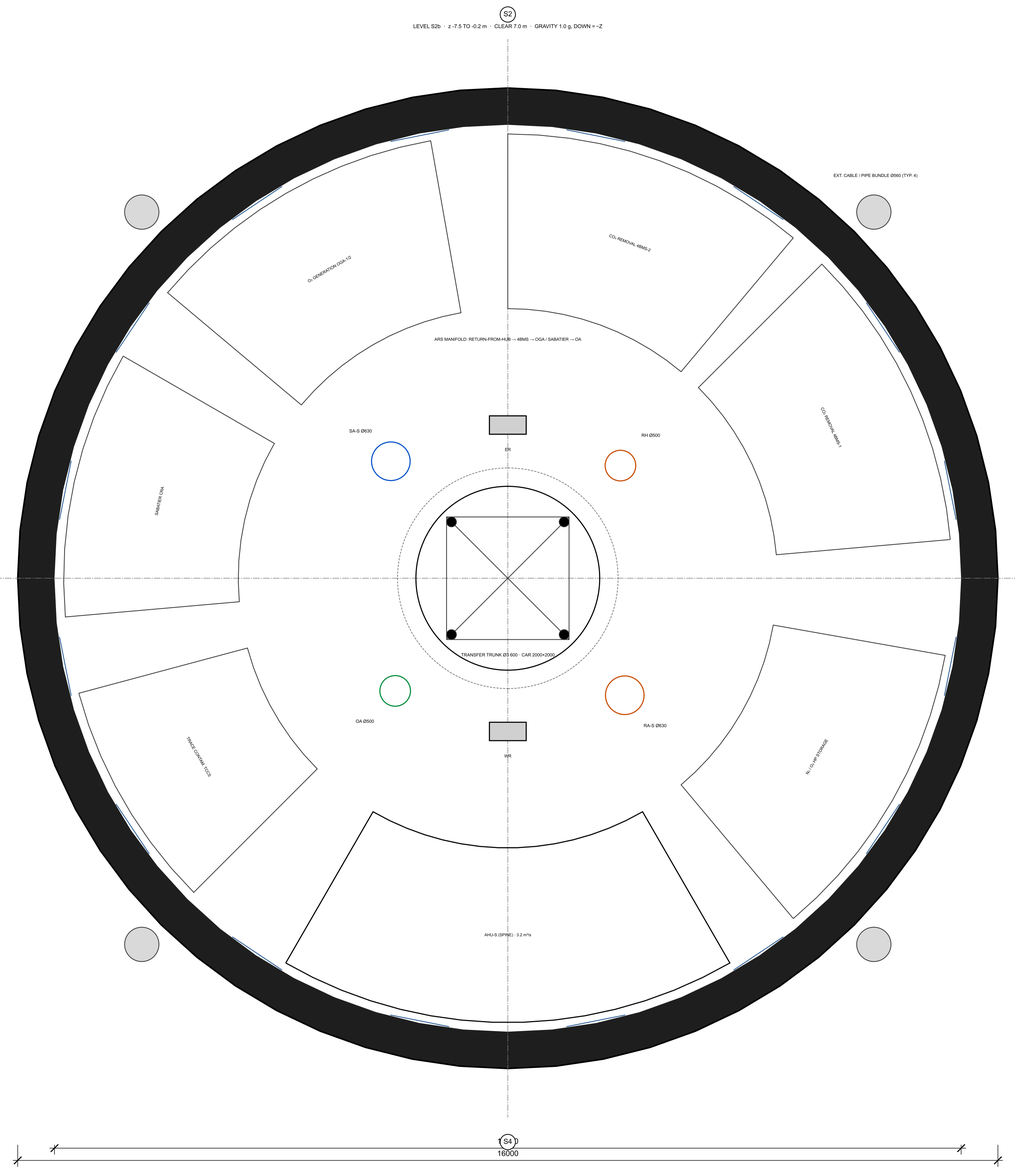
Level S0 · Level S1 · plans viewed from +Z

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| **CREATED BY** | **APPROVED BY** | **DATE OF ISSUE** | **PROJECTION** |
| Claude (AI) | — | 2026-10-17 | |
| **IDENTIFICATION NUMBER** | | **REV** | **SHEET** |
| **SF-OUT-A-401** | | **A** | **A-401** |




## 1 LEVEL S2A · ECLSS PLANT · WATER & THERMAL

SCALE 1:50




## 2 LEVEL S2B · ECLSS PLANT · AIR REVITALISATION

SCALE 1:50

1. GRAVITY 1.0 g DOWN = −Z, AS IN THE RING: EVERY SPINE LEVEL IS AN ORDINARY FLOOR. S3a (z −200) IS LEVEL WITH THE SPOKES AND THE RING DECK A, SO THE HUB IS ONE CONTINUOUS FLOOR WITH ALL FOUR SPOKES.
2. CENTRAL TRANSFER TRUNK Ø3 600 THROUGH EVERY DECK; THE TRANSFER CAR RUNS ON FOUR RAILS AT r 1 300.
3. RISERS AT EVERY LEVEL: SA-S / RA-S Ø630 (SPINE AHU), OA / RH Ø500 (HUB ↔ ECLSS), ELECTRICAL (ER), WATER (WR).
4. 16 PORTHOLES Ø1 400 PER LEVEL AT MID-HEIGHT (NOT ON S3a, WHICH HAS THE SPOKE PORTALS).

LEGAL OWNER / PROJECT

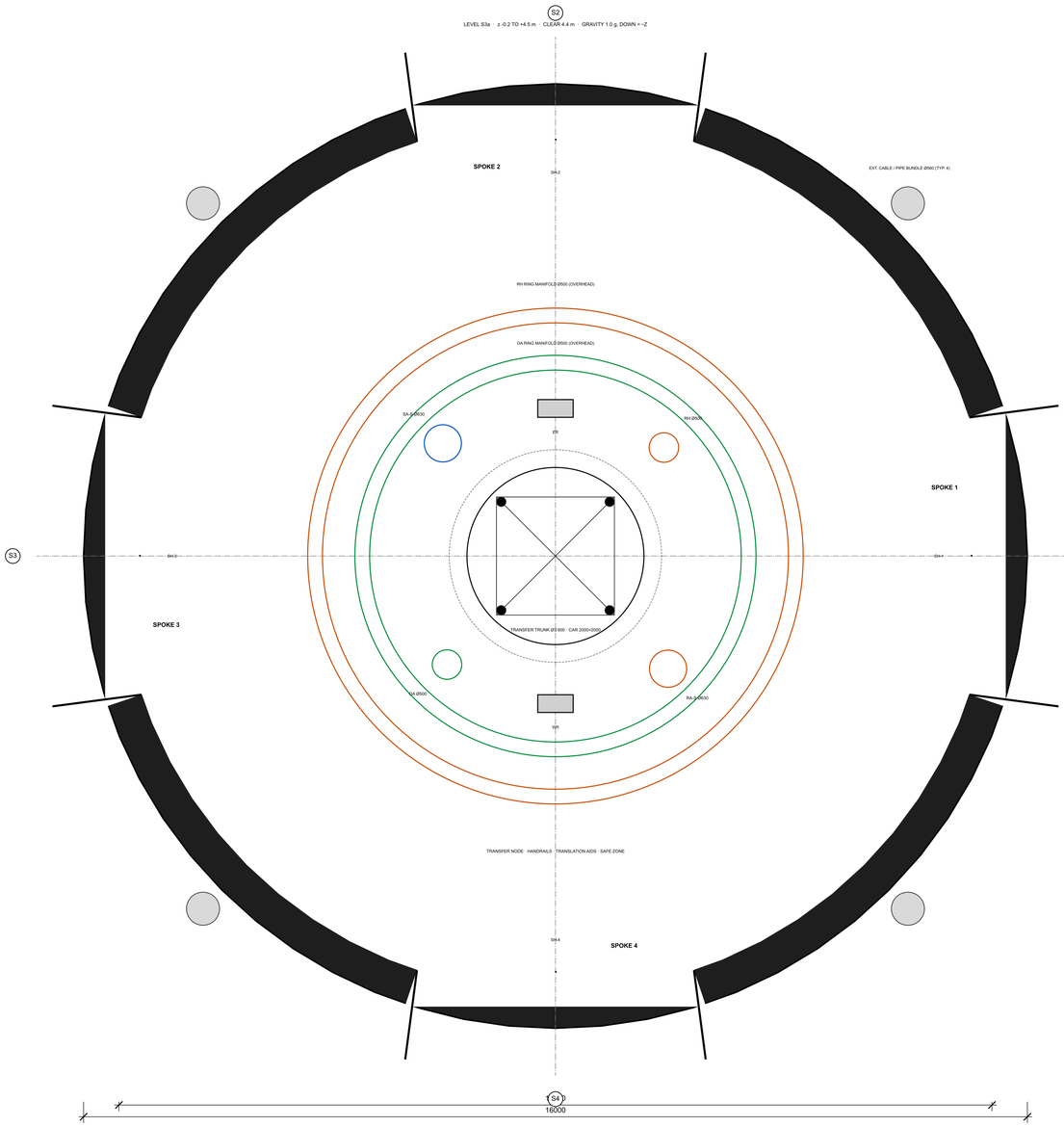
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE

# SPINE LEVELS

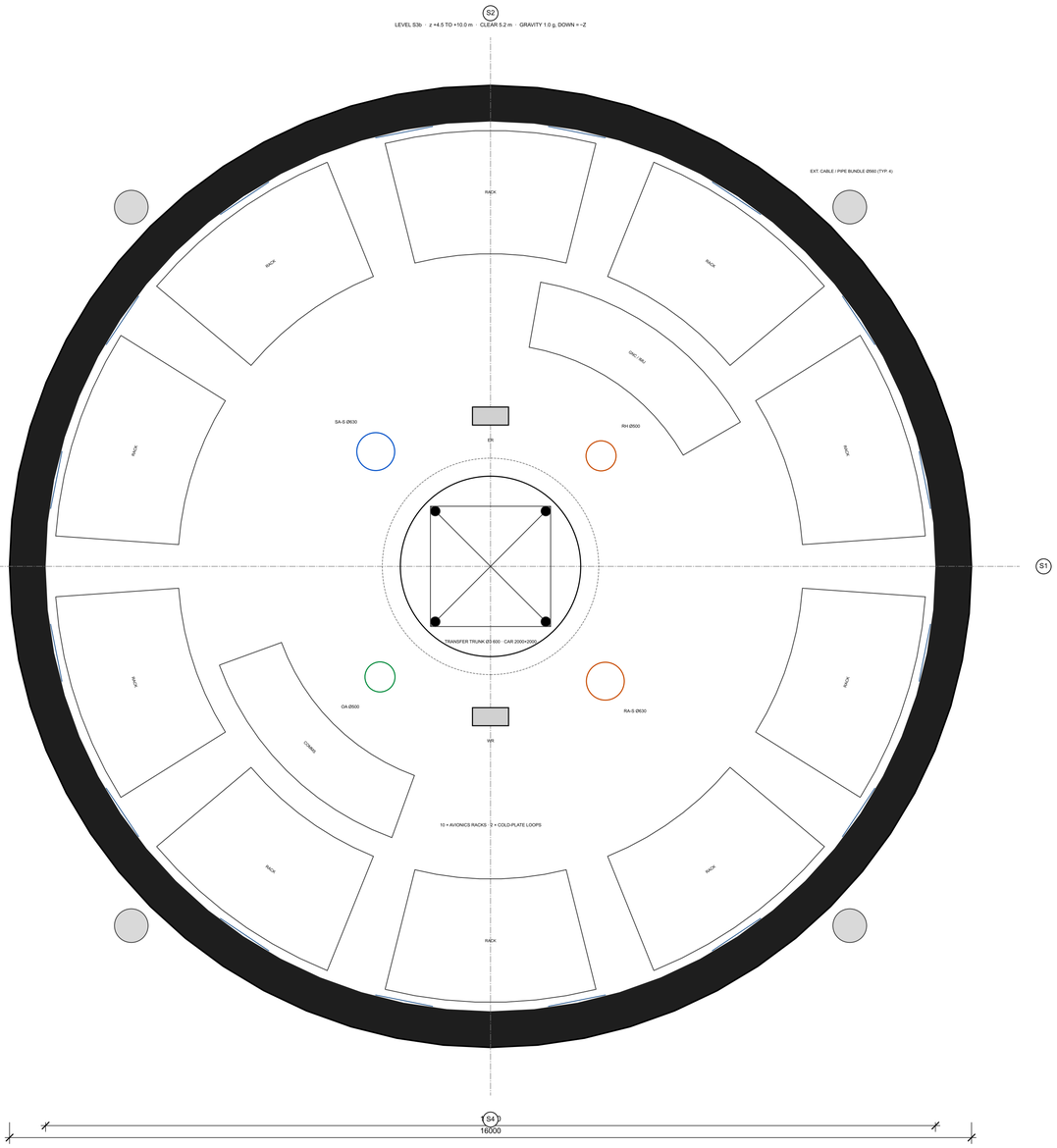
Level S2a · Level S2b · plans viewed from +Z

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| **CREATED BY** | **APPROVED BY** | **DATE OF ISSUE** | **PROJECTION** |
| Claude (AI) | — | 2026-10-17 | |
| **IDENTIFICATION NUMBER** | | **REV** | **SHEET** |
| **SF-OUT-A-402** | | **A** | **A-402** |




## 1 LEVEL S3A · HUB TRANSFER NODE · SPOKE JUNCTION

SCALE 1:50




## 2 LEVEL S3B · AVIONICS · DATA · GNC

SCALE 1:50

1. GRAVITY 1.0 g DOWN = −Z, AS IN THE RING: EVERY SPINE LEVEL IS AN ORDINARY FLOOR. S3a (z −200) IS LEVEL WITH THE SPOKES AND THE RING DECK A, SO THE HUB IS ONE CONTINUOUS FLOOR WITH ALL FOUR SPOKES.
2. CENTRAL TRANSFER TRUNK Ø3 600 THROUGH EVERY DECK; THE TRANSFER CAR RUNS ON FOUR RAILS AT r 1 300.
3. RISERS AT EVERY LEVEL: SA-S / RA-S Ø630 (SPINE AHU), OA / RH Ø500 (HUB ↔ ECLSS), ELECTRICAL (ER), WATER (WR).
4. 16 PORTHOLES Ø1 400 PER LEVEL AT MID-HEIGHT (NOT ON S3a, WHICH HAS THE SPOKE PORTALS).

LEGAL OWNER / PROJECT

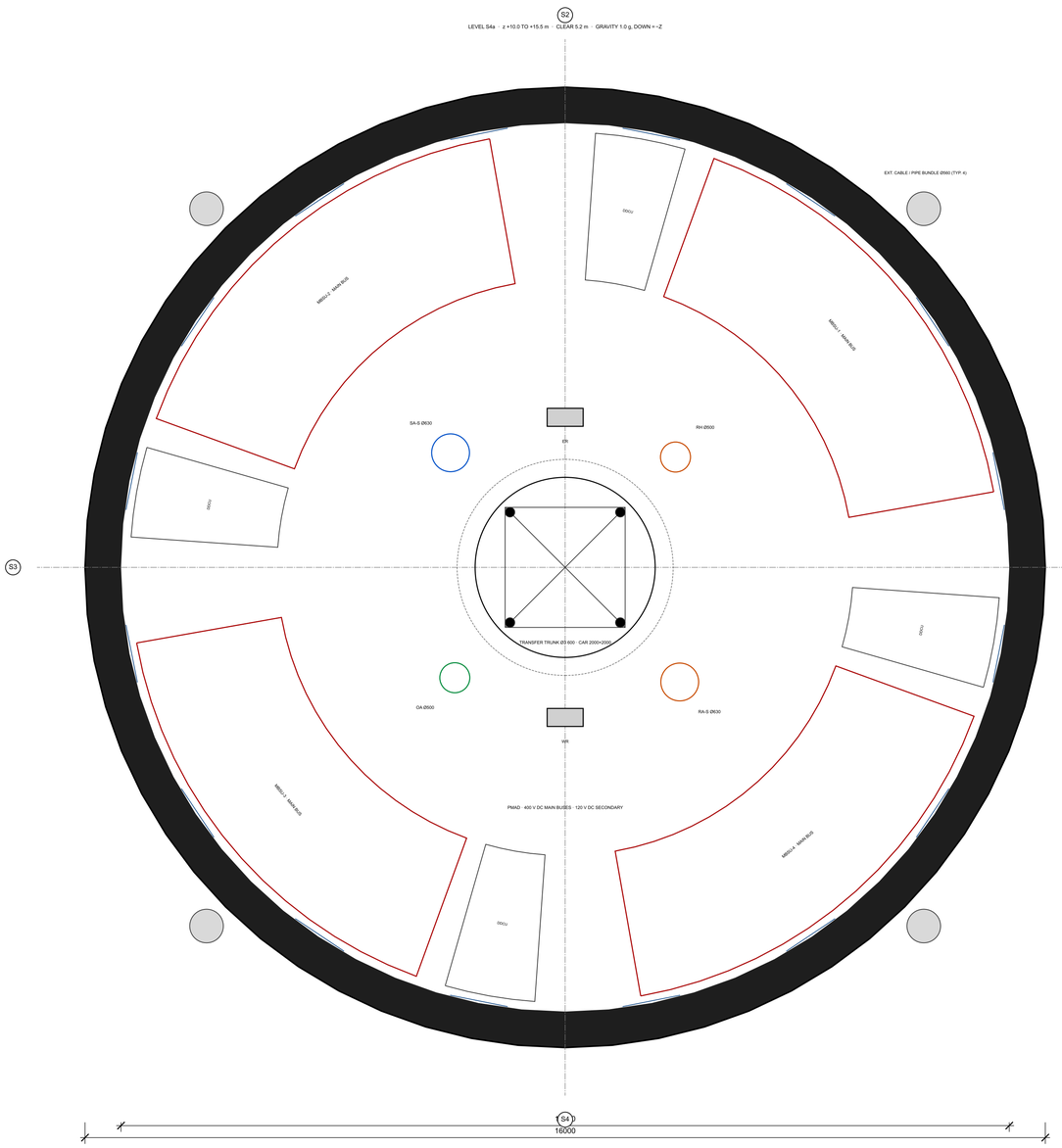
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE

# SPINE LEVELS

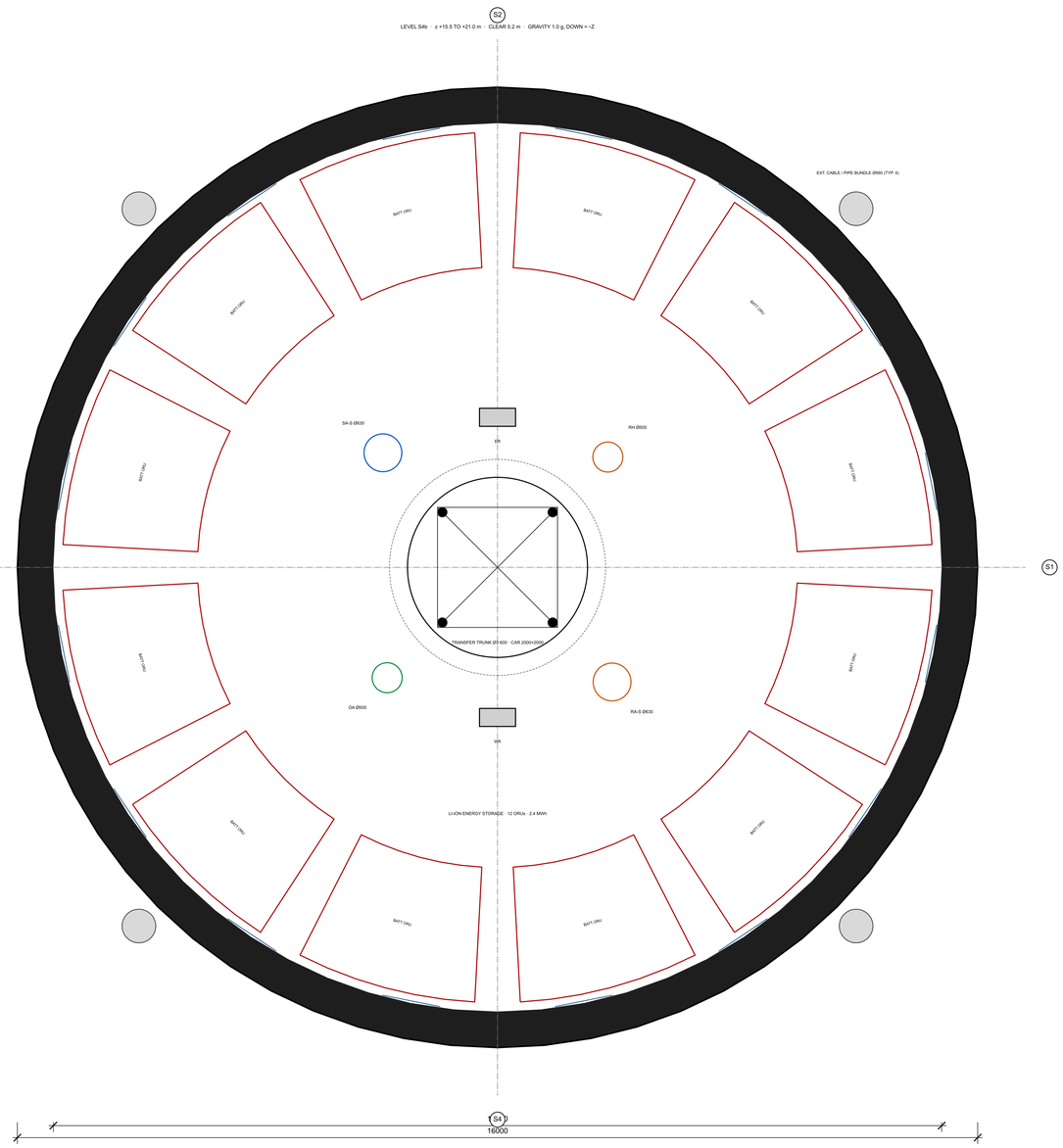
Level S3a · Level S3b · plans viewed from +Z

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| **CREATED BY** | **APPROVED BY** | **DATE OF ISSUE** | **PROJECTION** |
| Claude (AI) | — | 2026-10-17 | |
| **IDENTIFICATION NUMBER** | | **REV** | **SHEET** |
| SF-OUT-A-403 | | A | A-403 |




## 1 LEVEL S4A · POWER MANAGEMENT & DISTRIBUTION

SCALE 1:50




## 2 LEVEL S4B · ENERGY STORAGE

SCALE 1:50

1. GRAVITY 1.0 g DOWN = −Z, AS IN THE RING: EVERY SPINE LEVEL IS AN ORDINARY FLOOR. S3a (z −200) IS LEVEL WITH THE SPOKES AND THE RING DECK A, SO THE HUB IS ONE CONTINUOUS FLOOR WITH ALL FOUR SPOKES.
2. CENTRAL TRANSFER TRUNK Ø3 600 THROUGH EVERY DECK; THE TRANSFER CAR RUNS ON FOUR RAILS AT r 1 300.
3. RISERS AT EVERY LEVEL: SA-S / RA-S Ø630 (SPINE AHU), OA / RH Ø500 (HUB +- ECLSS), ELECTRICAL (ER), WATER (WR).
4. 16 PORTHOLES Ø1 400 PER LEVEL AT MID-HEIGHT (NOT ON S3a, WHICH HAS THE SPOKE PORTALS).

LEGAL OWNER / PROJECT

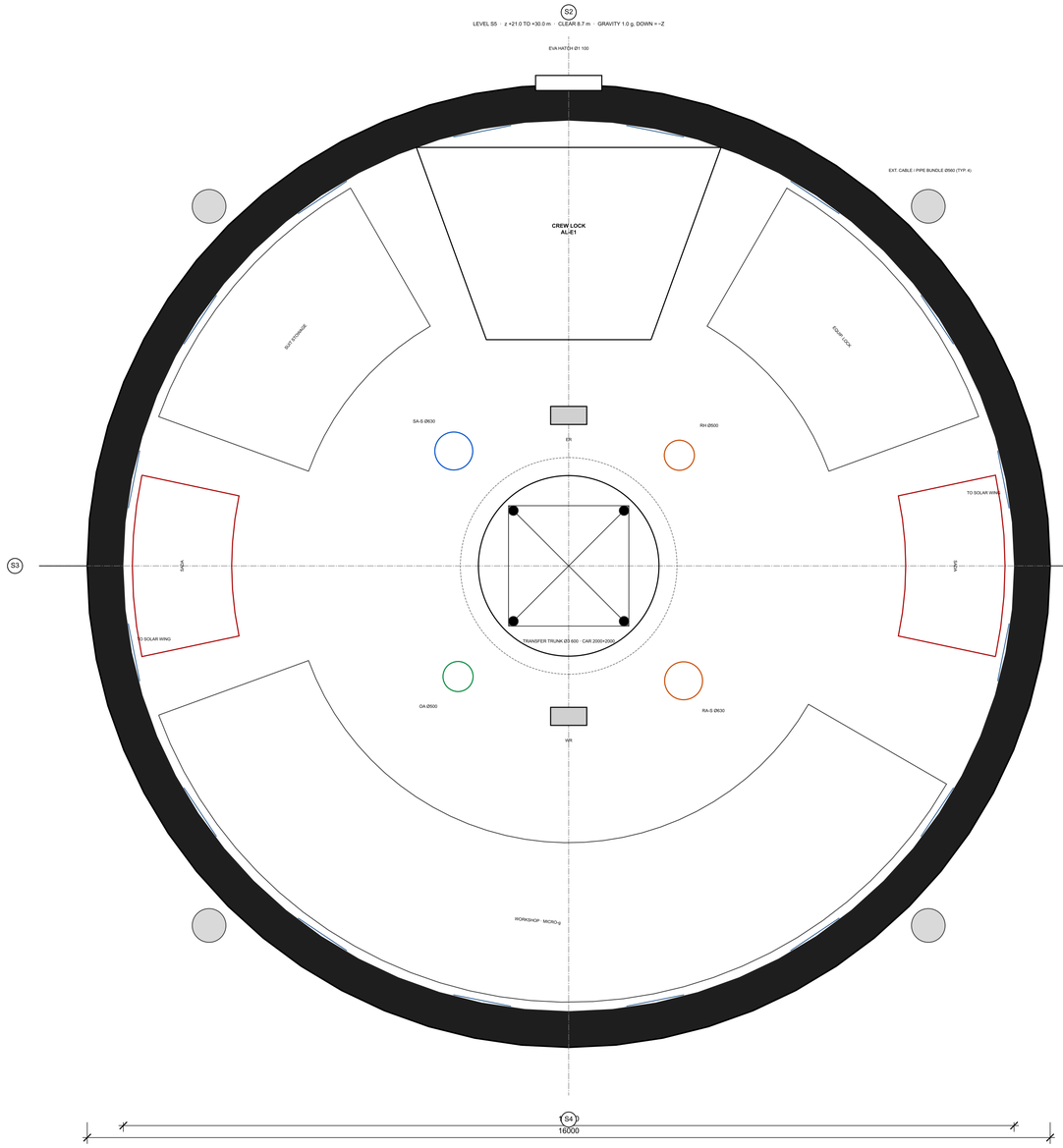
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE

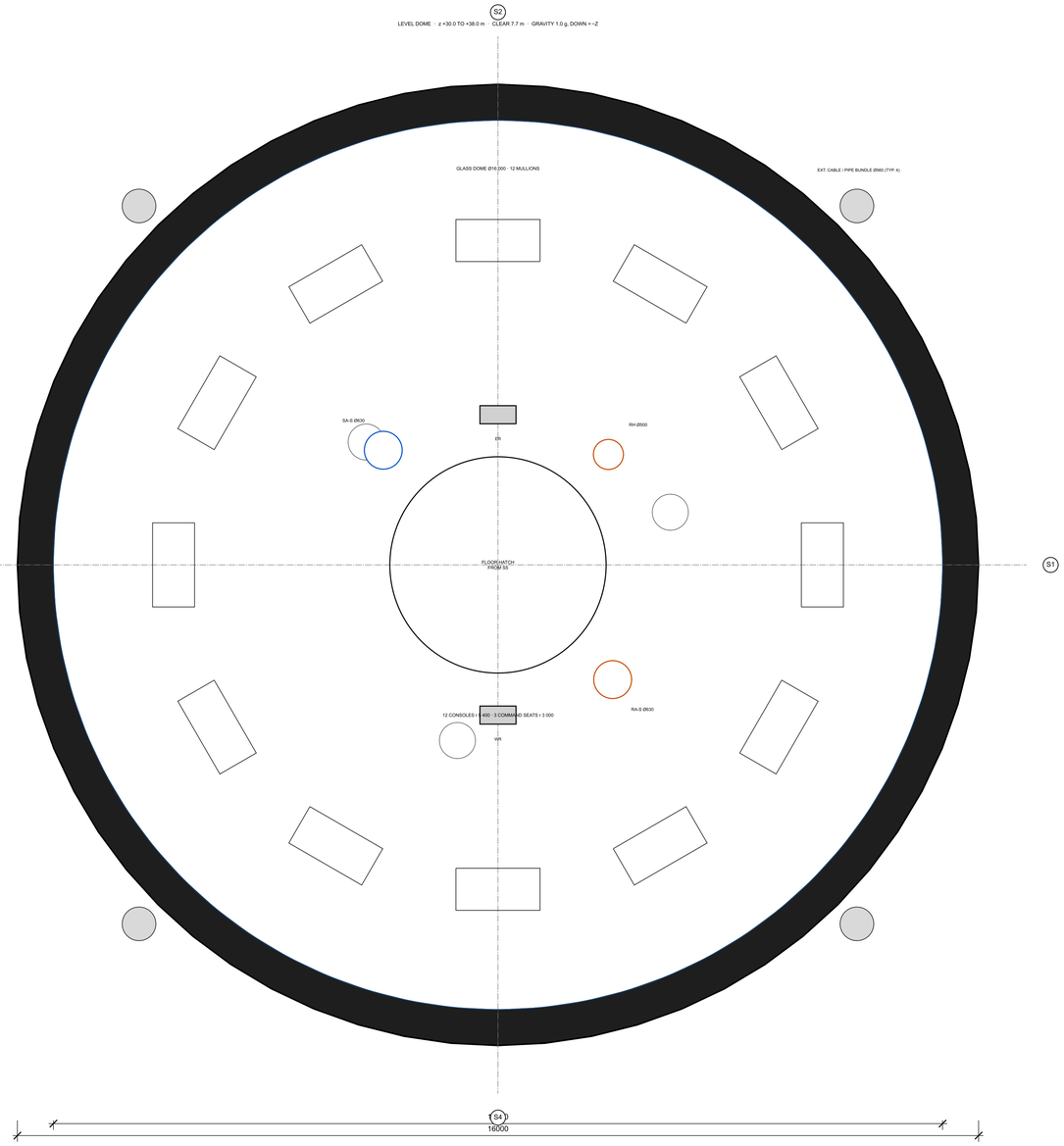
# SPINE LEVELS

Level S4a · Level S4b · plans viewed from +Z

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| **CREATED BY** | **APPROVED BY** | **DATE OF ISSUE** | **PROJECTION** |
| Claude (AI) | — | 2026-10-17 | |
| **IDENTIFICATION NUMBER** | | **REV** | **SHEET** |
| **SF-OUT-A-404** | | **A** | **A-404** |




1 **LEVEL S5 · EVA · SOLAR ARRAY DRIVES**
SCALE 1:50



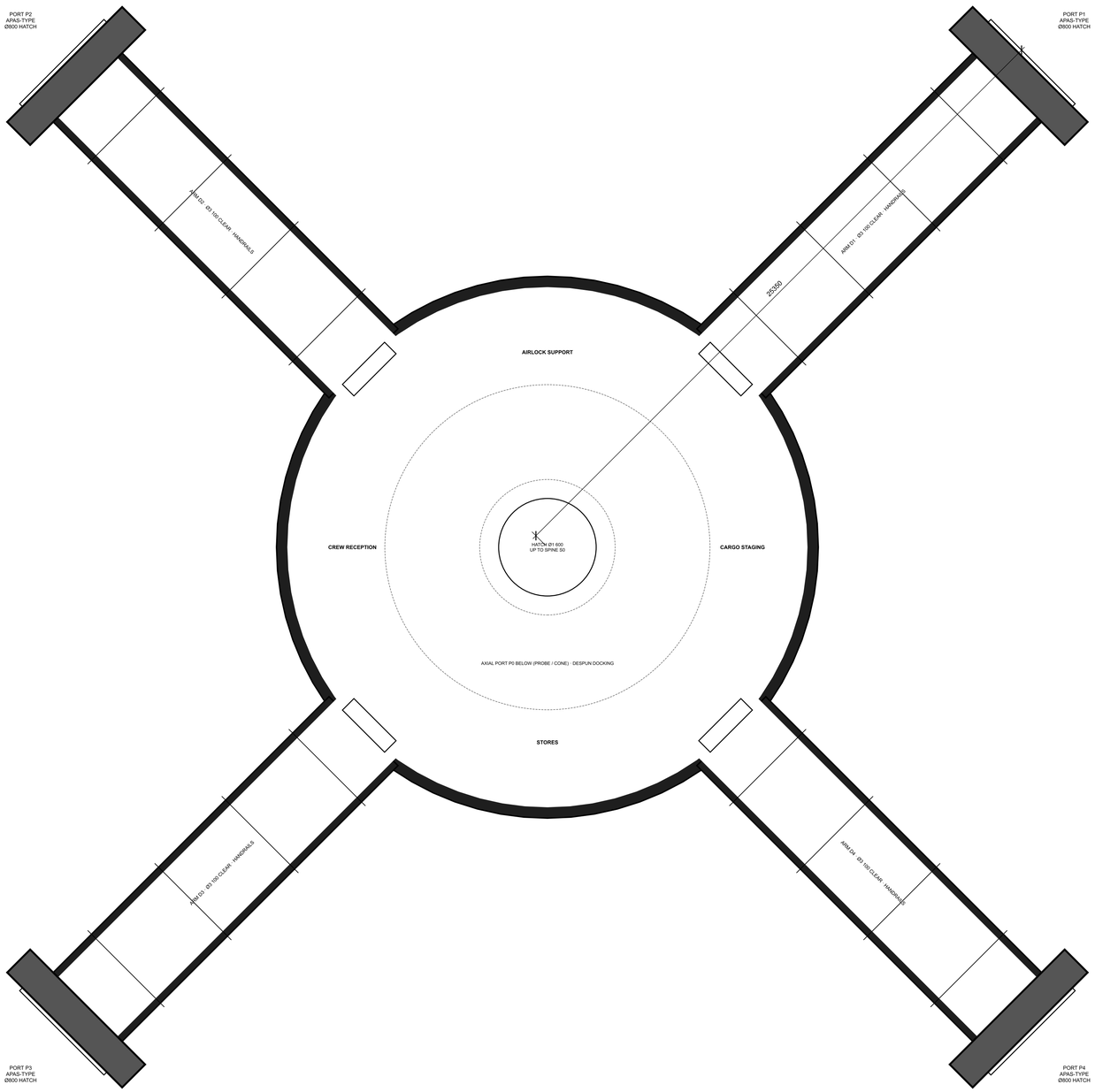

2 **LEVEL DOME · COMMAND DECK**
SCALE 1:50

1. GRAVITY 1.0 g DOWN = –Z, AS IN THE RING: EVERY SPINE LEVEL IS AN ORDINARY FLOOR. S3a (z −200) IS LEVEL WITH THE SPOKES AND THE RING DECK A, SO THE HUB IS ONE CONTINUOUS FLOOR WITH ALL FOUR SPOKES.
2. CENTRAL TRANSFER TRUNK Ø3 600 THROUGH EVERY DECK; THE TRANSFER CAR RUNS ON FOUR RAILS AT r 1 300.
3. RISERS AT EVERY LEVEL: SA-S / RA-S Ø630 (SPINE AHU), OA / RH Ø500 (HUB ↔ ECLSS), ELECTRICAL (ER), WATER (WR).
4. 16 PORTHOLES Ø1 400 PER LEVEL AT MID-HEIGHT (NOT ON S3a, WHICH HAS THE SPOKE PORTALS).

LEGAL OWNER / PROJECT
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE
**SPINE LEVELS**
Level S5 · Level DOME · plans viewed from +Z

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-405 | | A | A-405 |




## NOTES

1. THE HUB TURNS WITH THE STATION; PORTS P1–P4 AT r 25 350 SEE 0.45 g. VISITING CRAFT MATCH ROTATION BEFORE CONTACT. P0 (AXIAL) IS THE PRIMARY PORT FOR LARGE OR UNPOWERED VEHICLES.
2. EACH ARM: Ø3 600 TUBE, Ø3 100 CLEAR. PRESSURE HATCH AT BOTH ENDS. RING FRAMES @ 3 500.
3. HUB VESTIBULE Ø19 200 CLEAR × 5 200: CARGO STAGING, CREW RECEPTION, AIRLOCK SUPPORT, STORES.
4. VENTILATED FROM THE SPINE AHU (SA-S / RA-S EXTENDED DOWN THROUGH S0).

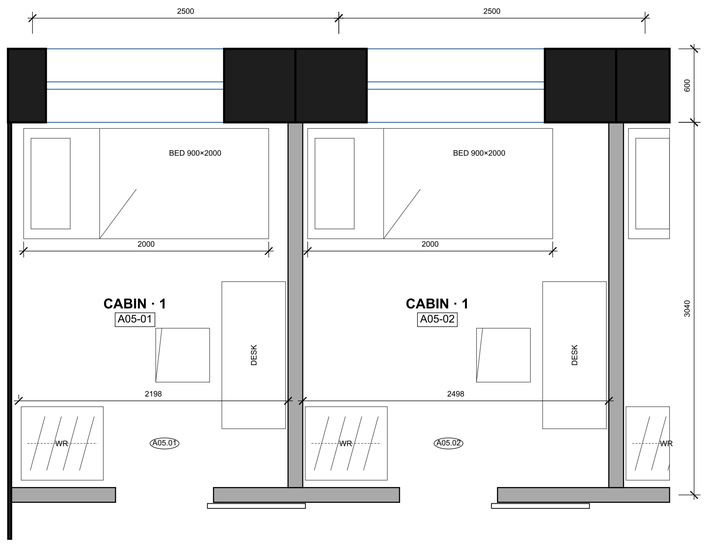
**1** **DOCKING HUB PLAN**
SCALE 1:100

20000

LEGAL OWNER / PROJECT
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

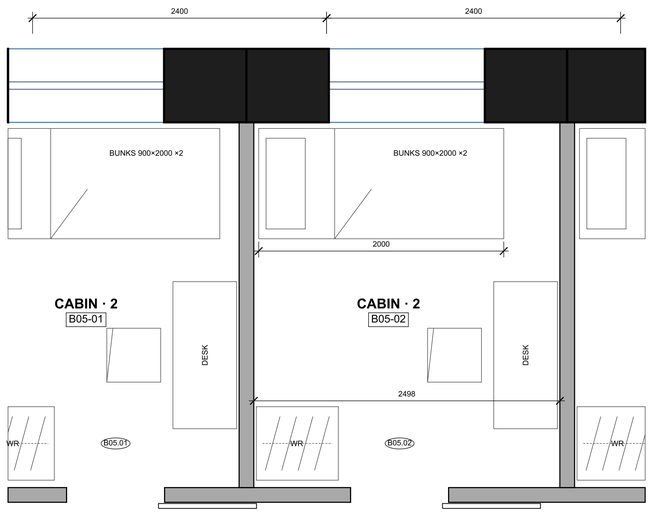
TITLE
**DOCKING HUB**
Plan at z −29 000 (arm axis)

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| **SF-OUT-A-406** | | **A** | **A-406** |



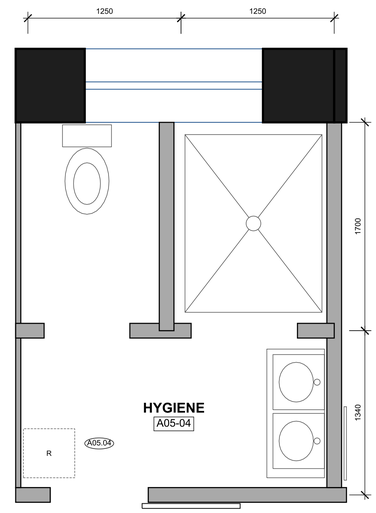

**CORRIDOR**
A05-00

## 1 SINGLE CABINS · DECK A (C05)
SCALE 1:25

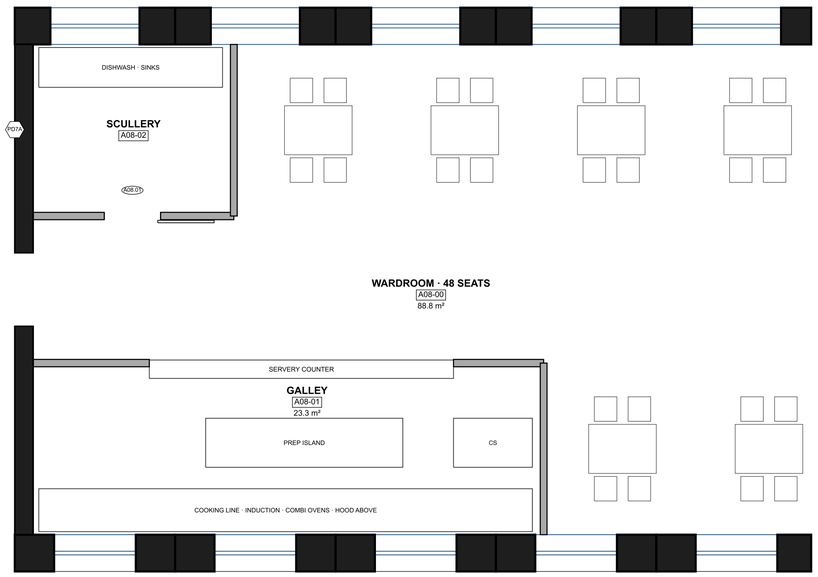



**CORRIDOR**
B05-00

## 2 DOUBLE CABINS · DECK B (C05)
SCALE 1:25




## 3 HYGIENE MODULE · DECK A
SCALE 1:25




**WARDROOM · 48 SEATS**
A08-00
88.8 m²

## 4 GALLEY AND WARDROOM · DECK A (C08)
SCALE 1:50

### FIT-OUT NOTES

SINGLE CABIN 2 500 × 2 980 (7.4 m²): BED 900×2 000 UNDER THE WINDOW; WARDROBE 670×600, FOLD-DOWN DESK 520×1 200, SLIDING DOOR 800, CEILING DIFFUSER, DOOR UNDERCUT 25.
DOUBLE CABIN 2 400 × 2 980 (7.2 m²): BUNKS 900×2 000 × 2 HIGH (1 000 BETWEEN), NO WINDOW.
HYGIENE MODULE 2 500 × 2 980: WC 1 250×1 700, SHOWER 1 250×1 700 (MIST, RECIRCULATING), BASIN LOBBY; EXHAUST 25 L/s EACH WC AND SHOWER; FLOOR DRAINS TO GREYWATER.
GALLEY: INDUCTION LINE + COMBI OVENS UNDER A FILTERED HOOD (EA 0.25 m³/s), PREP ISLAND, SERVERY COUNTER TO THE WARDROOM; SCULLERY OPPOSITE; COLD STORE ON DECK B (C08) ABOVE.
LOOSE FURNITURE IS SECURED (GRAVITY-LOSS EMERGENCIES); WORKTOP HEIGHT 900.

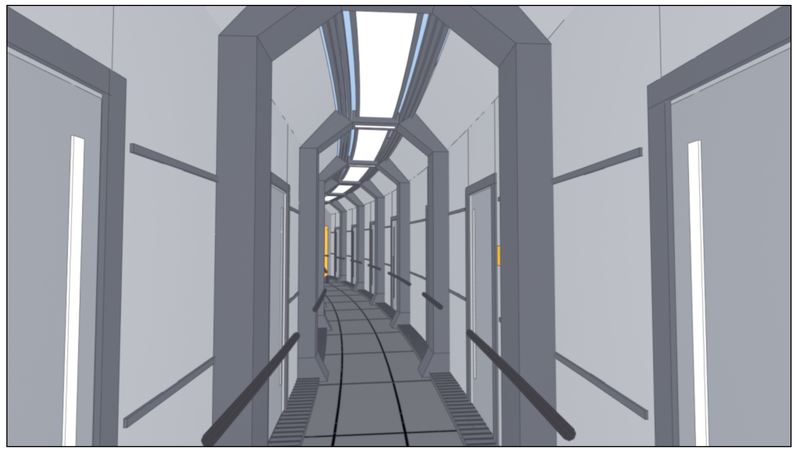
### DOOR TYPES (TAGS: COMPARTMENT.NUMBER, e.g. A05.03)

| TYPE | LEAF | DESCRIPTION | HEIGHT | No. |
|---|---|---|---|---|
| D1 | HINGED 800 | HINGED, SOLID CORE, CLOSER, VISION PANEL WHERE PUBLIC | 2 100 | 16 |
| D2 | SLIDING 1000 | SLIDING POCKET, Al HONEYCOMB, 25 UNDERCUT, PRIVACY LATCH | 2 100 | 23 |
| D3 | SLIDING 1200 | SLIDING POCKET, Al HONEYCOMB, 25 UNDERCUT, PRIVACY LATCH | 2 100 | 12 |
| D4 | SLIDING 800 | SLIDING POCKET, Al HONEYCOMB, 25 UNDERCUT, PRIVACY LATCH | 2 100 | 76 |
| PD | PRESSURE 1200 / 1000 | SLIDING PRESSURE DOOR IN BULKHEAD, FAIL-CLOSED, 2 kPa | 2 100 / 2 000 | 24 |

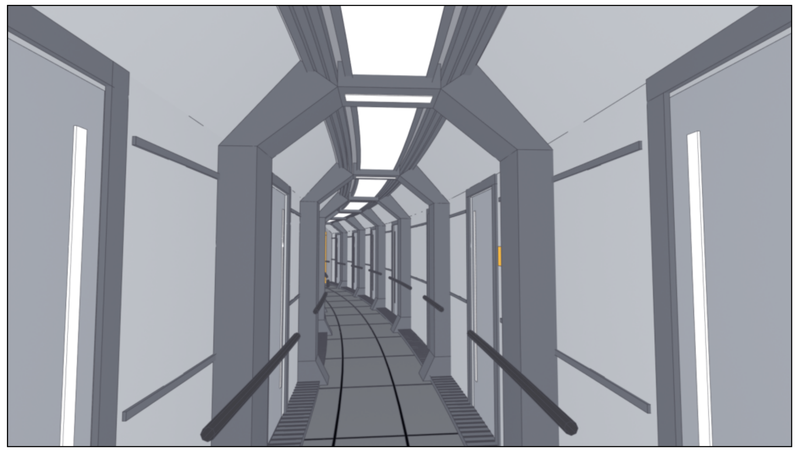
LEGAL OWNER / PROJECT
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE
# ENLARGED PLANS
Single cabin · double cabin · hygiene module · galley

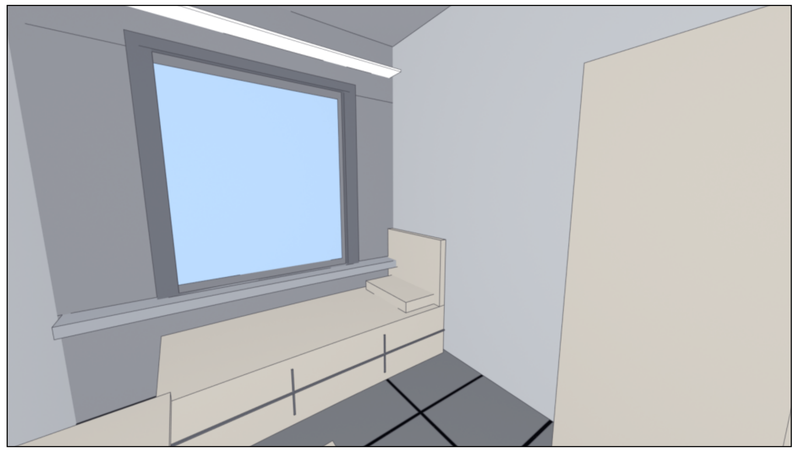
| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | AS SHOWN | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-501 | | A | A-501 |



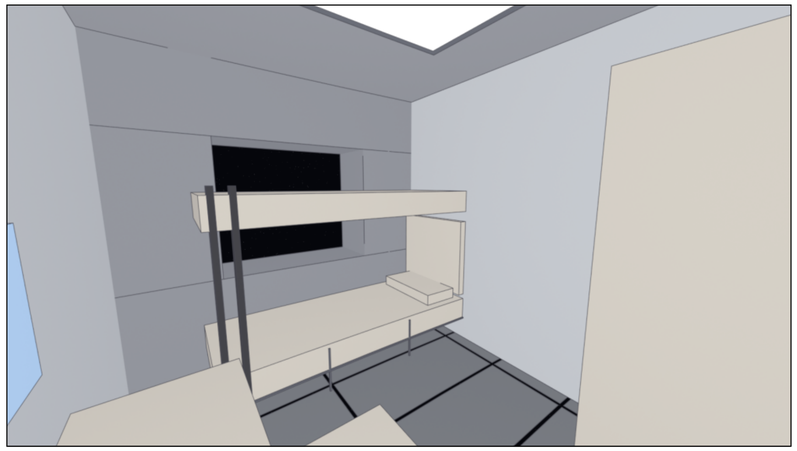
1 DECK A CORRIDOR · C05 · SKYLIGHT OVERHEAD (A-215)



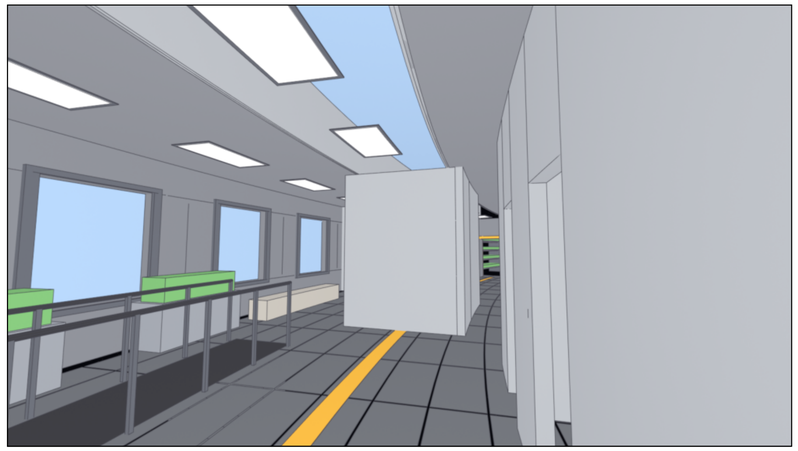
2 DECK B CORRIDOR · C05 · 2 600 CLEAR (A-215)



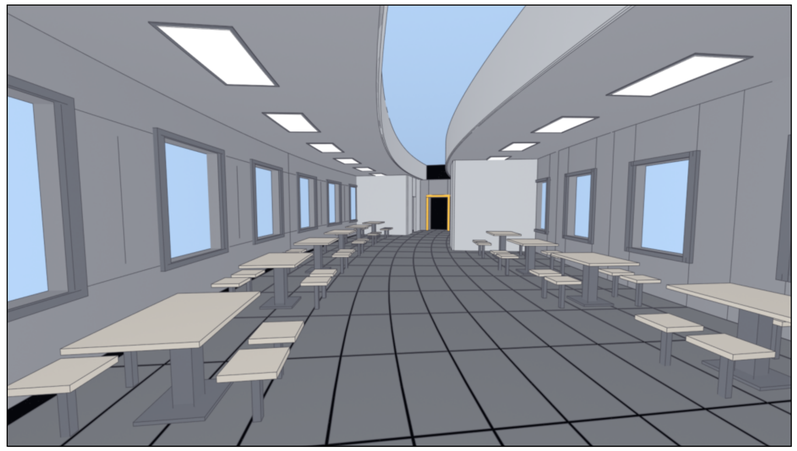
3 SINGLE CABIN · C05 · BED UNDER ITS OWN WINDOW W1 (A-215)



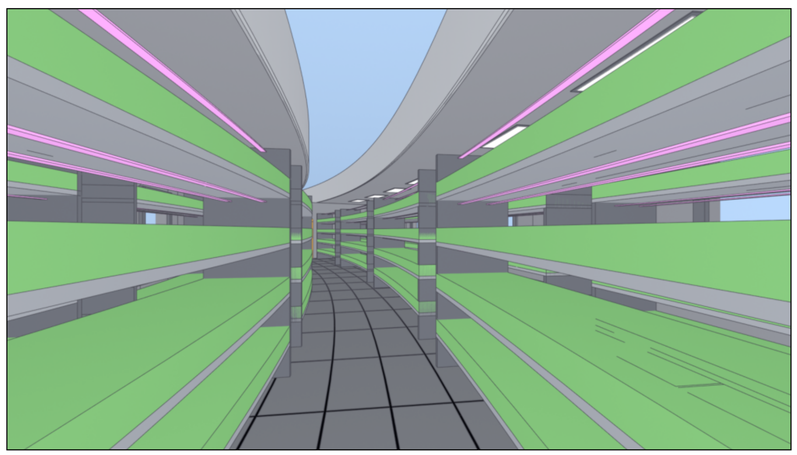
4 DOUBLE CABIN B05 · BUNKS, OWN WINDOW W2 (A-215)



5 SPOKE 2 CONCOURSE · C04 · STAIR DOWN TO DECK B, LIFT (A-214)



6 WARDROOM · C08 · 48 SEATS, WINDOWS BOTH SIDES (A-218)



7 HYDROPONICS BAY 1 · C02 · 4-TIER RACKS, GROW LIGHTS (A-212)

## DESIGN LANGUAGE

1. PORTAL FRAMES ON EVERY STRUCTURAL FRAME (3.75°, 2 840 AT DECK A): POSTS 300×200, CHAMFERED HEAD AND FOOT.
2. OCTAGONAL CORRIDOR: 420 SOFFIT CHAMFERS CARRY THE CABLE TRAYS; DECK A HAS THE ROOF SKYLIGHT DOWN ITS MIDDLE.
3. LINEAR FLOOR GRILLES ALONG BOTH WALLS = THE SUBFLOOR RETURN-AIR PLENUM (M-20x).
4. CONTINUOUS HANDRAILS AT +950 (LOW g): BROKEN AT DOORS, BRACKETED OFF THE FRAMES.
5. AMBER = SAFETY: PRESSURE-DOOR SURROUNDS, LIFT LANDINGS, ROOM-NUMBER PLATES, WAYFINDING STRIPE.
6. PALETTE: DARK GUNMETAL FRAMES, MID-GREY PANELS, WARM UPHOLSTERY, COOL WHITE LIGHT; PLANTS AT WINDOWS.
7. REV B: FLAT FLOORS, GRAVITY DOWN = −Z; THE CORRIDOR BENDS AWAY SIDEWAYS WITH THE RING (r 40 000).

## REV B · ONE BUILD WITH THE EXTERIOR

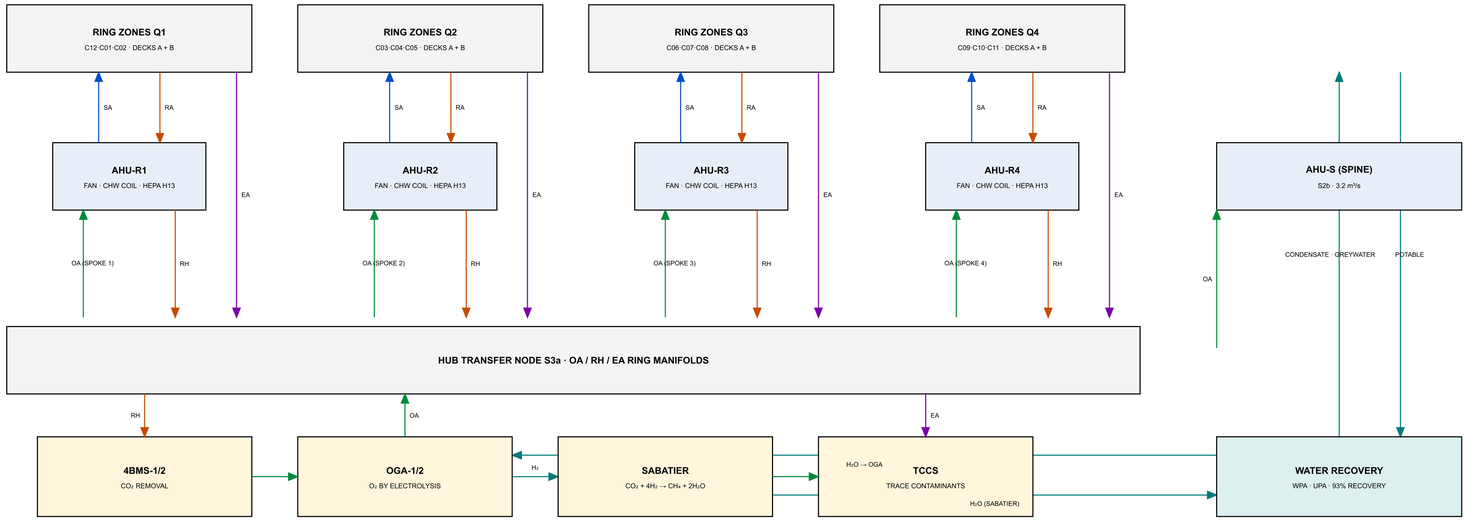
A. EVERY WINDOW IN THESE VIEWS IS A WINDOW ON THE SHELL: build.py READS docs/blueprints/openings.json (172 OUTER, 88 HUB, 96 SKYLIGHTS).
B. ONE CABIN PER FRAME BAY, EACH WITH ITS OWN WINDOW; DECK B CABINS ON THE OUTER WALL SEE OUT TOO (W2).
C. THE SPOKE FLOOR IS DECK A AND THE SPINE HUB DECK (z −200).
D. OPEN: GLAZE THE CONCOURSE-SIDE PARTITIONS OF THE JUNCTION ROOMS?

LEGAL OWNER / PROJECT
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE
# INTERIOR CONCEPT SKETCHES
Rev B: built in Blender from the plans · flat decks, curved walls · design language

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Architecture | ISSUED FOR MODELLING | NTS | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-A-601 | | A | A-601 |

## AIR AND WATER FLOW SCHEMATIC · NOT TO SCALE




### SYSTEM DESCRIPTION

1. RECIRCULATION: EACH RING AHU MOVES ITS ZONES' AIR OVER A CHILLED-WATER COIL (COOL, DEHUMIDIFY) AND HEPA H13 FILTERS.
2. REVITALISATION: A FIXED FLOW (12 L/s PER PERSON SERVED) LEAVES EACH AHU UP ITS SPOKE (RH) TO THE HUB, IS STRIPPED OF $CO_2$ (4BMS) AND TRACE CONTAMINANTS (TCCS), TOPPED UP WITH $O_2$ (OGA), AND COMES BACK DOWN THE SPOKE AS OA.
3. $O_2$: 0.84 kg/PERSON/DAY × 120 = 101 kg/DAY, FROM = 113 kg/DAY OF WATER ELECTROLYSED (OGA-1 + OGA-2 STANDBY).
4. $CO_2$: 1.0 kg/PERSON/DAY × 120 = 120 kg/DAY; THE SABATIER RECOVERS = 44 kg/DAY OF WATER; METHANE IS VENTED.
5. WATER: CONDENSATE, GREYWATER AND URINE TO WATER RECOVERY (93 %); POTABLE BUFFER IN S2a AND TANKS PWT-1/2 (C03 DECK B).
6. HEAT: AHU COILS AND COLD PLATES ON CHWS/CHWR LOOPS TO THE ATCS PUMPS (S2a) AND THE RADIATOR FINS (±Y, z −12 000).
7. EXHAUST (EA) FROM WCs, SHOWERS, GALLEY, LAB FUME HOODS AND THE WELD BOOTH GOES ONLY TO TCCS.
8. EMERGENCY: ON SMOKE OR DEPRESSURISATION THE ZONE'S FSDs CLOSE, ITS AHU STOPS, AND THE SAFE-HAVEN SCRUBBER STARTS.

LEGAL OWNER / PROJECT
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE
# ECLSS AND AIR DISTRIBUTION SCHEMATIC
Ring AHUs · spokes · hub manifolds · revitalisation · water recovery

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Mechanical (HVAC) | ISSUED FOR MODELLING | NTS | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-M-001 | | A | M-001 |

## ZONE AIRFLOW (6 AIR CHANGES / HOUR)

| ZONE | DECK A m³ | SA A m³/s | DECK B m³ | SA B m³/s | TOTAL m³/s |
|---|---|---|---|---|---|
| C01 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| C02 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| C03 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| C04 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| C05 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| C06 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| C07 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| C08 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| C09 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| C10 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| C11 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| C12 | 521 | 0.868 | 423 | 0.705 | 1.573 |
| TOTAL | 6250 | 10.42 | 5078 | 8.46 | 18.88 |

## AHU SCHEDULE

| TAG | LOCATION | SERVES | SA m³/s | OA m³/s | ESP Pa | FAN kW | COIL kW | FILTER |
|---|---|---|---|---|---|---|---|---|
| AHU-R1 | C01 DECK B | C12 C01 C02 | 4.72 | 0.32 | 750 | 5.9 | 57 | H13 |
| AHU-R2 | C04 DECK B | C03 C04 C05 | 4.72 | 0.32 | 750 | 5.9 | 57 | H13 |
| AHU-R3 | C07 DECK B | C06 C07 C08 | 4.72 | 0.32 | 750 | 5.9 | 57 | H13 |
| AHU-R4 | C10 DECK B | C09 C10 C11 | 4.72 | 0.32 | 750 | 5.9 | 57 | H13 |
| AHU-S | SPINE S2b | SPINE · DOME · HUB | 3.20 | — | 600 | 3.2 | 31 | H13 |

## DAMPER AND TERMINAL SCHEDULE

| TYPE | DESCRIPTION | QTY |
|---|---|---|
| FSD | FIRE/SMOKE/PRESSURE ISOLATION DAMPER AT A BULKHEAD · MOTORISED · FAIL-CLOSED · 2 kPa SEAL | 60 |
| CD-300 | CEILING SQUARE DIFFUSER 300×300, 4-WAY, OPPOSED-BLADE DAMPER | ≈ 290 |
| RG | RETURN GRILLE 500×150 (DECK A FLOOR) · 400×400 (DECK B CEILING) | ≈ 260 |
| EG-200 | EXHAUST GRILLE 200×200 WITH ODOUR FILTER (WC, SHOWER) | ≈ 60 |
| SH-1..4 | SPOKE ISOLATION HATCH Ø1 000 AT BOTH ENDS OF EACH SPOKE | 8 |

## ECLSS SIZING (120 PERSONS)

| QUANTITY | PER PERSON / DAY | STATION / DAY |
|---|---|---|
| $O_2$ CONSUMED | 0.84 kg | 101 kg |
| $CO_2$ PRODUCED | 1.00 kg | 120 kg |
| DRINKING + FOOD WATER | 2.5 kg | 300 kg |
| HYGIENE WATER | 7.0 kg | 840 kg |
| METABOLIC HEAT | 11.8 MJ | 1 416 MJ (16.4 kW) |
| REVITALISED AIR EXCHANGE | 12 L/s | 1.44 m³/s |
| WATER RECOVERY 93 % · MAKE-UP | — | ≈ 74 kg |

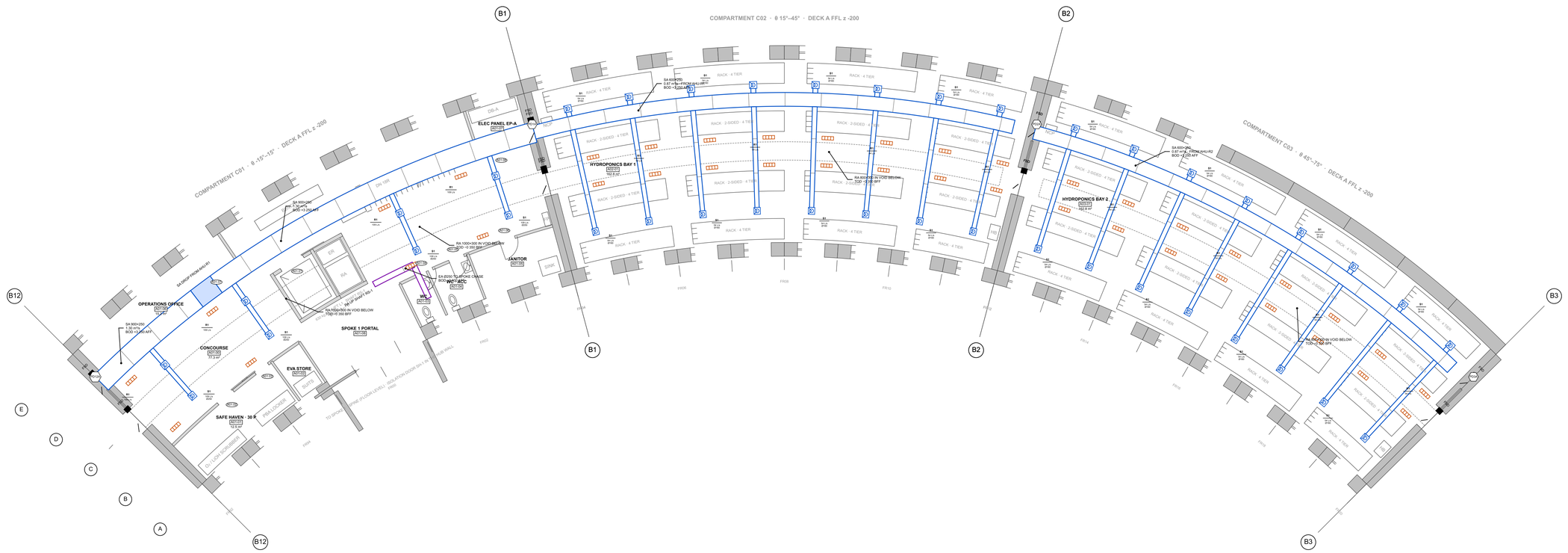
## CALCULATION NOTES

ZONE VOLUME = (DEVELOPED LENGTH − 600 BULKHEAD) × 8 000 × CLEAR HEIGHT (3 000 DECK A · 2 200 DECK B).
SA = VOLUME × 6 / 3 600. FAN kW = Q × ESP / $\eta_t$ = 0.60. COIL kW = SENSIBLE AT ΔT 10 K (ß K SPINE).
OA PER AHU = 12 L/s × 27 PERSONS (108 DESIGN + 10 % SPREAD OVER 4 AHUs).
DUCTS SIZED FOR ≤ 6 m/s IN MAINS. METABOLIC, EQUIPMENT AND LIGHTING HEAT GO TO THE RADIATORS THROUGH THE CHILLED-WATER LOOPS.

LEGAL OWNER / PROJECT
**STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT**

TITLE
**SCHEDULES AND CALCULATIONS**
AHU · damper · terminal schedules; airflow and ECLSS sizing

| DISCIPLINE | DOCUMENT STATUS | SCALE | SIZE / UNITS |
|---|---|---|---|
| Mechanical (HVAC) | ISSUED FOR MODELLING | NTS | A1 · mm |
| CREATED BY | APPROVED BY | DATE OF ISSUE | PROJECTION |
| Claude (AI) | — | 2026-10-17 | |
| IDENTIFICATION NUMBER | | REV | SHEET |
| SF-OUT-M-002 | | A | M-002 |

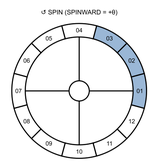



## 1 DECK A VENTILATION · SA IN THE 1 000 ROOF VOID · RA IN THE INTERSTITIAL VOID BELOW

SCALE 1:100

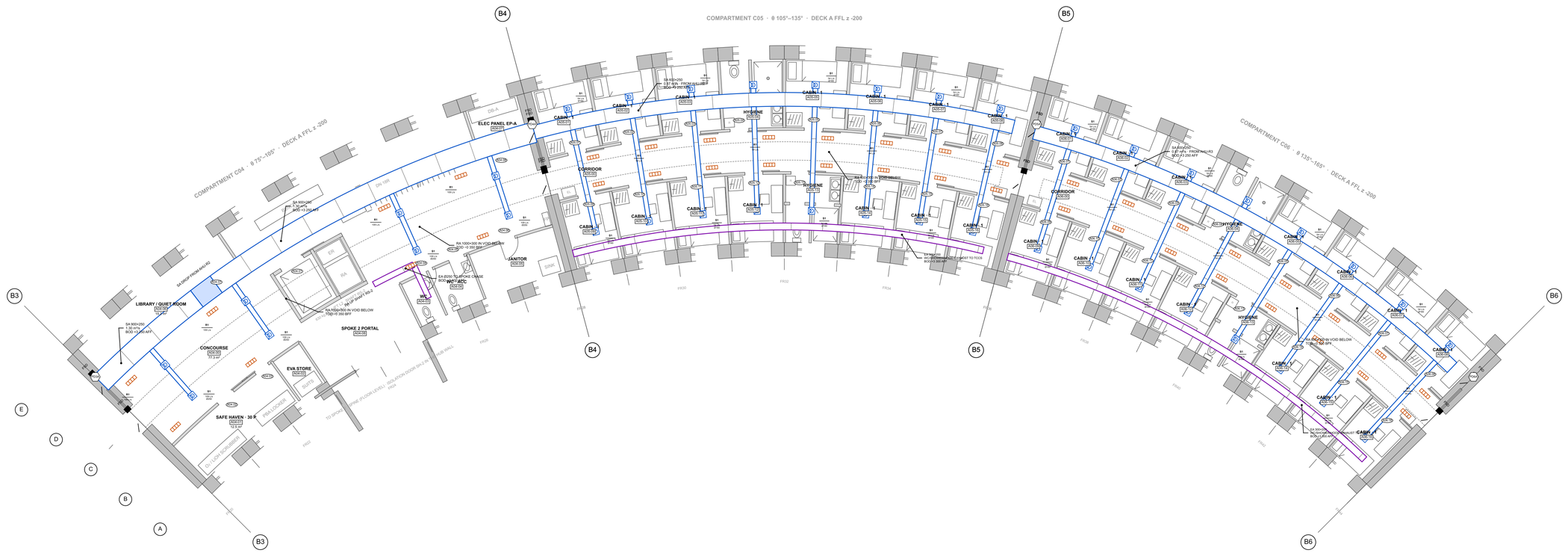
### VENTILATION NOTES

1. EACH SPOKE-JUNCTION AHU (AHU-R1..R4, IN THE DECK B PLANT ROOM) SERVES ITS OWN COMPARTMENT AND THE ONE EITHER SIDE, BOTH DECKS.
2. DECK A: SUPPLY IN THE 1 000 ROOF VOID (BESIDE THE SKYLIGHT WELLS) TO CEILING DIFFUSERS; RETURN THROUGH FLOOR GRILLES INTO DUCTS IN THE 400 INTERSTITIAL VOID BELOW, BACK UNDER THE CORRIDOR TO SHAFT RS-*n* AND DOWN TO THE AHU.
3. DECK B: SUPPLY AND RETURN BOTH IN THE 400 INTERSTITIAL VOID OVERHEAD, TO CEILING DIFFUSERS AND CEILING RETURN GRILLES.
4. EVERY DUCT CROSSING A BULKHEAD HAS A FIRE/SMOKE/PRESSURE DAMPER (FSD), FAIL-CLOSED.
5. WC / SHOWER / GALLEY / LAB / WELD EXHAUST (EA) IS DUCTED SEPARATELY TO THE SPOKE CHASE AND THE HUB TCCS — NEVER RECIRCULATED.
6. REVITALISED AIR (OA) COMES ALONG THE SPOKE FROM THE HUB ECLSS; AN EQUAL FLOW (RH) GOES BACK ALONG IT FOR $CO_2$ REMOVAL.
7. SIZES WIDTH × DEPTH IN mm; FLOWS FROM M-002 (6 ACH). VELOCITY ≤ 6 m/s IN MAINS, ≤ 3 m/s IN BRANCHES.




KEY PLAN · VIEWED FROM +Z (DOME)

| | | | |
|---|---|---|---|
| LEGAL OWNER / PROJECT: STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT | | | |
| TITLE: RING VENTILATION · DECK A · QUADRANT 1<br>C01–C03 · ductwork | | | |
| DISCIPLINE: Mechanical (HVAC) | DOCUMENT STATUS: ISSUED FOR MODELLING | SCALE: AS SHOWN | SIZE / UNITS: A1 · mm |
| CREATED BY: Claude (AI) | APPROVED BY: — | DATE OF ISSUE: 2026-10-17 | PROJECTION |
| IDENTIFICATION NUMBER: SF-OUT-M-201 | | REV: A | SHEET: M-201 |

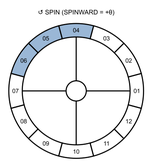



## 1 DECK A VENTILATION · SA IN THE 1 000 ROOF VOID · RA IN THE INTERSTITIAL VOID BELOW

SCALE 1:100

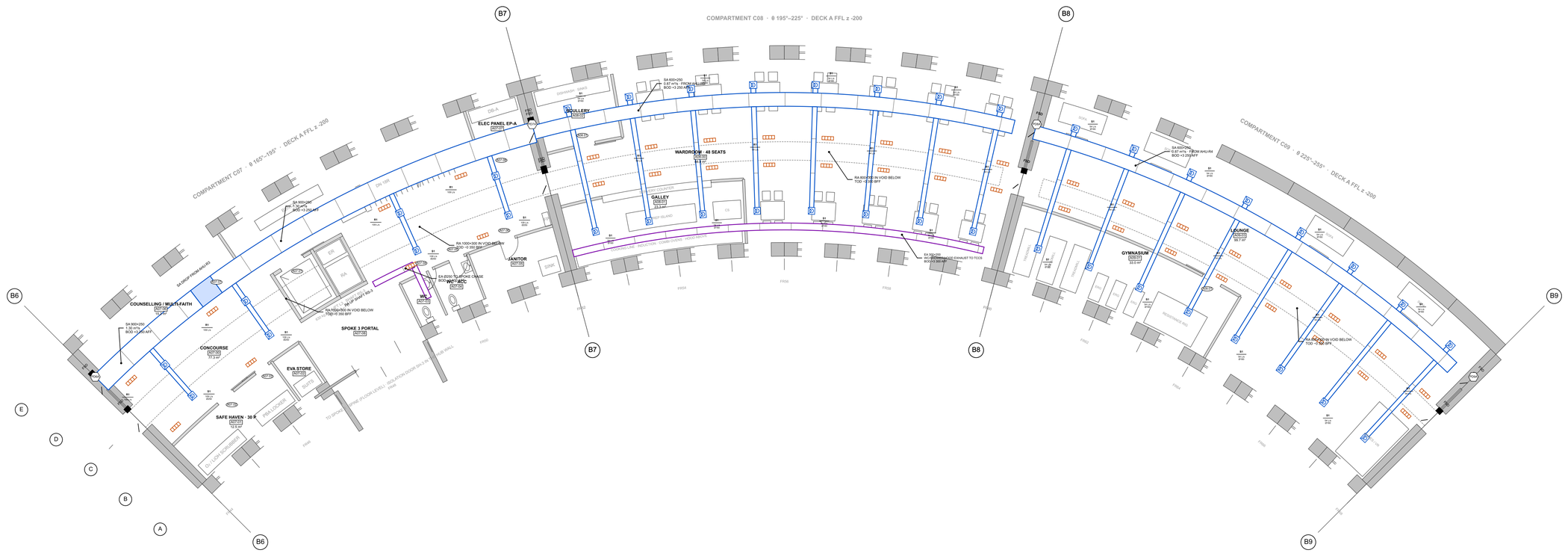
### VENTILATION NOTES

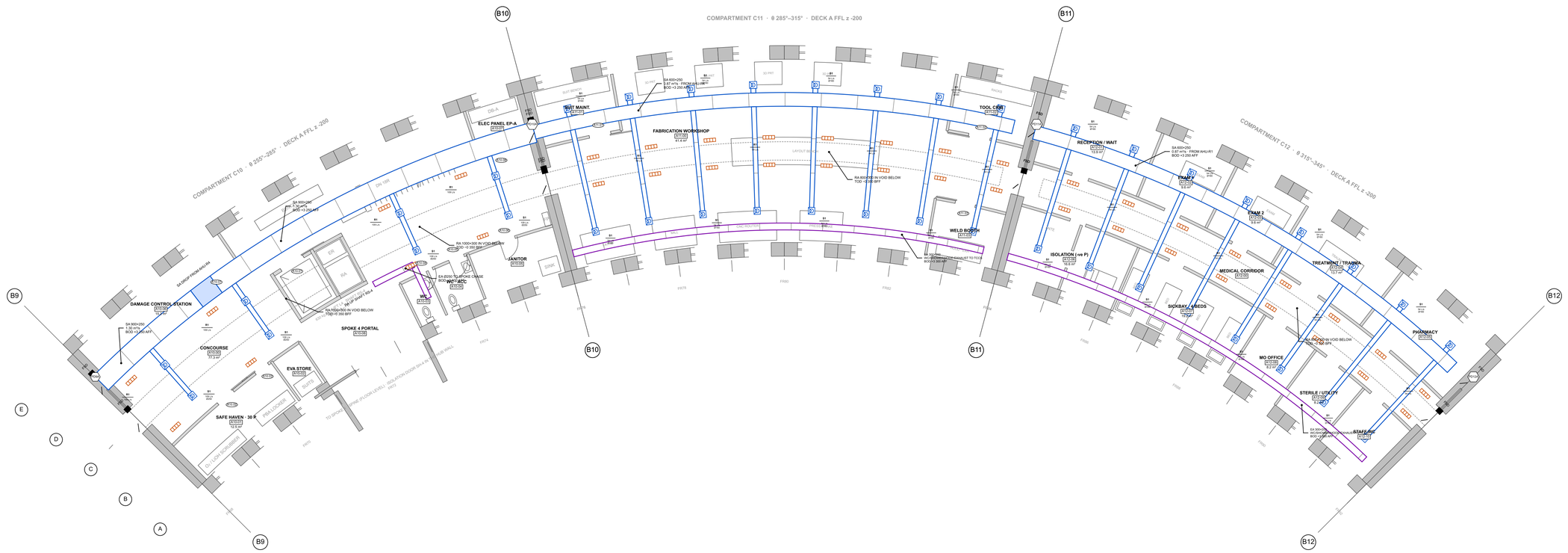
1. EACH SPOKE-JUNCTION AHU (AHU-R1..R4, IN THE DECK B PLANT ROOM) SERVES ITS OWN COMPARTMENT AND THE ONE EITHER SIDE, BOTH DECKS.
2. DECK A: SUPPLY IN THE 1 000 ROOF VOID (BESIDE THE SKYLIGHT WELLS) TO CEILING DIFFUSERS; RETURN THROUGH FLOOR GRILLES INTO DUCTS IN THE 400 INTERSTITIAL VOID BELOW, BACK UNDER THE CORRIDOR TO SHAFT RS-*n* AND DOWN TO THE AHU.
3. DECK B: SUPPLY AND RETURN BOTH IN THE 400 INTERSTITIAL VOID OVERHEAD, TO CEILING DIFFUSERS AND CEILING RETURN GRILLES.
4. EVERY DUCT CROSSING A BULKHEAD HAS A FIRE/SMOKE/PRESSURE DAMPER (FSD), FAIL-CLOSED.
5. WC / SHOWER / GALLEY / LAB / WELD EXHAUST (EA) IS DUCTED SEPARATELY TO THE SPOKE CHASE AND THE HUB TCCS — NEVER RECIRCULATED.
6. REVITALISED AIR (OA) COMES ALONG THE SPOKE FROM THE HUB ECLSS; AN EQUAL FLOW (RH) GOES BACK ALONG IT FOR $CO_2$ REMOVAL.
7. SIZES WIDTH × DEPTH IN mm; FLOWS FROM M-002 (6 ACH). VELOCITY ≤ 6 m/s IN MAINS, ≤ 3 m/s IN BRANCHES.

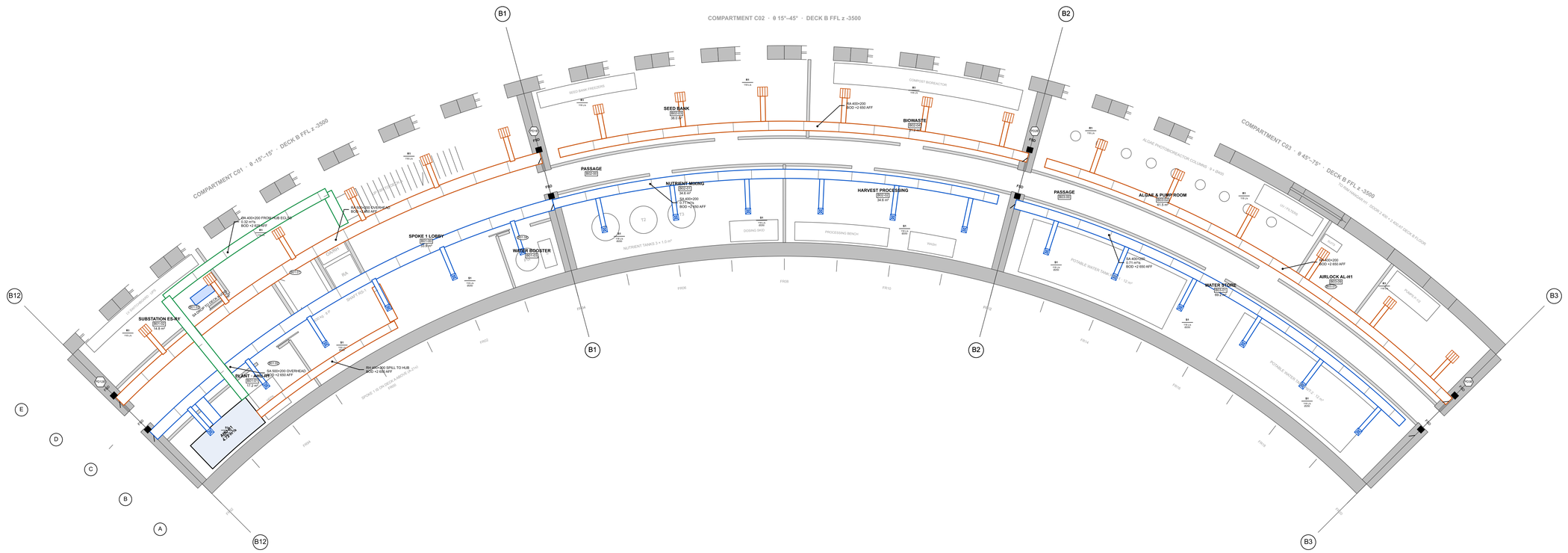



KEY PLAN · VIEWED FROM +Z (DOME)

| | | | |
|---|---|---|---|
| LEGAL OWNER / PROJECT: STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT | | | |
| TITLE: RING VENTILATION · DECK A · QUADRANT 2<br>C04–C06 · ductwork | | | |
| DISCIPLINE: Mechanical (HVAC) | DOCUMENT STATUS: ISSUED FOR MODELLING | SCALE: AS SHOWN | SIZE / UNITS: A1 · mm |
| CREATED BY: Claude (AI) | APPROVED BY: — | DATE OF ISSUE: 2026-10-17 | PROJECTION |
| IDENTIFICATION NUMBER: SF-OUT-M-202 | | REV: A | SHEET: M-202 |


COMPARTMENT C08 · θ 195°–225° · DECK A FFL z -200
COMPARTMENT C07 · θ 165°–195° · DECK A FFL z -200
COMPARTMENT C09 · θ 225°–255° · DECK A FFL z -200
B6
B7
B8
B9
ELEC PANEL EP-A
A207.01
GALLEY
A207.01
WARDROOM - 48 SEATS
A207.02
GYMNASIUM
A207.01
LOUNGE
A207.02
COUNSELLING / MULTI-FAITH
A207.02
CONCOURSE
A207.03
SPOKE 3 PORTAL
A207.03
EVA STORE
A207.03
SAFE HAVEN - 30 P
A207.01
JANITOR
A207.05
WC
A207.05
SA 600×250
0.87 m³/s · FROM AHU-R4
BOD +3.250 AFF
SA 900×250
0.30 m³/s
BOD +3.250 AFF
SA DROP FROM AHU-R3
RA 1000×500 IN VOID BELOW
TOD −0.350 BFF
RA 800×300 IN VOID BELOW
TOD +0.100 BFF
RA 1500×300 IN VOID BELOW
TOD +0.350 BFF
RA 600×300 IN VOID BELOW
TOD −0.100 BFF
EA 600×250
0.87 m³/s · FROM AHU-R4
BOD +3.250 AFF
EA 600×250 TO UPPER CHASE
BOD +3.250 AFF
EA 1500×250
WC / SHOWER / GALLEY / LAB / WELD EXHAUST TO TCCS
BOD +3.350 AFF
SA 600×250
0.87 m³/s · FROM AHU-R4
BOD +3.250 AFF
SHAFT RS-3
SHAFT RS-4
SHAFT RS-5
SHAFT RS-6
SHAFT RS-7
SHAFT RS-8
SHAFT RS-9
SHAFT RS-10
SHAFT RS-11
SHAFT RS-12
SHAFT RS-13
SHAFT RS-14
SHAFT RS-15
SHAFT RS-16
SHAFT RS-17
SHAFT RS-18
SHAFT RS-19
SHAFT RS-20
SHAFT RS-21
SHAFT RS-22
SHAFT


COMPARTMENT C11 · θ 285°–315° · DECK A FFL z -200
COMPARTMENT C10 · θ 255°–285° · DECK A FFL z -200
COMPARTMENT C12 · θ 315°–345° · DECK A FFL z -200
B9
B10
B11
B12
DAMAGE CONTROL STATION
A12120
CONCOURSE
A12030
EVA STORE
A12032
SAFE HAVEN - 30 P
A12031
SPOKE 4 PORTAL
A12033
ELEC PANEL EP-A
A12021
JANITOR
A12036
WC
A12034
FABRICATION WORKSHOP
A11020
WELD BOOTH
A11031
TOOL CRIB
A11030
SUIT MAINT.
A11021
RECEPTION / WAIT
A12022
EXAM 1
A12023
EXAM 2
A12024
MEDICAL CORRIDOR
A12120
ISOLATION (+ve P)
A12027
SICKBAY - 4 BEDS
A12025
MO OFFICE
A12026
STERILE / UTILITY
A12028
TREATMENT / TRAUMA
A12029
PHARMACY
A12030
STAFF WC
A12035
SA 900×250
0.30 m³/s · FROM AHU-R1
BOD +3 250 AFF
SA 600×250
0.87 m³/s · FROM AHU-R1
BOD +3 250 AFF
SA 600×250
0.87 m³/s · FROM AHU-R1
BOD +3 250 AFF
SA ROOF FROM AHU-R4
RA 1000×500 IN VOID BELOW
TOD −0 350 BFF
RA 800×500 IN VOID BELOW
TOD −0 350 BFF
RA 700×300 IN VOID BELOW
TOD −0 350 BFF
RA 600×300 IN VOID BELOW
TOD −0 350 BFF
EA 600×300
BOD +3 350 AFF
EA 600×300
BOD +3 350 AFF
EA 300×200
BOD +3 350 AFF
EA 600×300
BOD +3 350 AFF
EA 600×300
BOD +3 350 AFF
EA 300×200

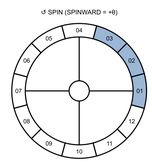



## 1 DECK B VENTILATION · SA AND RA IN THE 400 INTERSTITIAL VOID OVERHEAD

SCALE 1:100

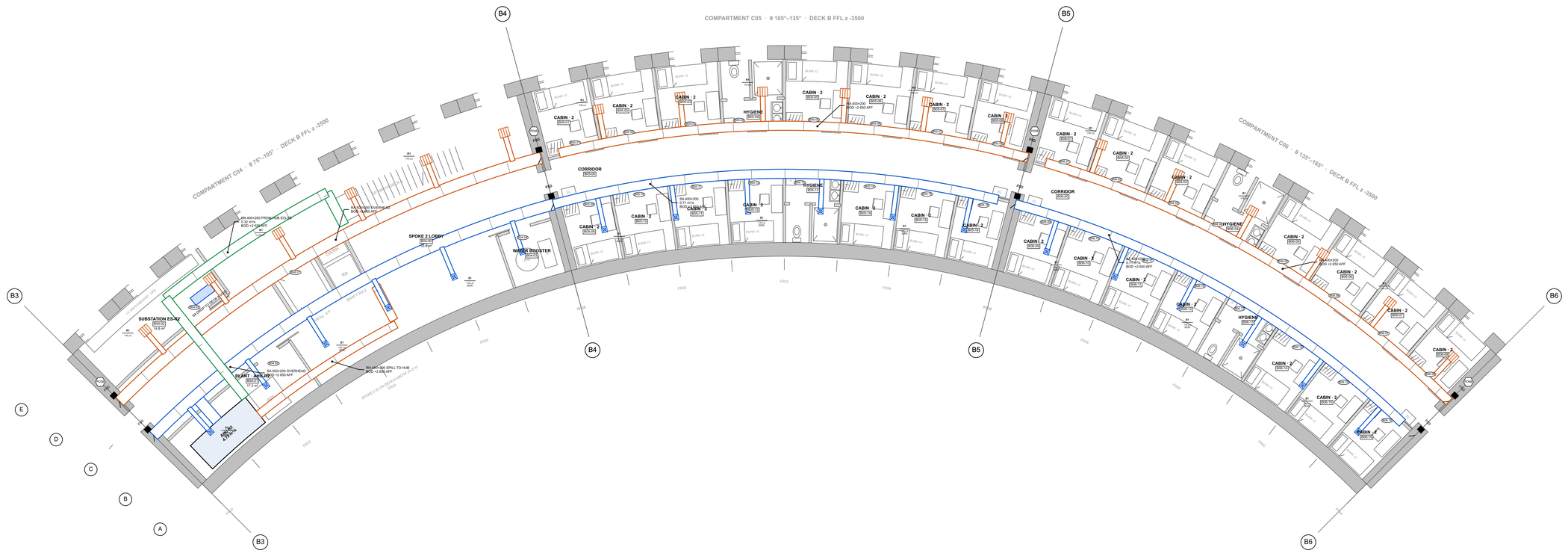
### VENTILATION NOTES

1. EACH SPOKE-JUNCTION AHU (AHU-R1..R4, IN THE DECK B PLANT ROOM) SERVES ITS OWN COMPARTMENT AND THE ONE EITHER SIDE, BOTH DECKS.
2. DECK A: SUPPLY IN THE 1 000 ROOF VOID (BESIDE THE SKYLIGHT WELLS) TO CEILING DIFFUSERS; RETURN THROUGH FLOOR GRILLES INTO DUCTS IN THE 400 INTERSTITIAL VOID BELOW, BACK UNDER THE CORRIDOR TO SHAFT RS-*n* AND DOWN TO THE AHU.
3. DECK B: SUPPLY AND RETURN BOTH IN THE 400 INTERSTITIAL VOID OVERHEAD, TO CEILING DIFFUSERS AND CEILING RETURN GRILLES.
4. EVERY DUCT CROSSING A BULKHEAD HAS A FIRE/SMOKE/PRESSURE DAMPER (FSD), FAIL-CLOSED.
5. WC / SHOWER / GALLEY / LAB / WELD EXHAUST (EA) IS DUCTED SEPARATELY TO THE SPOKE CHASE AND THE HUB TCCS — NEVER RECIRCULATED.
6. REVITALISED AIR (OA) COMES ALONG THE SPOKE FROM THE HUB ECLSS; AN EQUAL FLOW (RH) GOES BACK ALONG IT FOR $CO_2$ REMOVAL.
7. SIZES WIDTH × DEPTH IN mm; FLOWS FROM M-002 (6 ACH). VELOCITY ≤ 6 m/s IN MAINS, ≤ 3 m/s IN BRANCHES.




KEY PLAN · VIEWED FROM +Z (DOME)

| | | | |
|---|---|---|---|
| LEGAL OWNER / PROJECT: STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT | | | |
| TITLE: RING VENTILATION · DECK B · QUADRANT 1<br>C01–C03 · ductwork | | | |
| DISCIPLINE: Mechanical (HVAC) | DOCUMENT STATUS: ISSUED FOR MODELLING | SCALE: AS SHOWN | SIZE / UNITS: A1 · mm |
| CREATED BY: Claude (AI) | APPROVED BY: — | DATE OF ISSUE: 2026-10-17 | PROJECTION |
| IDENTIFICATION NUMBER: SF-OUT-M-205 | | REV: A | SHEET: M-205 |

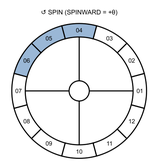



## 1 DECK B VENTILATION · SA AND RA IN THE 400 INTERSTITIAL VOID OVERHEAD

SCALE 1:100

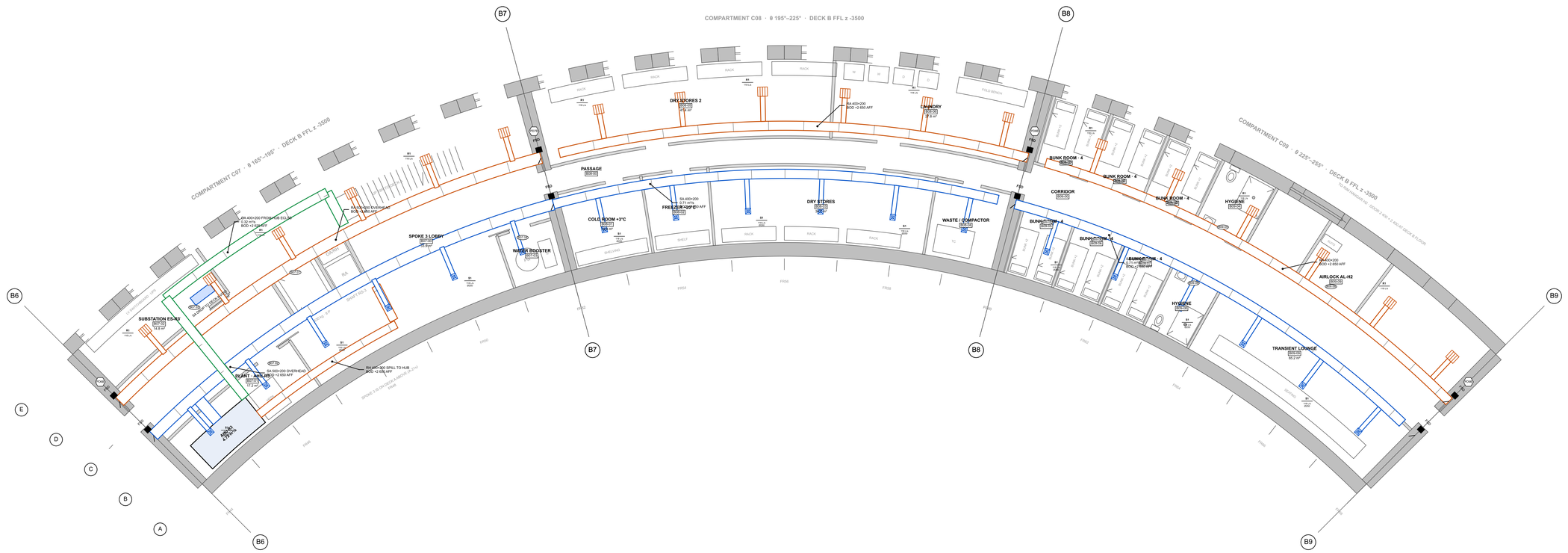
### VENTILATION NOTES

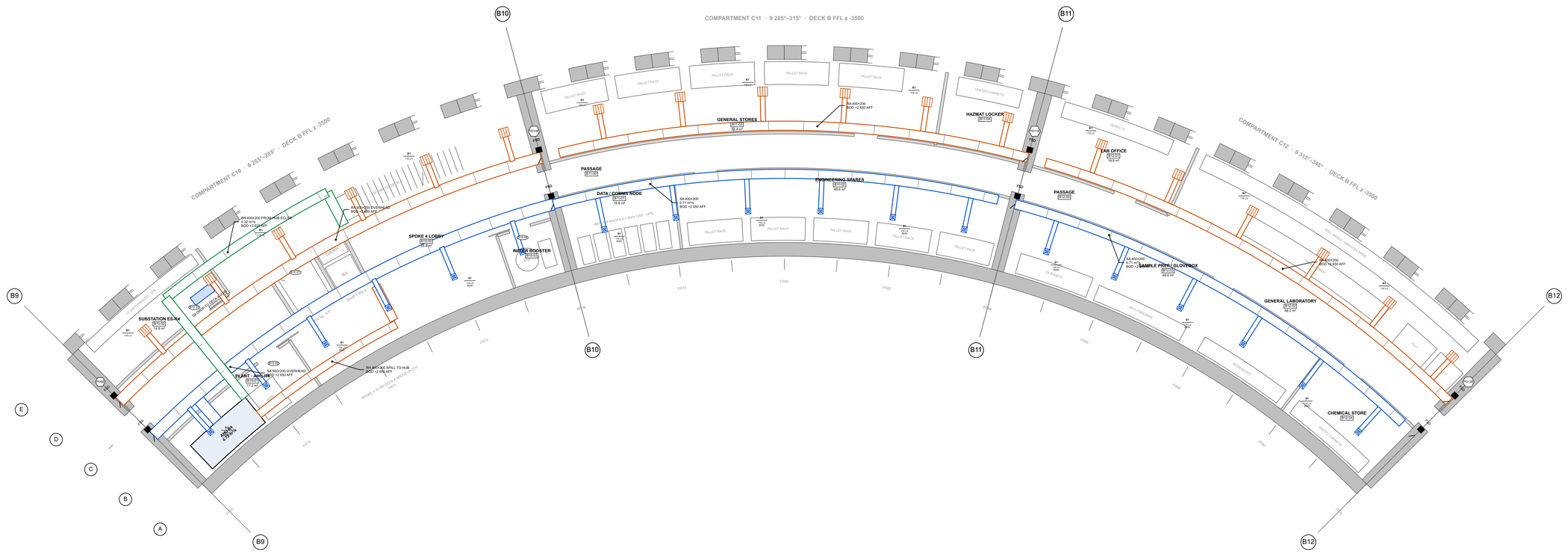
1. EACH SPOKE-JUNCTION AHU (AHU-R1..R4, IN THE DECK B PLANT ROOM) SERVES ITS OWN COMPARTMENT AND THE ONE EITHER SIDE, BOTH DECKS.
2. DECK A: SUPPLY IN THE 1 000 ROOF VOID (BESIDE THE SKYLIGHT WELLS) TO CEILING DIFFUSERS; RETURN THROUGH FLOOR GRILLES INTO DUCTS IN THE 400 INTERSTITIAL VOID BELOW, BACK UNDER THE CORRIDOR TO SHAFT RS-*n* AND DOWN TO THE AHU.
3. DECK B: SUPPLY AND RETURN BOTH IN THE 400 INTERSTITIAL VOID OVERHEAD, TO CEILING DIFFUSERS AND CEILING RETURN GRILLES.
4. EVERY DUCT CROSSING A BULKHEAD HAS A FIRE/SMOKE/PRESSURE DAMPER (FSD), FAIL-CLOSED.
5. WC / SHOWER / GALLEY / LAB / WELD EXHAUST (EA) IS DUCTED SEPARATELY TO THE SPOKE CHASE AND THE HUB TCCS — NEVER RECIRCULATED.
6. REVITALISED AIR (OA) COMES ALONG THE SPOKE FROM THE HUB ECLSS; AN EQUAL FLOW (RH) GOES BACK ALONG IT FOR $CO_2$ REMOVAL.
7. SIZES WIDTH × DEPTH IN mm; FLOWS FROM M-002 (6 ACH). VELOCITY ≤ 6 m/s IN MAINS, ≤ 3 m/s IN BRANCHES.

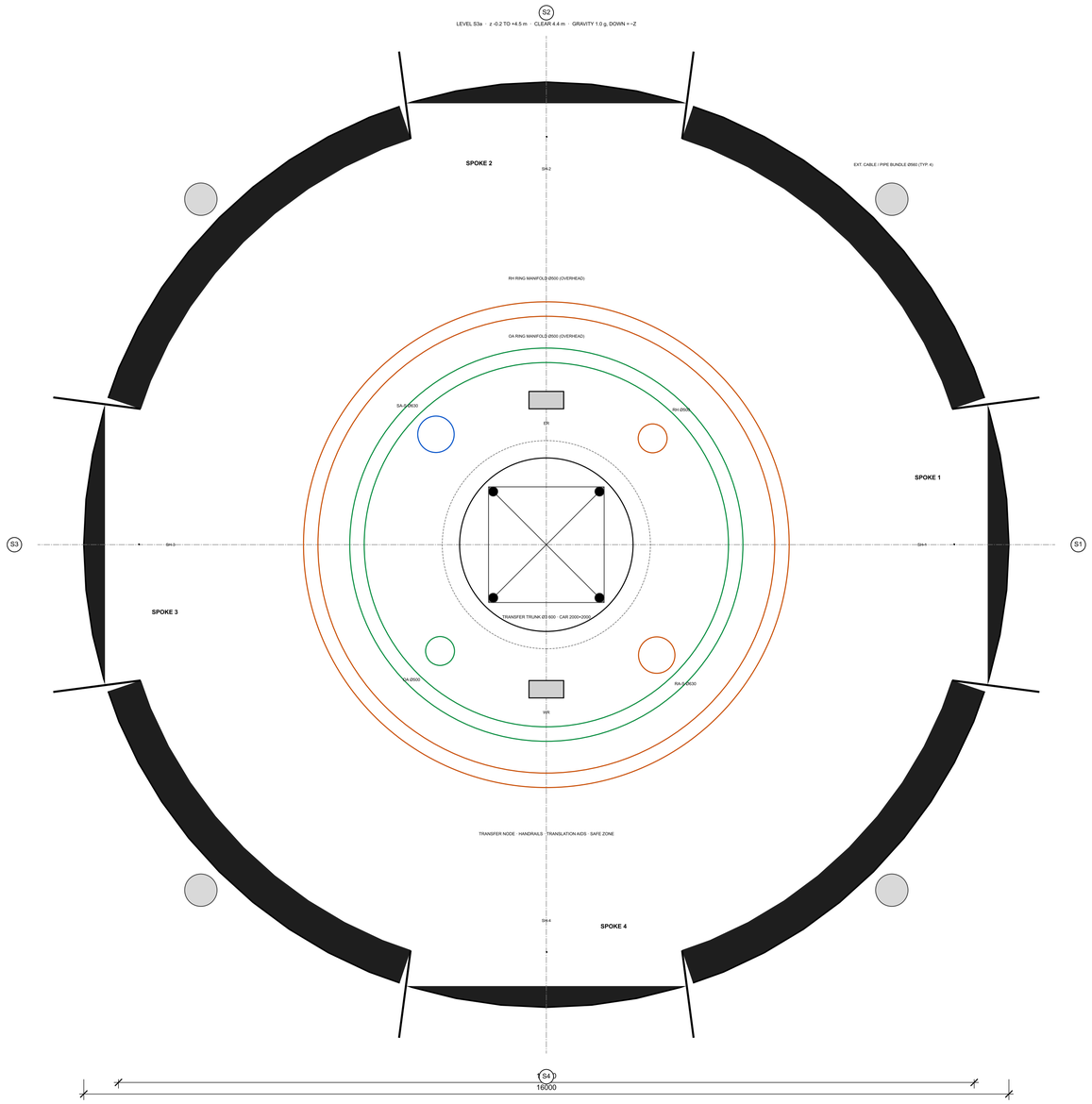



KEY PLAN · VIEWED FROM +Z (DOME)

| | | | |
|---|---|---|---|
| LEGAL OWNER / PROJECT: STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT | | | |
| TITLE: RING VENTILATION · DECK B · QUADRANT 2<br>C04–C06 · ductwork | | | |
| DISCIPLINE: Mechanical (HVAC) | DOCUMENT STATUS: ISSUED FOR MODELLING | SCALE: AS SHOWN | SIZE / UNITS: A1 · mm |
| CREATED BY: Claude (AI) | APPROVED BY: — | DATE OF ISSUE: 2026-10-17 | PROJECTION |
| IDENTIFICATION NUMBER: SF-OUT-M-206 | | REV: A | SHEET: M-206 |

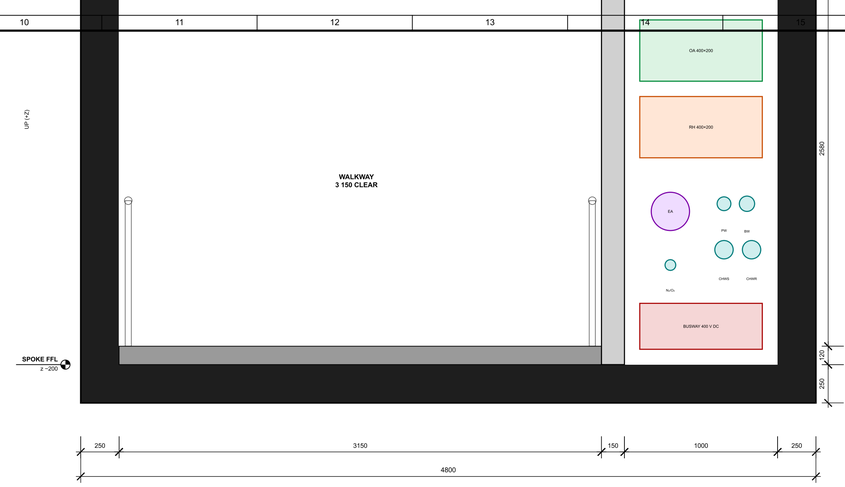

COMPARTMENT C08 · θ 195°–225° · DECK B FFL z -3500
COMPARTMENT C07 · θ 165°–195° · DECK B FFL z -3500
COMPARTMENT C09 · θ 225°–255° · DECK B FFL z -3500
B6
B7
B8
B9
PASSAGE
805.01
SPOKE 3 LOBBY
807.01
WATER BOOSTER
807.02
COLD ROOM +3°C
805.01
FREEZER -20°C
805.02
DRY STORES
805.03
DRY STORES 2
805.04
WASTE / COMPACTOR
805.05
LAUNDRY
805.06
CORRIDOR
805.07
BUNK ROOM · 4
805.08
BUNK ROOM · 4
805.09
BUNK ROOM · 4
805.10
BUNK ROOM · 4
805.11
BUNK ROOM · 4
805.12
HYGIENE
805.13
HYGIENE
805.14
AIRLOCK AL-H2
805.15
TRANSIENT LOUNGE
805.16
SUBSTATION ES-H2
807.03
PLANT · AHU-R1
807.04
PLANT · AHU-R2
807.05
PLANT · AHU-R3
807.06
PLANT · AHU-R4
807.07
SHAFT RS-n
SHAFT ES-n
SHAFT SA-n
SHAFT RA-n
SHAFT EA-n
SHAFT AL-n
SHAFT AL-H2
SHAFT SA-H2
SHAFT RA-H2
SHAFT EA-H2
SHAFT SA-H2 DECK B FFL z -3500
SHAFT RA-H2 DECK B FFL z -3500
SPOKE 3 TO DECK B SHAFT AL-H2
SA 400×200 OVERHEAD
BOD +2 600 AFF
RA 400×200 OVERHEAD
BOD +2 600 AFF
EA 400×200 OVERHEAD
BOD +2 600 AFF
SA 500×200 OVERHEAD
BOD +2 600 AFF
RH 400×300 SPILL TO HUB
BOD +2 600 AFF
SA 400×200
BOD +2 600 AFF
EA 400×200
BOD +2 600 AFF
SA 400×200
BOD +2 600 AFF
RA 400×200
BOD +2 600 AFF
EA 400×200
BOD +2 600 AFF
SA 400×200 FROM HUB EQ-R4
0.50 m³/s
BOD +2 600 AFF
RACK
SHELF
SHELVES
FOLD BENCH
TC
W
M
D
FR45
FR50
FR54
FR56
FR58
FR60
FR62
FR64
FR66
FR68
FR70
FR72
FR74
FR76
FR78
FR80
FR82
FR84
FR86
FR88
FR90
FR92
FR94
FR96
FR98
FR100
FR102
FR104
FR106
FR108
FR110
FR112
FR114
FR116
FR118
FR120
FR122
FR124
FR126
FR128
FR130
FR132


COMPARTMENT C11 · θ 285°–315° · DECK B FFL z -3500
COMPARTMENT C10 · θ 255°–285° · DECK B FFL z -3500
COMPARTMENT C12 · θ 315°–345° · DECK B FFL z -3500
B10
B11
B9
B12
GENERAL STORES
B1T105
92.4 m²
HAZMAT LOCKER
B1T106
LAB OFFICE
B1T121
19.8 m²
PASSAGE
B1T100
DATA / COMMS NODE
B1T101
16.8 m²
ENGINEERING SPARES
B1T102
48.8 m²
PASSAGE
B1T103
SAMPLE PREP / GLOVEBOX
B1T104
48.8 m²
GENERAL LABORATORY
B1T123
88.9 m²
SPOKE 4 LOBBY
B1T107
45.8 m²
WATER BOOSTER
B1T122
SUBSTATION ES-N-4
B1T120
14.8 m²
PLANT - AHU-04
B1T124
17.2 m²
CHEMICAL STORE
B1T125
SA 400×200 ROD +2 600 AFF
RA 400×200 OVERHEAD ROD +2 600 AFF
SA 500×200 OVERHEAD ROD +2 600 AFF
RH 400×300 SPEL TO HUB ROD +2 600 AFF
EA 400×200 FROM HUB EQ.SP. 0.50 m³/s ROD +2 600 AFF
PALLET RACK
SHELVING
VENTED CABINETS
WORKBENCH
SHAFT RS-4
SHAFT EA-4
SHAFT RA-4
SPOKE 4 DUCT CROSSING A BULKHEAD
FR74
FR75
FR80
FR82
FR84
FR86
FR88
FR90
FR92
FR94
FR96
FR98
FR100
FR102
FR104
FR106
FR108
FR110
FR112
FR114
FR116
FR118
FR120
FR122
FR124
FR126
FR128
FR130
FR132
FR134
FR136
FR138
FR140
FR142
FR144
FR146
FR148
FR150
FR152
FR154
FR156
FR158




## 1 HUB TRANSFER NODE S3A · MANIFOLDS

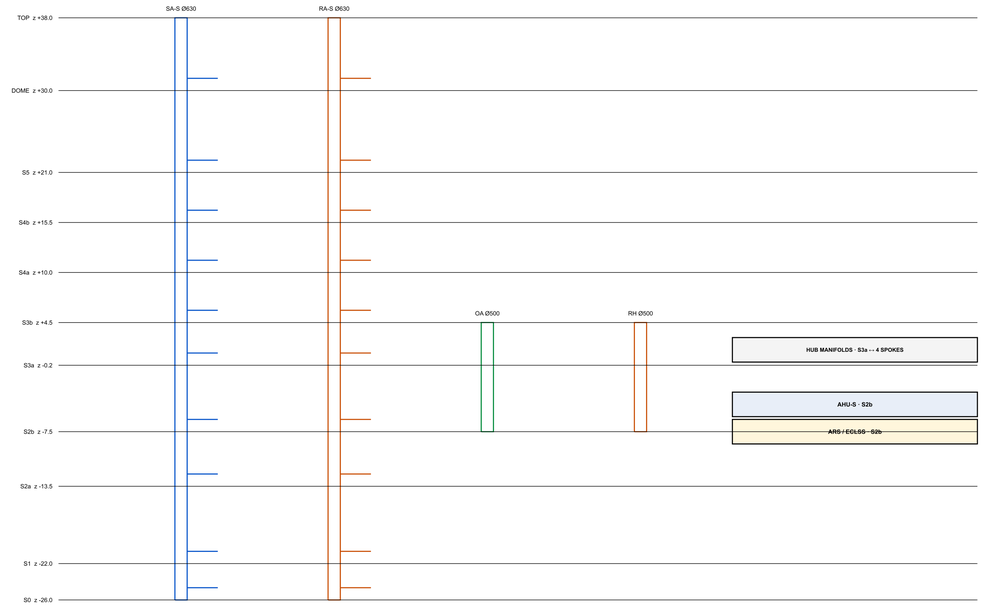
SCALE 1:50




## 2 SPOKE CHASE (TYPICAL)

SCALE 1:20

### SPINE RISER DIAGRAM · NOT TO SCALE




| | | | |
|---|---|---|---|
| LEGAL OWNER / PROJECT: STARFORGE · OUTPOST STARBASE (TIER 0) · INTERIOR FIT-OUT | | | |
| TITLE: **HUB NODE · SPOKE CHASE · SPINE RISERS** — Ventilation at the hub transfer node and in the spokes | | | |
| DISCIPLINE: Mechanical (HVAC) | DOCUMENT STATUS: ISSUED FOR MODELLING | SCALE: AS SHOWN | SIZE / UNITS: A1 · mm |
| CREATED BY: Claude (AI) | APPROVED BY: — | DATE OF ISSUE: 2026-10-17 | PROJECTION |
| IDENTIFICATION NUMBER: **SF-OUT-M-301** | | REV: **A** | SHEET: **M-301** |